

OPERATOR'S MANUAL

AUTOLUMO A2000 PLUS

Automatic Luminescence Immunoassay Analyzer

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AUTOLUMO

AUTOBIO LABTEC INSTRUMENTS CO., LTD.

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Contact

If you have any questions related to your AutoLumo A2000 Plus, Automatic Luminescence Immunoassay Analyzer, please contact us by email with a short description of your question to the address below. Any suggestions for improvements to our products and services are gladly acceptable.



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In Vitro Diagnostic Medical Device



This product is used for *In Vitro* Diagnostic Purpose.

**Waste Electrical and Electronic Equipment (WEEE)**

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For more information concerning the correct disposal of this product, please contact your local authority or the dealer.

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CHAPTER 1 INTRODUCTION

1.1 System Description and Intended Use

1.1.1 System Description

System consists of an instrument and a control unit, of which the instrument performs all the functions of the automatic sample and analysis operation, and the control unit through the client software to control the instrument. The whole automation process includes the patient sample information input to the test results output.

1.1.2 Intended Use

Automatic Luminescence Immunoassay Analyzer is an *in vitro* diagnostic equipment, based on the principle of magnetic particle chemiluminescence. The device is used in conjunction with AUTOBIO's series magnetic particle chemiluminescent reagents, can measure for qualitative and quantitative determination of clinical immunization programs which derived from human serum, plasma or other body fluid samples, including tumor markers, hormone, pathogen antigen or antibody class project.

Contraindications: None

1.2 Manual Content

This manual is mainly for guiding the user to complete relevant operation and general maintenance. It includes the following contents: a detailed description of the instrument features and general operating concepts, specifications, operating principles of quality control, calibration functions, operating techniques, emergency contingency measures, product marking and maintenance measures.

1.3 Additive Supplies

AutoLumo A2000 Plus system provides the following additional documents:

An installation CD of Automatic Luminescence Immunoassay Analyzer;

A software operation manual;

1.4 Technical Support

If you need any technical supports, please call our service hotline, [86]-400-056-9995, or contact our service support representatives. For Any technical support, the series number of instrument is necessary to provide.

1.5 Manufacturer

Autobio Labtec Instruments Co., Ltd.

No.199, 15th Ave, National Eco & Tech Zone, Zhengzhou 450016, China

Tel: [86]-400-056-9995

1.6 Service Provider

Autobio Diagnostics Co., Ltd.

No.199, 15th Ave, National Eco & Tech Zone, Zhengzhou 450016, China

Tel: [86]-400-056-9995

1.7 Product Information

Product Name	:	Automatic Luminescence Immunoassay Analyzer
Product Model	:	AutoLumo A2000 Plus
Production Date	:	Refer to the label
Service Life	:	7 years (The service life of the instrument is related to the operating environment and frequency of usage. Regular maintenance can prolong the service life. If there is any question, please contact with after-sales service provider.)

1.8 Precaution & Safety Symbol

1.8.1 Safety Features

The safety symbols on the instrument remind you of caution. Read contents of the symbols and instructions of this manual carefully when running the instrument. If any questions, please contact our technical service department of Autobio Diagnostic Co., Ltd.

1.8.2 Protective Device

AutoLumo A2000 Plus has protective device inside to avoid the user from harm. When open the upper cover of the instrument, the protective device will stop all moving parts. If open the upper cover when the instrument is working, the device will stop all movements in process.

1.8.3 Safety Symbol

In case of operating **AutoLumo A2000 Plus** not according to the accurate safe operation procedure, there are some areas may cause harms to body or instrument damages. In consideration of safety, we paste the below security symbols on these areas.



CAUTION:

Please check 1.9.3 part of operator's manual. When you see the following symbols on the instrument, you can understand the meaning of those symbols.

Symbol	Description
 General Warning Sign:	Remind you of reading important information on the label or in Operator's manual, or of noticing potential danger existed along with another symbol nearby.
 Electrical Safety:	Attend the potential risk of electric shock, especially for the high voltage parts, such as Photomultiplier (PMT), and power module. It's too danger to open these parts for checking and maintenance without trained professionally.
 Biological Hazard:	Indicate that there is the risk of infectious human serum or blood products left on these areas of the instrument or consumables & Accessories. It's necessary to clean these areas according to the correct regulation of Lab or dispose of these consumables & accessories according to local official regulation.
 Biological Risks:	Biological Risks.

Sharp Element: 	Attend the sharp element which can hurt operator. When running instruments, don't put your hand or any body parts in these areas.
Crushing Of Hands: 	Watch your hand when opening or closing the cover of the instrument.
	Loading of reaction vessels can not exceed the warning line. It may cause risks of abnormal supply of RV.
Keep away from water: 	Keep away from water. Please reading operator's manual before operation.
Emergency Sample Position 	When the instrument is running, if there any emergency sample, open the sample carrier cover, put the sample carrier with emergency samples into the position of this sign.
Emergency Stop 	Show the Emergency Stop position. During the operating of instrument, if press the button, the instrument will stop running.
Preorde & Turnplate Function  Button	In the state of ready for work or running, press the button can change the reagent. See the details of 8.3.1

1.8.4 Biological hazards

The material derived from human, animal, tissue culture or *in vitro* culture must be handled in accordance with the principles of potential infection risk. When dealing with biological hazardous substances, operator should always wear suitable protective outfit, such as approved disposable gloves, waterproof lab coats and safety goggles. Handle biological harmful substances according to the process of equipment biological hazards.

The following items must be carried out as potential biohazards substances:

All *in vitro* diagnostic equipment, pretreatment equipment, patient specimens, serum based calibration reagents, quality control (QC) products and waste materials.

Touch these potentially harmful biological materials: such as injectors, tubes, waste containers, storage, sample needle, reagent needle, cleaning tank, reaction plate cover, etc.

Users should contact the manufacturer if the instrument needs to be scrapped, and it cannot be handled as a regular item.

1.9 System Configuration and Feature

1.9.1 Space Requirement

The size of the instrument & weight of each parts are shown in the below sheet.

Instrument: when the cover and drawer of the instrument are closed	Length=1374 mm Height=1200 mm Width=950 mm
Instrument: when the cover and drawer of the instrument are opened	Length =1374 mm Height =1700 mm Width =1530 mm
Ventilation , safe and efficient operation, maintenance space required	Back=340 mm Top=800 mm Front=800 mm
Space requirement for System control flat (including outer computer and go-cart)	Width=600 mm Height=1700 mm Depth= 950 mm

1.9.2 Weight of Instrument

Instrument without any consumables and accessories	390 kg
Computer	Refer to manufacturer document

1.9.3 Performance Parameters

Name	Automated Chemiluminescence Immunoassay Analyzer
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Assay method	Magnetic particle chemiluminescent immunoassay (CMIA)
Analysis method	Stochastic analysis test
Number of samples	A batch of 100 samples (can be cyclic appended), there is a priority channel for emergency samples.
The number of analysis items at one time	24
The number of reagent slots	24
Test speed	200 tests / hour
Add sample method	Teflon needle (with liquid level sensing and clot detection)
Sample carrier	Sample rack (5 samples / rack)
Sample container	Blood collection tubes, special sample containers
Sample handling capability	192 incubation position, five times magnetic cleaning
Sample adding amount	5 μ L-150 μ L
Sample tube/cup dead volume	150 μ L
Sample dilution function	Inner dilution (Maximum 500 times dilution)
RV(Reaction Vessels) supply	Continue supply
The fastest time of sampling	21 minutes
The slowest time of sampling	40 minutes
Carry contamination	carrying contamination rate of the sample needle $\leq 10^{-6}$
Repeatability	Intra-batch measurement repeatability (CV, %) $\leq 8\%$
Linearity	Intra-batch linear correlation coefficient (r) ≥ 0.99
Stability	The relative bias of 8h and 4h is not more than $\pm 10\%$
Data processing method	Four parameters, linear regression, logit-log, point to point
Sound pressure level	Maximum sound power is 70 dB
Temperature scope of reagent area	4 $^{\circ}$ C $\sim$ 10 $^{\circ}$ C
Temperature scope of reaction area	Accuracy: 37 $^{\circ}$ C $\pm$ 0.5 $^{\circ}$ C; Fluctuation rate is no more than $\pm 1^{\circ}$ C of central value.

Pipettor precision	Accuracy: 50 μ l $\pm$ 2.5; Precision rate (50 μ l): CV $\leq$ 2%
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**CAUTION:**

User must use the following accessory which are combined with instrument.

Accessory for safe operation of the equipment are described as follow:

Accessory Name	Description
Power cord	Supplied by Autobio Labtec Instruments Co., Ltd.

**CAUTION:**

Change of analytical performance of the device:

When a message of error or a change of result comes out, please contact our service support engineer.

1.10 Environment Requirement

AutoLumo A2000 Plus has to be operated indoor, and the environment should meet the following standards.

Temperature	Normal Working Temperature: 10°C ~30°C
Humidity, RH (No agglutination condition)	Normal Working Humidity: $\leq$ 85% Optimal Working Humidity: 50%~70%
Maximum altitude	2000m
Atmospheric pressure	85kPa ~106kPa
Brightness	Avoid direct Sunshine and Ultraviolet light
Operating ambient temperature change rate	2°C per 30 mins
Pollution degree	Based on EN 61010-1-2010, AutoLumo A2000 Plus applies to the pollution degree: class II.
Electromagnetic interference	Far from Electromagnetic interference
Other instruments interference	Far from instruments and equipments with high power, strong amplitude, strong light
Water quality requirements	Sterility deionized water: < 10 cfu/mL Conductivity: $\leq$ 5 μ s/cm (at 25°C) Hydraulic pressure: 50~340 kpa (0.5~3.4 kg/cm ²) Water consumption: 10L/h

1.11 Electrical Requirement

AutoLumo A2000 Plus has two separate power supply: one is for instrument and the other is for outer computer, printer, etc. Each power should meet the specific requirements.

1.11.1 Instrument

Power supply should meet the following requirements in order to protect the instrument from damage:

Input power supply	AC 100V—230V, 50/60Hz, Single phase
Power cord Attributes	Special (only for power supply of AutoLumo A2000 Plus)
Connector of Power cord	AC 250V,10A
Plug of Power cord	AC 250V,16A
Power cord	300/500V
Power cord length	Less than 2m
Voltage Fluctuation	No more than $\pm 10\%$
Maximum resistance of Safety Grounding between instrument and the laboratory	No more than 0.1Ω
Overvoltage category	II



CAUTION:

- 1) Please use Power cord which supplied by our company.
- 2) Do not replace Power cord of the instrument by inadequately rated cords.

1.11.2 Computer

Items	Requirements
Processor	Intel Pentium 4 1.8G
Internal storage	1G
Hard disk	80G
Safety certificate	Compliance with CE certificate



CAUTION:

The computer should conform to the specification of current version of IEC/EN 60950 -1.

1.11.3 Power Consumption

PARTS	POWER CONSUMPTION
Instrument	800 VA
Computer	See computer manual

1.11.4 Standby Power (UPS)

No Standby Power Supply (UPS) provided along with AutoLumo A2000 Plus. If required, the UPS with Local Earthing Isolator and Low Battery Indicator is recommended. For more information about UPS, please contact our technical support department.

UPS Configuration Requirement:

Minimum Power Output	2400 VA
Output Voltage	100-230V AC
Output Frequency	50/60Hz
Output Wave	Sine Wave
Working time without load	1000 VA, 15 mins at least
Certificate	CE



CAUTION:

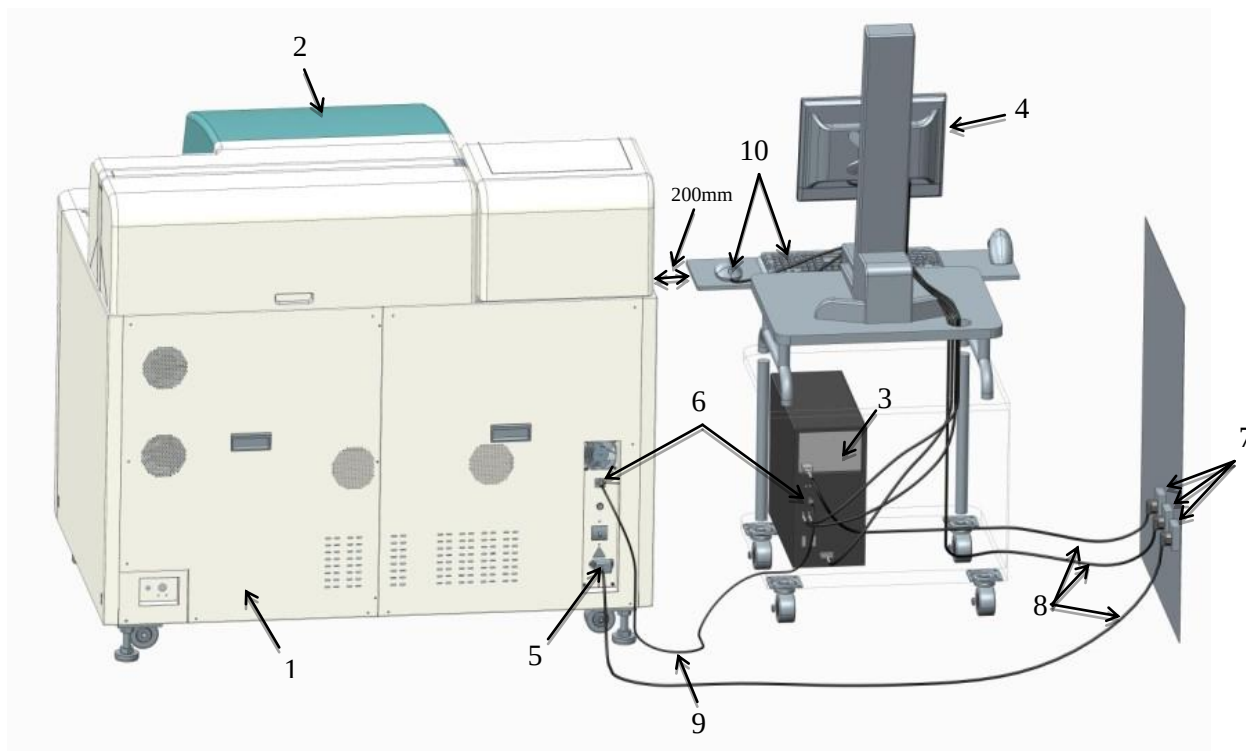
If you need to disconnect all parts of the instrument, the power supplies to the equipment and supplies energy should be disconnected firstly.

1.12 Installation

The **AutoLumo A2000 Plus** can be unpacked and installed only by authorized staff or technical person of AUTOBIO. Do not take out the instrument from its package without engineers presenting to the site.

Interconnection of instrument and other equipment:

Number	Name	Number	Name
1	Instrument	2	The cover of instrument
3	PC	4	LCD
5	Instrument power input	6	Network port
7	Mains supply outlet with protective earth	8	Power cord
9	Network cable	10	Input device



Interconnection instruction



CAUTION:

- 1) The ground of instrument installation position should be flat and hard, and its bearing capacity is not less than 300kg/m^2 . The dimension for floor should be: Length is not less than 1730 mm; Width is not less than 1380 mm.
- 2) The instrument must be installed at the position which is easy to operate power and emergency braking switch.
- 3) Contact with professional engineers before transporting or moving the instrument.
- 4) Make sure that the side shell must be completely installed before the instrument is powered on.
- 5) Conductor which is easy to be accessed or may cause a shock hazard should be grounding. Please connect the Power cord supplied by manufacturer to supply mains with a protective earth.

- 6) Failing to comply with the usage specified by the manufacturer may cause damage to the protective function of the instrument.
- 7) The instrument can be connected to the computer by communication cable. Please refer to the label on the instrument and corresponding that on computer.
- 8) Before using the instrument, please use screwdriver, hex wrench to ensure that all screws, fasteners are fixed tightly on the instrument.
- 9) During normal operation of the instrument, please ensure that all covers and doors of the instrument are closed.
- 10) If the door and all covers of the instrument need to be removed, please contact with the manufacturer. Service personnel or engineer will use specified screwdriver, hex wrench to remove them.
- 11) Drainage systems equipment.

Drainage systems parts	Requirements
Height of waste solution discharge	< 1m
Inner diameter of waste solution discharge	$\geq$ 12mm
Length of waste solution discharging pipe	$\leq$ 10mm

1.13 Warranty

AUTOBIO warrants that **AutoLumo A2000 Plus** delivered to you will be repaired in accordance with the warranty provisions in the purchase contract for A2000 Plus or reagents. Clients shall be responsible for execution of the protective maintenance and cleaning procedures. If malfunction caused by fail of execution of the above procedures within specified time, fees of repairment and spare parts shall be born by clients.

The following situations are not covered by warranty.

1. Loss of Warranty Label on the instrument.
2. Instrument serial number is not provided by our company or has been altered.
3. Installation, Power supply and operation of Instrument & Accessories fails to comply with this manual and its appendix.
4. Damages caused during delivery.
5. Damages caused by spattering of reagent, contact with corrosive substance or air.
6. Instrument is disassembled or spare parts are changed by any unauthorized personnel.
7. Damages caused by unclear internal voltage.
8. Damage caused by natural disasters or force majeure.
9. Software, hardware malfunction or damage caused by improper operation and computer viruses, etc.

Free repairs shall be supplied by AUTOBIO if there is any malfunction during the warranty period. AUTOBIO shall have the right to decide which spare parts should be changed, and the warranty period of spare parts should be same with that of the instrument.

Warranty service is non-transferable and does not apply for any defects caused by misuse, changes, improper delivery or maintenance by unauthorized engineer.



CAUTION:

The Instrument is designed according to related standards, and substantially conforms to EN 61010-1-2010 Safety Standards.

It's safe to operate according to this Operator's Manual. Don not try to make any changes to the instrument, otherwise, the below results will happen.

- No more quality guarantee supplied by the manufacturer.
- Not complying with the requirements of EN 61010-1-2010 Safety Standards.
- Potential safety hazards.

AUTOBIO will not be responsible for any damages or defects caused by using the instrument for any purpose other than the intended purpose, or repairment by any unauthorized person.

Use period of instruments associated with the instrument operation environment, frequency of use, etc., and regular maintenance can prolong life.

If any questions, please contact the service provider.

Parts List:

NO.	NAME	Quantity	Instruction for use
1	PVC tube	15m	Used for connection between distilled water and waste dilution, cut to suitable length according to installation situation at the site
2	Φ1×Φ3 Silicone tube	0.5m	Cut to suitable length according to maintenance
3	Screw lid	10Ps	Use when instrument is In short of screw lips
4	Fuse(2A)	2Ps	Change when the fuse is damaged
5	Fuse(1.3A)	3Ps	Change when the fuse is damaged
6	Liquid distribution	1Ps	Used for extra position distilled water bucket

1.14 Used in Combination with other Products

The instrument needs to be used in conjunction with the reagents of AUTOBIO Diagnostics Co., Ltd, please read reagent instructions carefully before use.



CAUTION:

If operation can not comply with the reagent instructions or attempting to use other company's reagents, any report results and responsibilities arise from above action will not be borne by AUTOBIO.

CHAPTER 2 SYSTEM SUMMARY

2.1 System Description

1) Structure and components: This product contains three parts: main instrument, computer(Optional) and software. The main instrument is composed by sample transportation system, sample adding system, reaction system, reagent storehouse, test system, bar code reader, communication interface and power device.

2) Automobile A2000 Plus System has two major subsystems: Functional instrument executing all sample process, & system control platform providing a human-machine interface. Interactive operation will be available at the several points as below:



(1)	Emergency Switch	(2)	Sample Loading platform	(3)	RV(Reaction Vessels) supplies station
(4)	Sample Pipettor	(5)	Reagent Pipettor 1	(6)	Reaction Tray
(7)	Reagent Carousel	(8)	Reagent Pipettor 2	(9)	Photomultiplier(PMT)
(10)	Waste	(11)	Wash Buffer	(12)	Distilled Water
(13)	System Wash	(14)	Solid Waste Container	(15)	LCD

(16)	Input Device	(17)	PC	
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2.2 System Status & Control

2.2.1 Status Indication

AutoLumo A2000 Plus has 4 Status Symbols at the left top of software interface. They can be controlled by 3 control buttons at the right bottom of software interface.

Status Indication	Description
 Ready	Ready: System is ready to use
 Running	Running: System is running. Assays are in progress and the instrument is operating in certain process.
 Pause	Pause: No new assay added, but current assays are still in the process.
 Not Ready	Not Ready: System is not ready for work. Execute system initialization firstly to check subsystem status and initialize motor, or reset the motive parts. Then, further operation can be continued only.

2.2.2 System Shortcut Buttons

Total 6 System Status Buttons are colorless during the normal running of instruments. View each Status interface by clicking each button. In the following cases, the button will become red or yellow, the left amount of consumables reach alert level, or unexpected problem with Sample processing, or event Logs error. The buttons will return back to be colorless until these problems or alarms got solved.

Status Button	Description	Alarm Color
	Access to "Abnormal Assay" lists shown on sample management interface by clicking. View which samples were detected abnormally. If the list got cleared, the button will become colorless.	RED indicates One or some Abnormal Assays happened in process.

	Access to “Work Pending” lists shown on sample management interface by click. View which samples are waiting in line for being tested. If the list got cleared, the button will become colorless.	YELLOW indicates some samples cannot be tested and still wait in line due to some reason.
	Access to “Reagent” management interface by click, where listing which reagent and its calibration are lack.	YELLOW indicates that the reagents need to load into reagent carousel and their assay calibrations need to be added to the system.
	Access to “Bulk” interface for Large volume supply of materials by click, showing the left volumes of Substrate, Washing solution, dilution solution and reaction cup as well as the left space of solid waste container and liquid waste container. If the residual availability volume of consumables is at normal level, the button is colorless.	YELLOW indicates Substrate will get expired or one liquid waste container gets full. RED indicates substrates are empty or expired or waste containers are full.
	Access to “Quality Control” interface by click, where we can set QC or check the result of QC. If the results of QC are normal, the button is colorless.	RED indicates that QC result is not in the expected range.
	Access to “System LOG” interface by click, where we can find the system LOG information, including malfunction removal information of warning and alert event. After viewing the information, the button will become colorless.	YELLOW indicates system generate a common alert event indicating that some situation needs immediate notice. RED indicates the system generates a serious alert event indicating a serious failure or error.

2.2.3 System Control Button

Control button	Description
----------------	-------------

 STOP	When system stays at RUNNING or PAUSE status, select the button for stop test operation. The system will stop processing and cancel the existing test by click the button. System Status: Running Pause  CAUTION When you select "STOP", all test requests will be cancelled, including the assay request waiting in line. Therefore, please confirm what you selected again. After executing "STOP", the systems need to be re-initialized for the following assay operation.
 PAUSE	When the system stays at RUNNING status, press for pausing test operation. By click the button, no new test will start after the scheduling test is completed. However, the sample in process will be continued. System Status: Running
 RESTART	While system stays at READY or PAUSE status, Press "RESTART", test will be started or recovered. . System Status: Ready Pause  NOTE At the system status of READY, Attend to the below situation: Don't restart the system while the amount of RV, the left space of solid waste container, left volume of dilution and substrate reach the alert level. Make sure the left volume of dilution is sufficient, otherwise, dilution adding operation can't be executed during test. Don't restart until the temperature of Substrate, Reaction Tray and Reagent Carousel reach to the normal range.

2.2.4 Other Buttons & Status Description

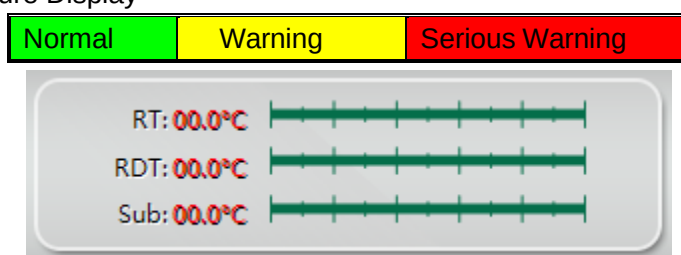
 Home	Return back to the home page
 Test Request	Directly into the test request interface

	Directly into the test result interface
	System HELP Description

1. Time Display



2. Temperature Display



2.3 System Mode

Mode	Description
	LIS HOST QUERY MODE: AutoLumo A2000 Plus may be interfaced to a LAB INFORMATION SYSTEM (LIS). Through the communication with LIS, all the information of samples as well as assay requests can be acquired by scanning the bar code of samples. Enable method: [System Setup] — [LIS Communication Setup] — [Receiving LIS Requests]—[LIS Interface]—Select[Open]
	STAND-ALONE MODE: All the information of sample and assay requests needs to be input by hand on the operational interface. Enable method: [System Setup] — [LIS Communication Setup] — [Receiving LIS Requests]—[LIS Interface]—Select[Close]



NOTE

Under LIS HOST QUERY MODE, system can read sample data and test request information input by hand, and can acquire these information from LIS system.



WARM TIP

As for the samples which system failed to scan its barcode under LIS HOST QUERY MODE, we may enter into STAND-ALONE MODE (LIS communication must be set "OPEN" in system setting interface), and input these samples' barcode by hand into the systems. Then, Open LIS communication, and perform the operation of acquiring LIS request. These samples information and assay requests will be acquired from LIS system.

2.4 Shutdown & Restart

During the running process of **AutoLumo A2000 Plus**, you need to shutdown PC or Instrument or both of them. Please follow the below steps of SHUTDOWN & RESTART.

WARNING

The below steps have to be followed, otherwise, the potential damage may happen to the instrument or system database.

2.4.1 Shutting down instrument for a while

System Status: READY NOT READY

CAUTION

Press switch on the left bottom of the instrument, the power (O position) will be off. The main interface of software on PC shows that the status of system is "NOT READY". It means the instrument powers off.

2.4.2 Shutting down the PC

There are two options to shut PC down.

1. User interface (UI) software of **AutoLumo A2000 Plus**.
2. Keyboard of PC.

NOTE

1. Use the user interface (UI) to execute standard shutdown
2. In the condition of long time do not use the instrument, select "Instrument of dormancy" on the interface of maintenance, make the instrument to enter power saving mode.

System Status: READY NOT READY

WARM TIP

Keyboard for shutdown can only be used when there are no other solutions.

1. Simultaneously Press **【Ctrl】 + 【Alt】 + 【Delete】** together
2. Select "SHUTDOWN"
3. Select one of option depending on whether you want to restart PC immediately or not.
4. Select "CONFIRM" on the POP-UP window
5. If keyboard doesn't work, please shut PC down by pressing Power Switch of PC for 10 seconds at least. Restart PC only after 20 seconds.

2.4.3 Shutting down instrument for long time

In case of relocation of instrument or shutdown of instrument for more than 5 days, Please shut **AutoLumo A2000 Plus** down. Before shutdown, please contact our technical support team firstly.



WARNING

During the process of the above performance, it's possible to touch the potential infectious materials. Please dispose of these biological hazards strictly according to the regulation of Lab. Attend to the protection of hand, eye and face especially.



NOTE

When the instrument is shut down, the cooling device of Reagent Carousel will be shut down either.

System Status: **READY** **NOT READY**

1. Run "ROUTINE CLEAN" program.
2. Take all the RV, sample tubes and racks out from Sample Supplies Unit and RV supplies Station.
3. Unpack all reagents out from reagents Carousel, put them to 2-8℃ warehouse.
4. Clean large volume waste solution container.
5. Clean Solid waste container.
6. Shut the instrument and PC down, and turn the power off.

2.4.4 Sleeping instrument

In "SLEEP" mode, the instrument stays at low power consumption status, and only provides the power to thermal insulation systems for keep working, and other parts stop working.

System Status: **READY** **NOT READY**

1. Select "MAINTENANCE" on the main interface.
2. Select "SLEEP MODE" on the "MAINTENANCE" interface.
3. Select "CONFIRM" on the POP-UP window

2.4.5 Restarting instrument

When the power of instrument is off, follow the below steps to restart **AutoLumo A2000 Plus**.

Facing the instrument, and move the bottom switch to I position.



WARM TIP

The system interface of software will stay at "OFF-LINE" status when the PC just starts.

1. Make sure upper cover of the instrument is at "Closed" status.
2. Turn on the power switch on the back side of the instrument and confirm that the emergency stop button is at back-out status.
3. Please perform "RESTART" if the computer is not running and start **AutoLumo A2000 Plus** system.
4. If the user interface software is running, perform "reinitialization" in the maintenance interface.
5. If the instrument fails to initialize, please contact with our Technical Support Department.
6. Don't load samples or open the cover of the instrument until the internal temperature of instrument reach the required level. Normally, it will take 15-20 minutes for the system to recover the temperature to the level required.
7. Make sure the system enters into "READY" mode. If the system cannot get ready after 60 minutes, please contact technical support team.
8. "Wash substrate" is suggested after "DAILY CLEAN" at the first time. Then, run the system regularly.

2.4.6 Restarting after long term shutdown

In case the instrument is shutdown for a long time, please restart **AutoLumo A2000 Plus** by the following steps.



WARNING

During the process of the above performance, it's possible to touch the potential infectious materials. Please dispose of these biological hazards strictly according to the regulation of Lab. Attend to the protection of hand, eye and face especially.

System Status: Not Applicable

1. Restart the instrument
2. Restart the computer and user interface software.



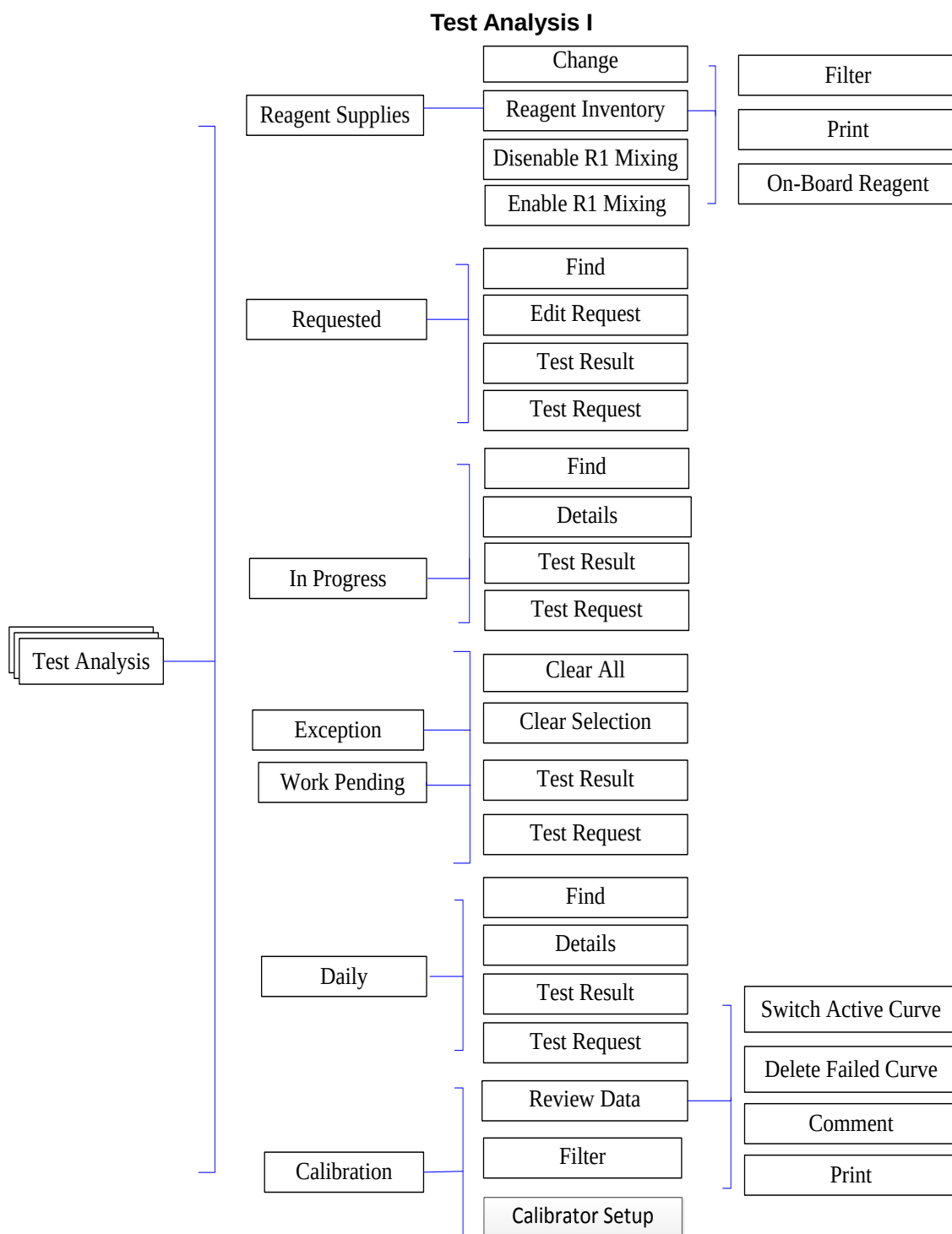
NOTE

1. It will take almost 1 hour for the system to recover the internal temperature to the required level due to long term shutdown.

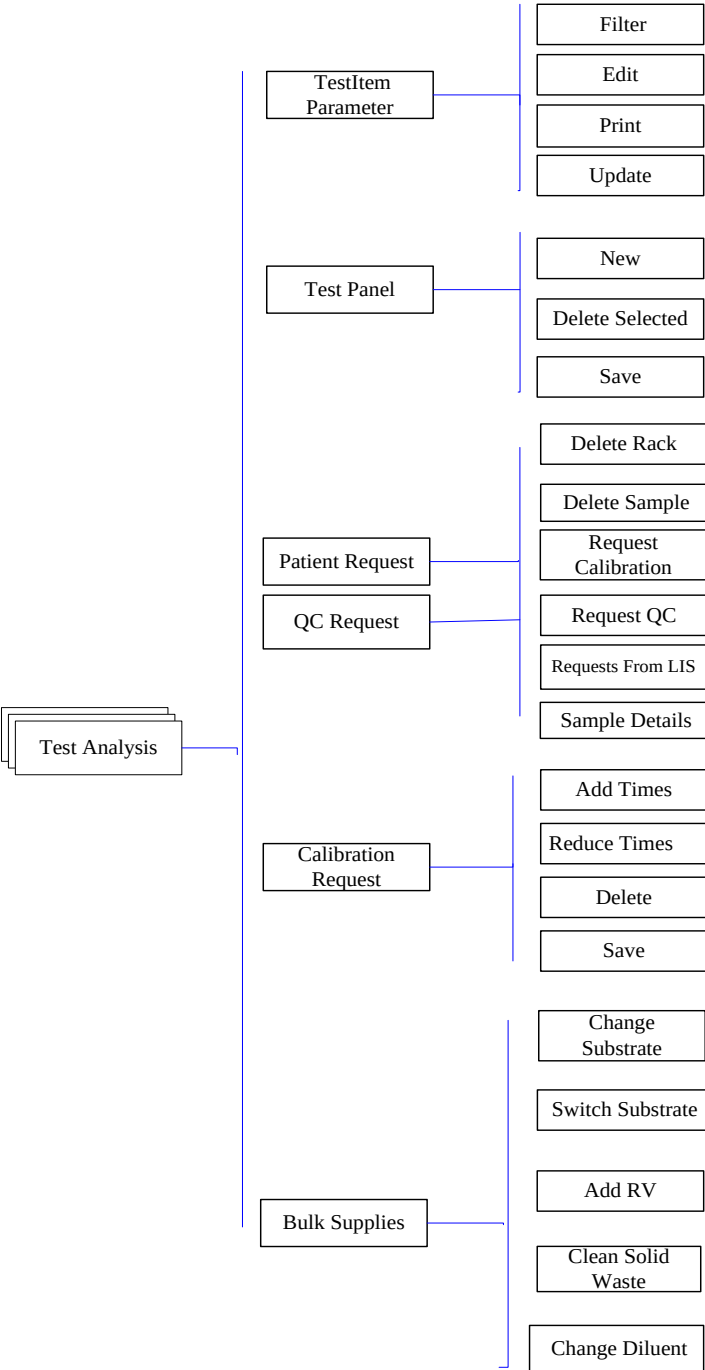
2. Don't load samples or open the cover of the instrument before the internal temperature reaches the required level.
3. Make sure of sufficient washing solution supplies. If necessary, replace with larger washing solution container.
4. Make sure of enough space of waste container left, and if necessary, clean waste solution container and solid waste container.
5. Check the amount of RV, and if necessary, add more RVs to cup supplier deck.
6. Load new substrate
7. Load the required reagents
8. Perform "DAILY CLEAN"
9. Clean the solid waste.

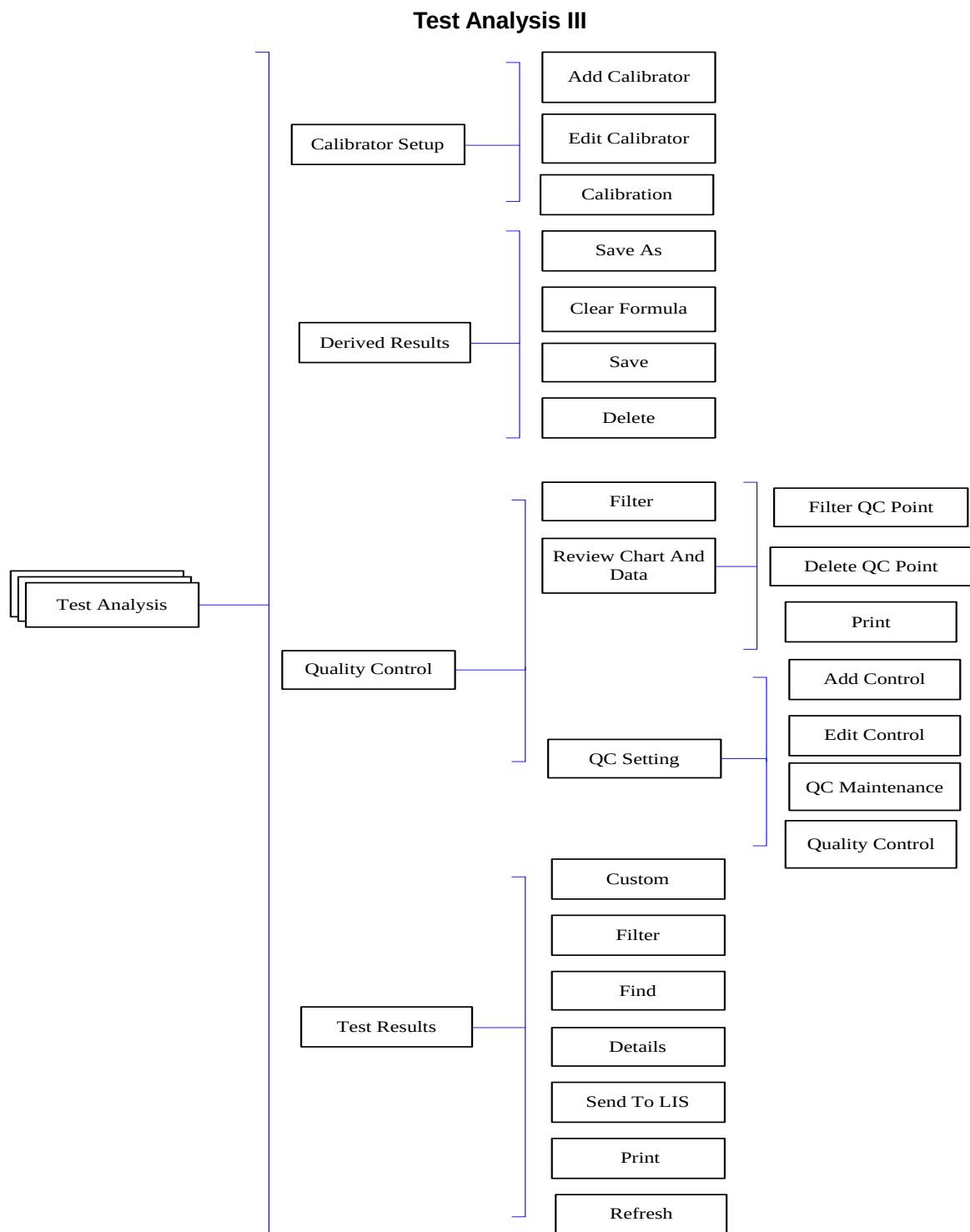
2.5 Function Navigation

2.5.1 Test Analysis

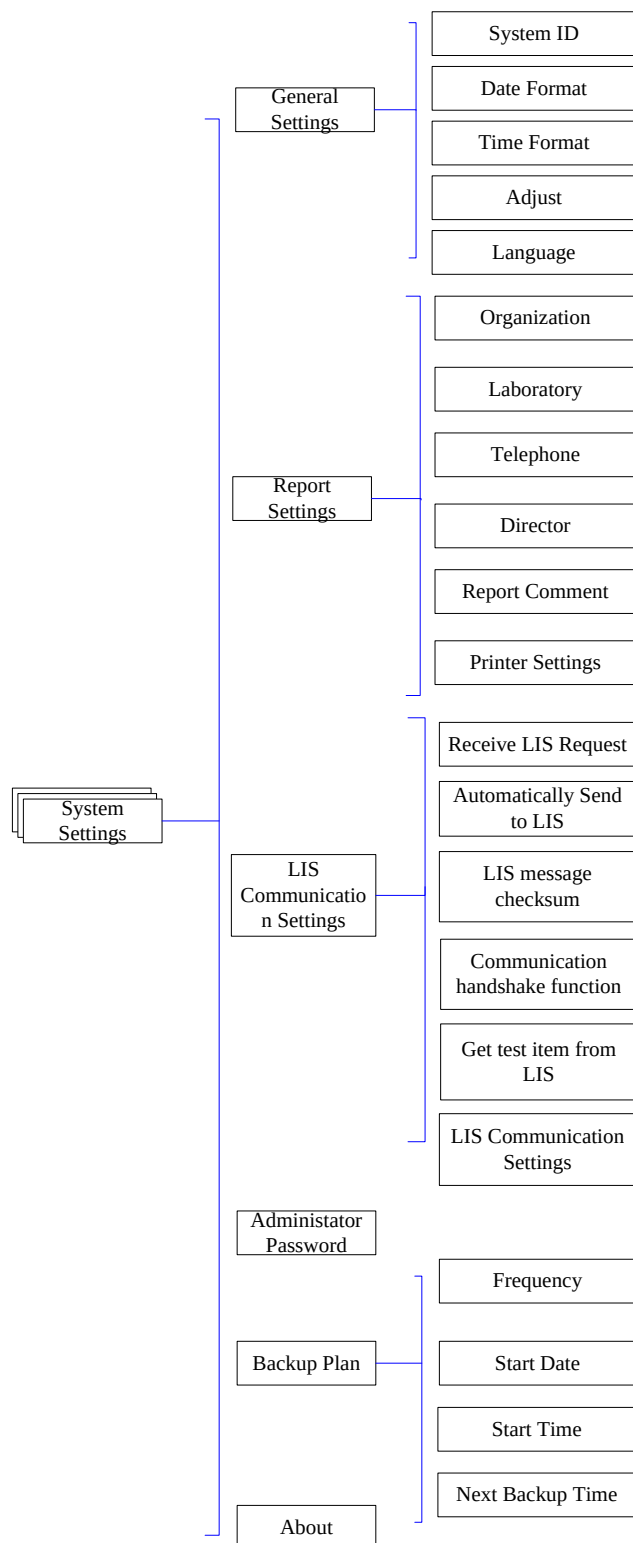


Test Analysis II

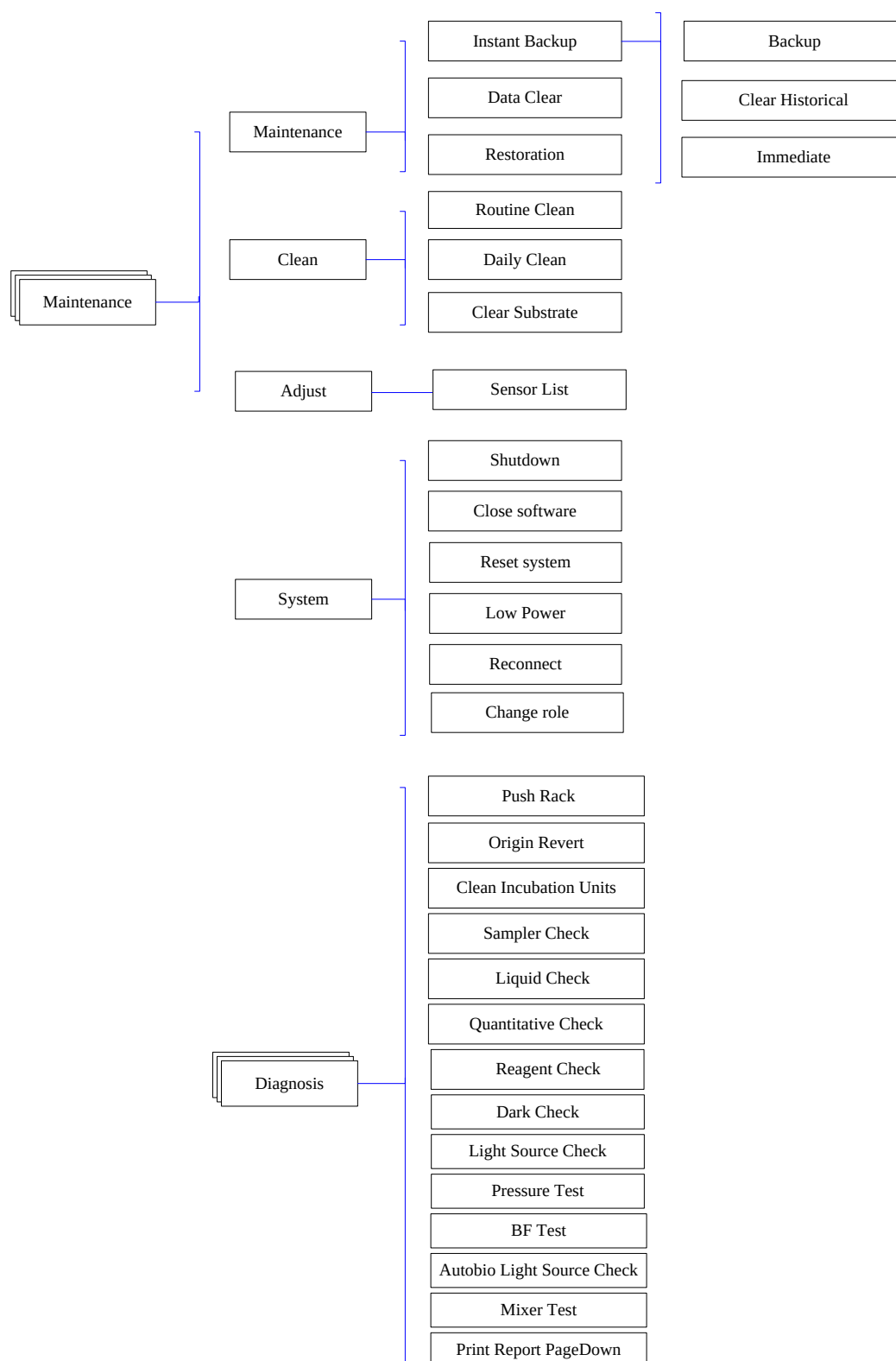




2.5.2 System Parameter Setting

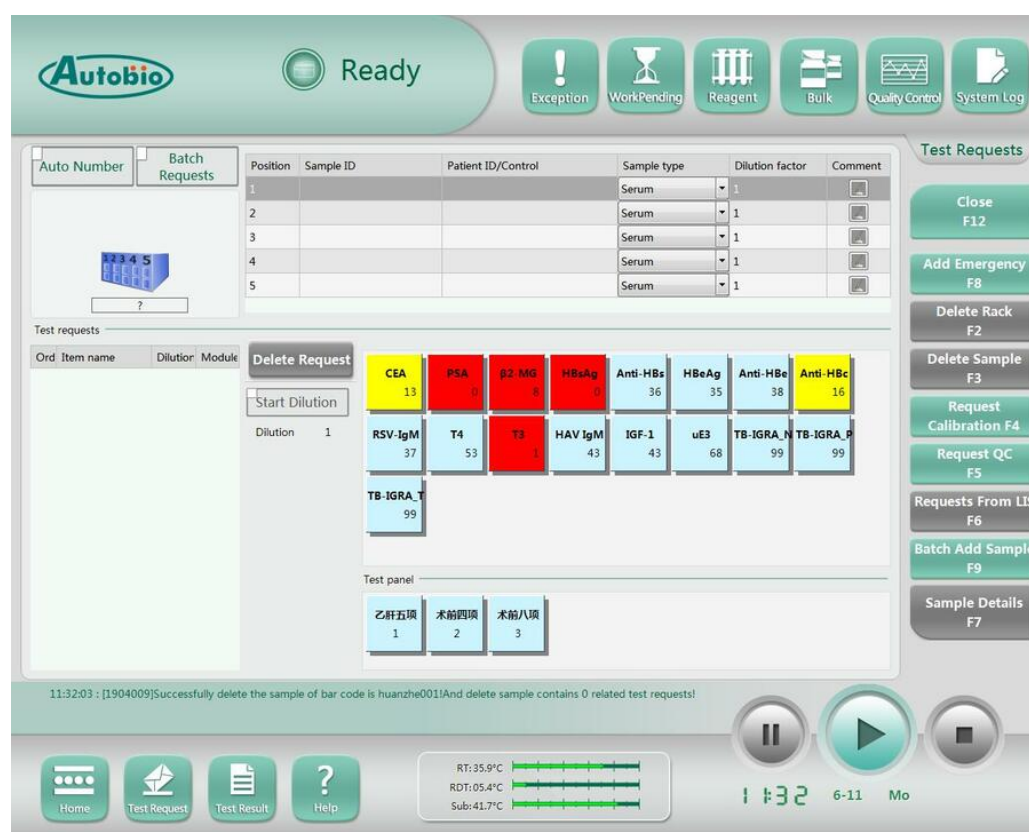


2.5.3 Maintenance and Diagnosis



CHAPTER 3 SPECIMEN OF PATIENTS

“Test & Analysis” icon on the main interface of **AutoLumo A2000 Plus** software provides access to “Test & Analysis” interface. Select “Test Requests” there to the functional areas for the specimen programming.



Interface Description:

NAME	DESCRIPTION
Sample Rack Code	Click “?” icon below the picture of Sample Rack on the left side to activate the input box.
Position	Fixed automatic numbering from 1 to 5.
Sample ID	Sample bar code or Number.
Patient ID/Control	Patient ID or control name.
Sample Type	Sample types include: Serum, Plasma & Urine etc., totally 10 sample types. NOTE: Currently, the instrument only supports serum, plasma and urine, if you need to do sample tests of other types, please contact our technical support team.
Dilution Factor	Default Factor is “1”. It's optional depending on your requirement.

Comment	Sample Description.
Auto Number	Automatically numbering the following sample series No. once initiating this function
Batch Requests	In case that a number of samples request the same tests, please select "Batch Request" function by ticking it, so that the system will perform the same tests for all the loaded samples automatically.
Close	Exit the current interface and display its previous "Test & Analysis" interface or directly back to the main interface if you didn't open "Test & Analysis" interface before.
Delete Rack	Delete the current sample rack as well as samples' information involved. When the Sample Rack No. is not correct or void, the icon is inactive.
Delete Sample	Delete the current sample information as well as test requests information it includes. When the sample doesn't exist, the icon is inactive.
Calibration Request	Switch to "Request Calibration" interface.
QC Request	Load Quality Control for tests.
Request from LIS	Acquire the information from LIS.
Sample Details	Show more information of the sample selected. If the sample doesn't exist, the icon is inactive.

**CAUTION**

1. Maximum 5 samples tube can be put into involved in each sample rack. Both information of sample racks and samples are empty while entering into "Test Requests" interface. Sample rack code need to be inputted by hand, once rack code gets identified by system, samples information on the rack will be collected automatically. If rack code cannot be identified, it will be regarded as a new one, and the samples information have to be re-added.
2. Please note that the required sample volume must be greater than 150 µl.

**WARNING**

During the process of the above performance, it's possible to touch the potential infectious materials. Please dispose of these biological hazards strictly according to the regulation of Lab. Attend to the protection of hand, eye and face especially. During loading and unloading of samples and reagents, operator must wear medical glove to protect themselves from biological hazards.

3.1 Adding Rack and Specimen

1. Click “?” icon below the picture of Sample Rack on the right side to activate the inputting column of sample Rack Code
2. Input sample rack code into the column, for example, “HZ01”, then, the sample ID column is activated too. Sample ID can be input by hand or automatically by scanning the sample rack bar code.

Auto Number	Batch	Position	Sample ID	Patient ID/Control	Sample Type	Dilution Factor	Comment
 HZ01		1			Serum	1	
		2			Serum	1	
		3			Serum	1	
		4			Serum	1	
		5			Serum	1	

3. After entering the sample number 1, you can enter the name of the medical record number/quality control product, setup the sample type, the external dilution factor, and whether there is any comment, etc. First of all, enter the name of the medical record number/quality control product.
4. Sample Type Setting: Serum
5. External Dilution Factor: 5

Auto Number	Batch Requests	Position	Sample ID	Patient ID/Control	Sample type	Dilution factor	Comment
 HZ01		1	huanzhe001		Serum	5	
		2			Serum	1	
		3			Serum	1	
		4			Serum	1	
		5			Serum	1	

6. Add the sample comment by clicking “Comment” button. Input the sample information to the pop-up window, and then click “OK” button. Thus, all the information of sample rack and samples had been input into the databases.
7. Put the sample and sample rack to their position on Sample Loading Platform. Then, Loading sample work is over.



CAUTION

1. Don't make any difference to input the information of samples and sample racks firstly or put the sample rack to its position on Sample Loading Platform firstly before test start. But, if test started, it's necessary to input the information of samples and sample racks firstly, and then, put the sample rack to Sample Loading Platform. Otherwise, the system cannot identify the samples added, so that test request is pending, and the result will be marked as “No Test” after test is over.
2. The Sample Rack Code inputted should be the same as actual Sample Rack Code loaded.
3. To ensure the reliability of test results, please be sure to the volume of

remaining sample before starting, be supposed to the volume of remaining sample $\geq 150\mu\text{l}$ (inherent volume flow rate of a sample cup) + the sample volume actual tests need.

**NOTE**

1. Sample Rack Code is composed of Identification Code and Series Number, which can be four or two characters. If the code is two characters, it should be the digit figure from 01-99. If four characters, the first two characters should be capital Letter, A-Z, and following characters should be the digit figure from 01-99.
2. Before inputting sample information, Sample ID has to be input firstly. Except for Sample ID, other information is optional based on request.
3. Sample ID and Patient ID have to be composed by capital letter (A-Z) and digit figure (0-9).
4. All the information of STAT samples is in RED.

3.2 Editing Patient Information

The procedure of editing patient information is similar as the procedure of editing specimen's information. Herein, briefly introduce the editing method according to the "Patient Info" interface.

1. Select patient sample that you want to edit, and click "Sample Details (or F7)" button. The "Patient Info" window will come out.
2. Edit all the patient information you want.

Patient info

Patient info

Patient ID: huanzhe001

Name: Surname:

Sex: ☒ Male ☐ Female Age:

Department: Ward:

Bed No.: Doctor:

Submission date: 2016-4-14 13:59:47 Inspector:

OK Cancel

3. Click "OK", then the new edited information will be saved automatically.

Click the “Sample Details” button again, the pop-up patient information is the saved information.

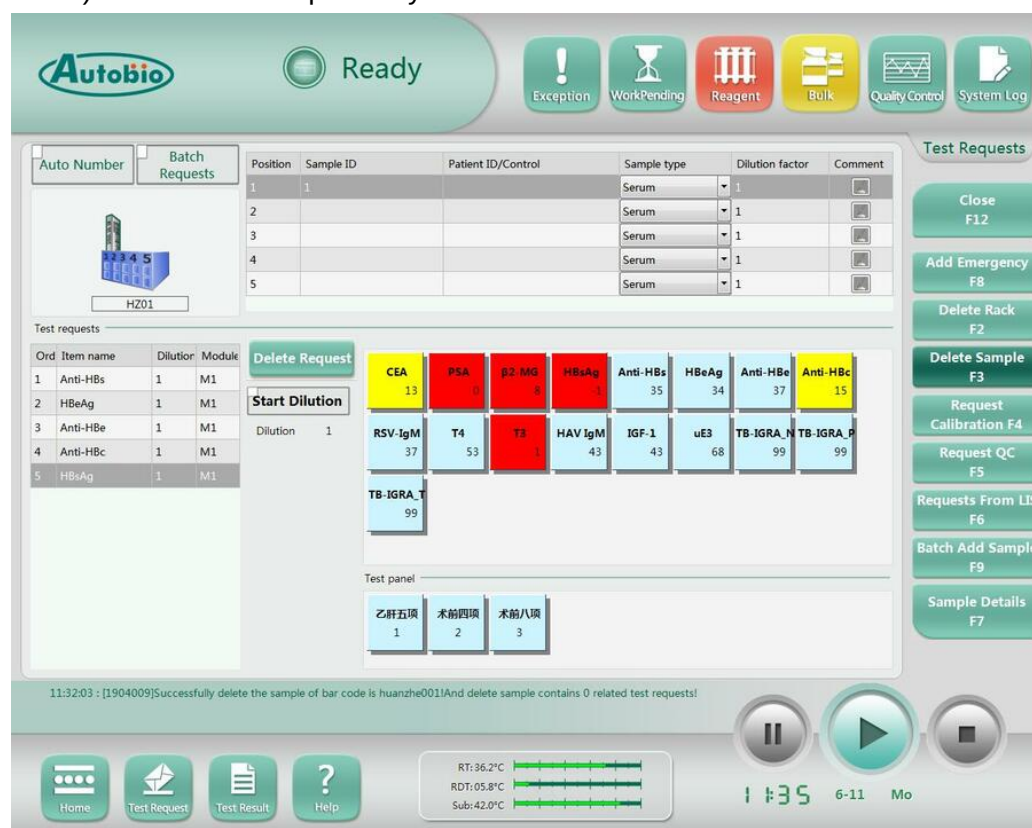
3.3 Editing Test Request

Select “Test Request” for current samples. One or more tests can be carried out for each sample.

1. Select the sample which needs test request.
2. Add test request in selection box of normative test or combination box and set Dilution Ratio.

3.4 Deleting Sample Racks and Samples

1. Deleting sample
 - a) Input Sample Rack code in the column.
 - b) Select the sample that you want to delete.



- c) Click “Delete Sample” icon on the right side of interface.
- d) Click “confirm” on the pop-up window, then, the sample and the test request information it included got deleted, Remove the samples from the sample rack.

2. Deleting Sample Rack
 - a) Input Sample Rack code in the column.
 - b) Click "Delete Rack" icon on the right side of interface.
 - c) Click "confirm" on the pop-up window, then, the Rack and the sample information it included got deleted.
 - d) Remove sample rack from Sample Loading Platform if necessary.

**CAUTION:**

The samples under the sampling line, the software will not be able to edit, nor can it be moved or removed.

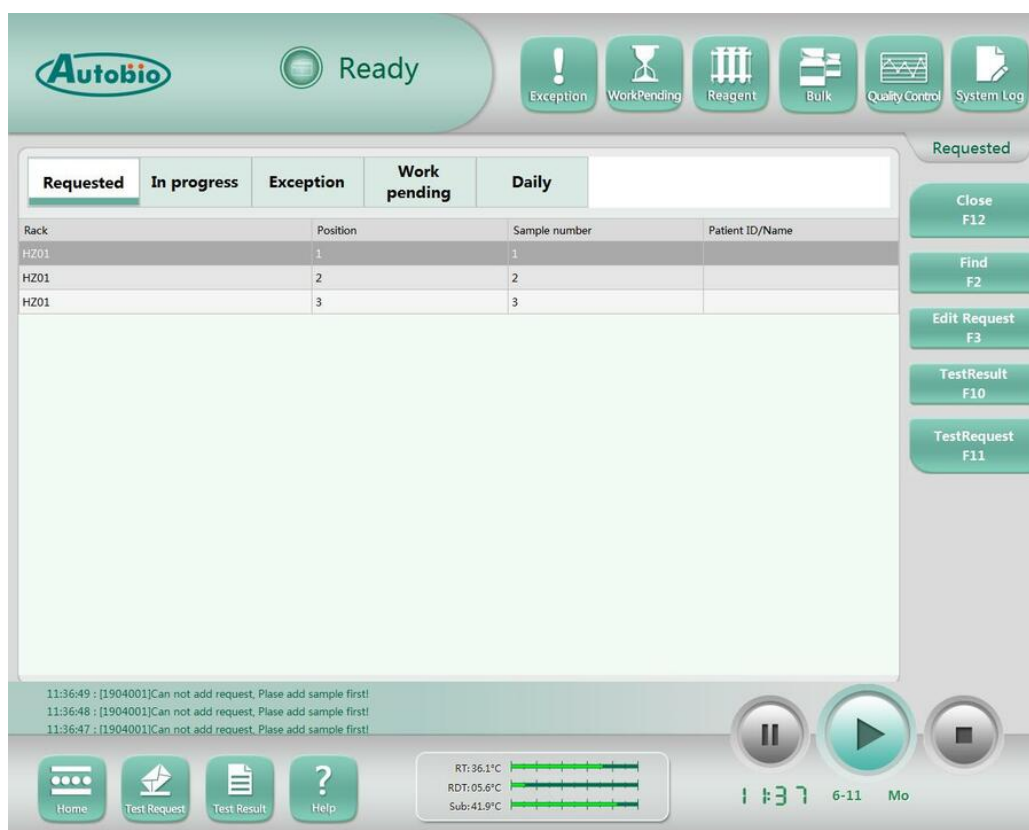
3.5 Requesting from LIS.

1. Load sample rack on the instrument.
2. Position each samples on Rack.
3. Input sample ID.
4. Select "Requests from LIS" icon.
5. The sample rack and the sample information it included will be covered after click "yes".

CHAPTER 4 SAMPLE MANAGEMENT

4.1 Requested

Enter “Test & Analysis” interface from the main menu by clicking its icon, and then select “Requested” button within the column of “Queue”. The sample Status is displayed on the “Requested” interface, which test had been ordered and scheduled for processing, but without being pipetted.



Interface Description

Button Name	Definition
Close (Labeled F12)	Exit the current interface and return back to its previous interface of “Test & Analysis”, “Maintenance” and “Diagnostic” or directly back to the main menu if you didn’t open “Test & Analysis” interface before.
Find(Labeled F2)	Search for a sample of specified Sample ID.
Edit Request (Labeled F3)	Switch to “Test Requests” interface for editing the samples selected.
Test Result (Labeled F10)	Switch to “Test Result” interface.
Test Request (Label F11)	Switch to “Test Request” interface for editing.

1. Find (Labeled F2)

Click “Find” icon, sample searching Dialog window come out, input the Sample ID for searching.

2. Edit Request (Labeled F3)

Select the sample, and click “Edit Request” icon. Then, “Test Request” interface comes out for editing the sample selected.

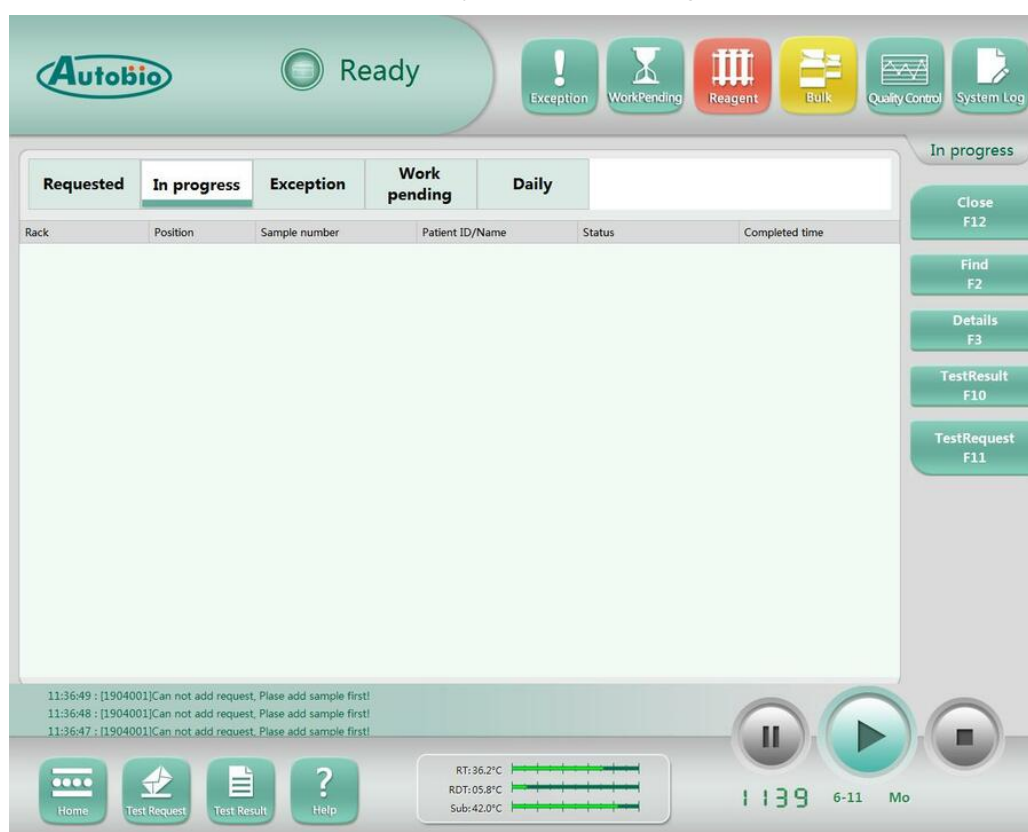


CAUTION:

Sample Rack Code and Sample Position cannot be edited unless being deleted and inputted again.

4.2 In Progress

Enter into “Test & Analysis” interface from the main menu interface by clicking its icon, and then select “In Progress” icon within the column of “Queue”. The samples which tests are in process are displayed on the “In Progress” interface.



Interface Description

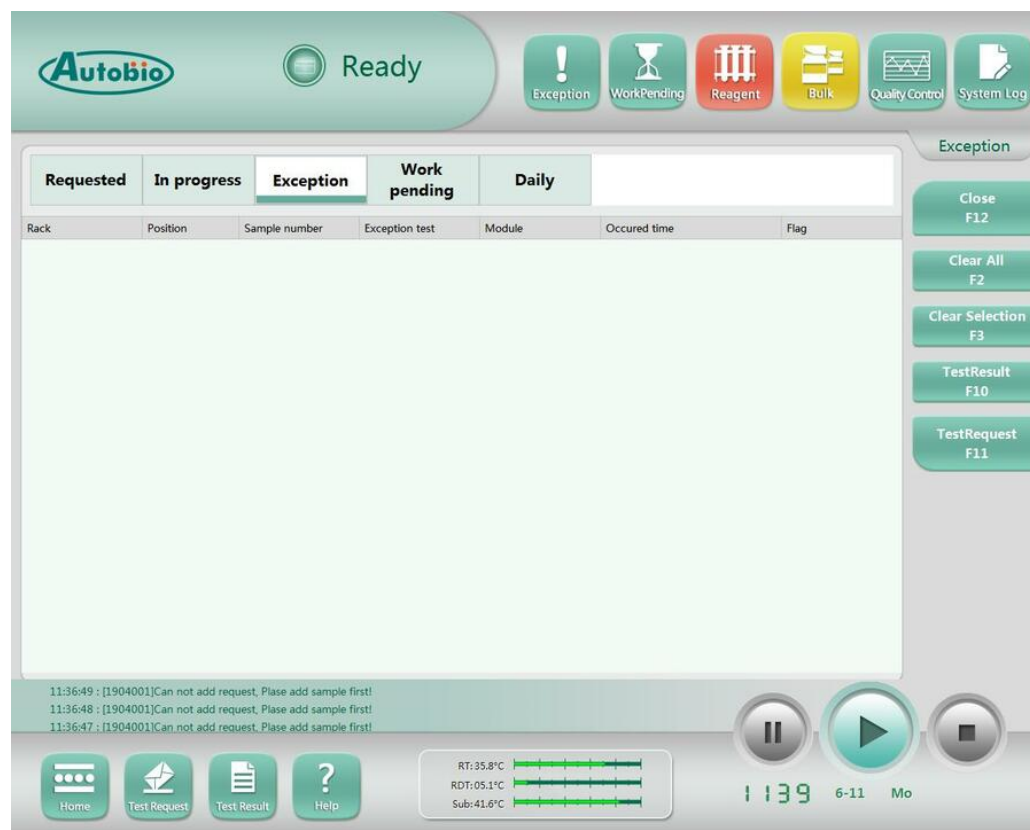
Button Name	Definition
Close (Labeled F12)	Exit the current interface and return back to its previous “Test & Analysis”, “Maintenance” and “Diagnostic” or

	directly back to the main menu if you didn't open "Test & Analysis" interface before.
Find(Labeled F2)	Search for a sample of specified Sample ID.
Details (Labeled F3)	Switch to "Test Requests" interface for editing the samples selected.
Test Result (Labeled F10)	Switch to "Test Result" interface.
Test Request (Labeled F11)	Switch to "Test Request" interface for editing.

1. Find (Labeled F2)
Click "Find" icon, sample searching Dialog window come out, input the Sample ID for searching.
2. Details(Labeled F3)
Select the sample, and click "Details" icon, then the details of selected sample are displayed in the pop-up window, including test name, finished time, status, mark and running speed.

4.3 Exception

Enter into "Test & Analysis" interface from the main menu interface by clicking its icon, and then select "Exception" icon within the column of "Queue". The samples which tests are exceptional are displayed on the "Exception" interface.



Interface Description

Button Name	Definition
Close (Labeled F12)	Exit the current interface and display its previous "Test & Analysis" interface or directly back to the main menu interface if you didn't open "Test & Analysis" interface before.
Clear All(Labeled F2)	Clear all Exception Tests on the "Exception" Interface.
Clear Selection(Labeled F3)	Clear Exception Tests selected.
Test Result (Labeled F10)	Switch to "Test Result" interface.
Test Request (Labeled F11)	Switch to "Test Request" interface for editing.

1. Clear all exception tests:

Click "Clear All" icon, and then all exception tests get deleted.

2. Clear Exception Tests selected

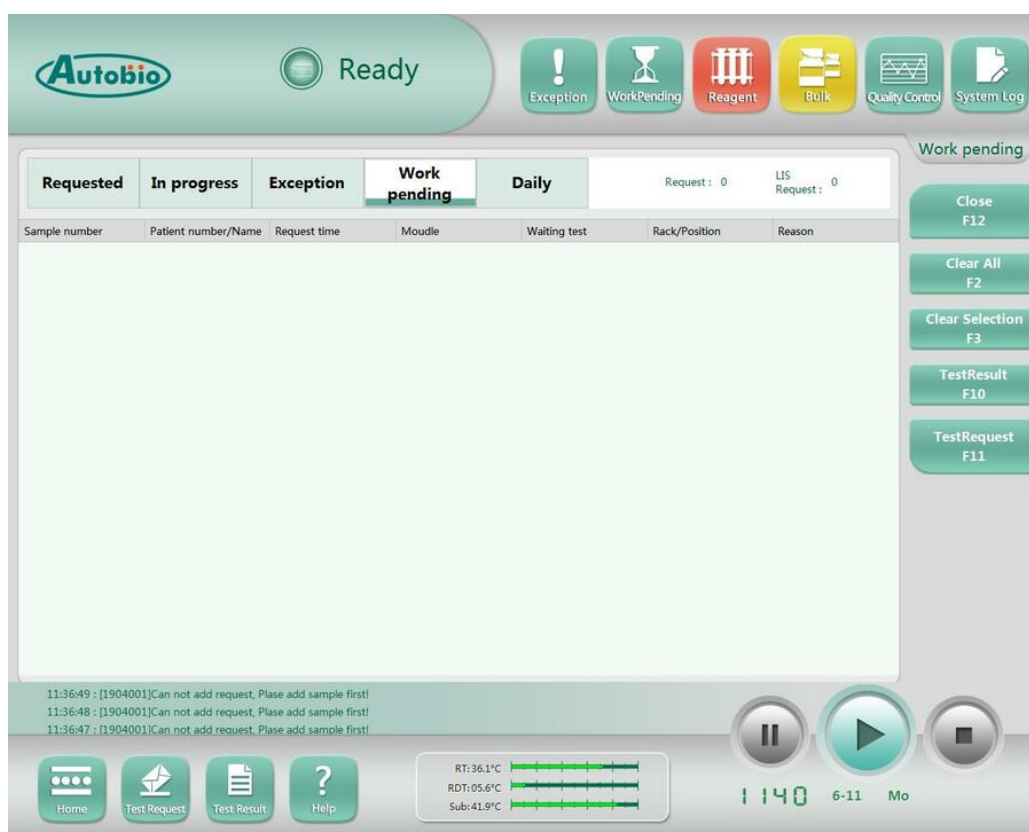
Select exception tests you want to delete, and click "Clear Selection" icon. Then, the selected exceptional tests get deleted.

**NOTE**

1. Information shown "Exception" interface include "Rack", "Position", "Sample ID", "Exception test", "Occurred Time", "Flag".
2. Refer to Appendix 1, the Description of "Fatal Flag" & "Non-Fatal Flag".

4.4 Work Pending

Enter into "Test & Analysis" interface from the main menu interface by clicking its icon, and then select "Work Pending" icon within the column of "Queue". The samples which tests assay errors happened in process are displayed on the "Work Pending" interface.



Interface Description

Button Name	Definition
Close (Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu interface if you didn’t open “Test & Analysis” interface before.
Clear All(Labeled F2)	Clear all the work pending.
Clear Selection(Labeled F3)	Select the work pending listed on the interface, and click “Clear Selection” button. Then, the selected work pending is deleted.
Test Result (Labeled F10)	Switch to “Test Result” interface.
Test Request (Labeled F11)	Switch to “Test Request” interface for editing.

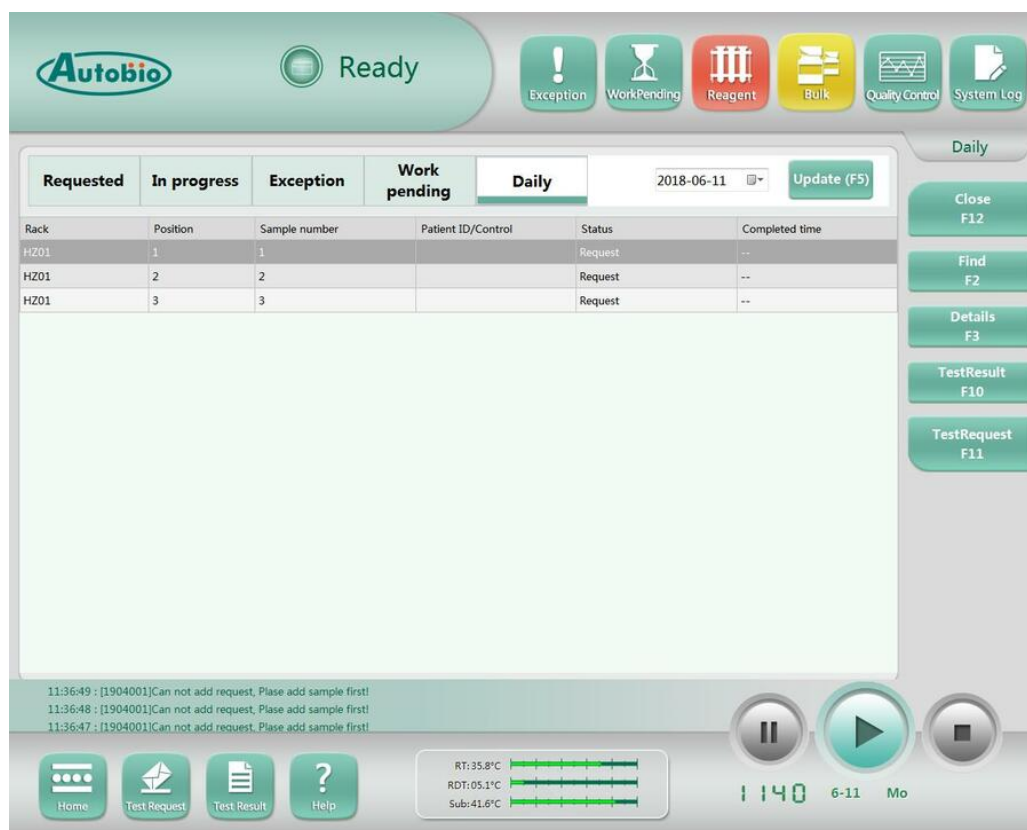


NOTE

1. “Work pending” interface shows all tests which assay errors happened in process, such as “CLT(Clots of sample pipettor)”, “QNS(Quantity of sample is not sufficient)”, “Supply Wait(Supplies not sufficient)”, “No Test(No Test Request)”.
2. “Work pending” icon will become yellow while “work pending” tests existed.

4.5 Daily Sample

Enter into “Test & Analysis” interface from the main menu interface by clicking its icon. Click “Daily” icon, the “Daily” window come out with all the samples information according to specified date.



Interface Description

Button Name	Definition
Close (Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu interface if you didn’t open “Test & Analysis” interface before.
Find(Labeled F2)	Search for the specified sample with its Sample ID.
Details (Labeled F3)	Show all the samples information on the day, including Rack, Position, Sample ID, Patient ID/Control, Status, and Completed Time.
Test Result (Labeled F10)	Switch to “Test Result” interface.
Test Request (Labeled F11)	Switch to “Test Request” interface for editing.



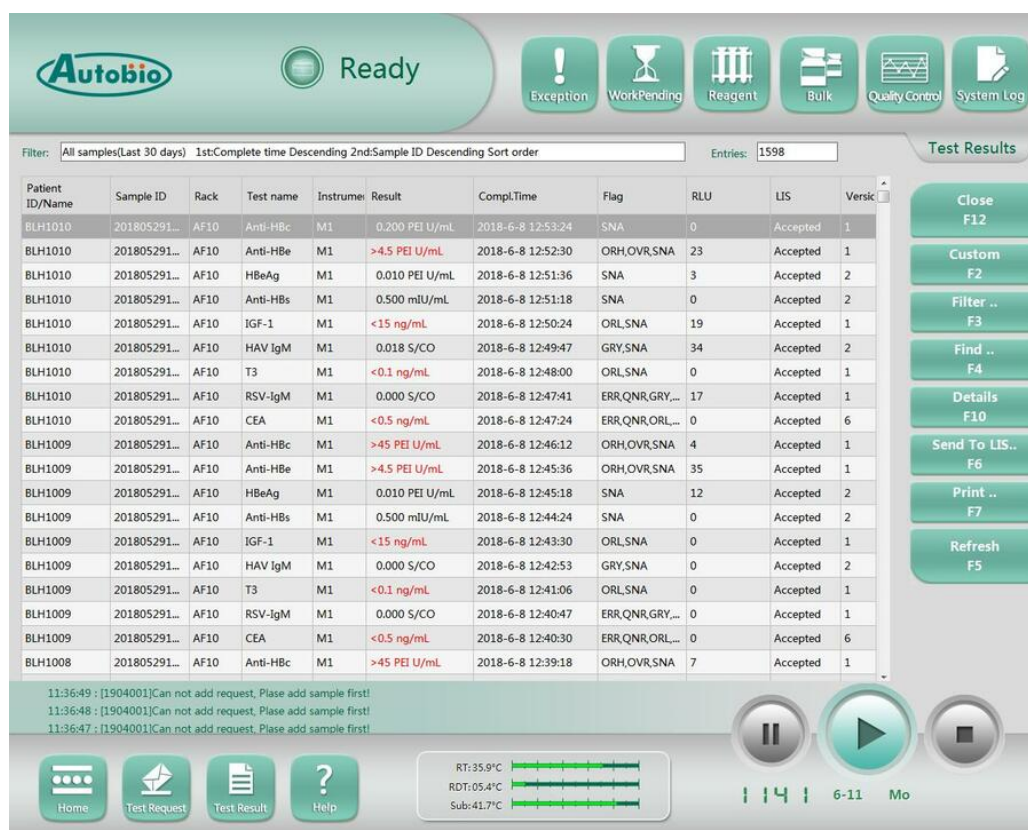
NOTE

Sample Status: Done, In Progress, Exception

Completed time indicates the time for all the tests to be done. If the time cannot be anticipated, the column will show "--"

CHAPTER 5 TEST RESULTS

Enter into “Test & Analysis” interface from the main menu interface by clicking its icon. Then select one of the option icons listed on the column of “Test Results”, indicating the type of result to review.



Interface Description

Button Name	Definition
Close (Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu interface if you didn’t open “Test & Analysis” interface before.
Custom(Labeled F2)	Determine which type of results’ information to show and display sequence.
Filter(Labeled F3)	Establish the filter condition, and filter test results.
Find(Labeled F4)	Search the test results from filtered results according to specified condition.
Details(Labeled F9)	Show all the detailed information of selected test result.
Send to LIS(Labeled F6)	Send the test result to LIS.
Print(Labeled F7)	Print the test result out.
Refresh(Labeled F5)	Refresh current test results.

5.1 Custom

Determine which type of results' information to show. Click "Custom" button, and select the desired options on pop-up "Custom setting" window as well as display sequence. After click "confirm" button, the results display on the interface according to Custom setting.

5.2 Filter

Click "Filter" icon to enter into "Filter" interface for setting of existed filter condition or setup of new filter condition. After setting filter condition, click "confirmation", the test results will show on the interface according to the filter conditions selected or established.

5.3 Find

Click "Find" icon to search the test results among filtered results according to specified condition.

5.4 Details

Select the test result, and click "Details" icon or double-click current row for more information of selected result, which will be displayed on the pop-up window.

5.5 Send to LIS

Select the test result, and click "send to LIS" icon to send the result to LIS.

5.6 Print

Print the selected test result out in report format. Click "Print" icon, and the results will be printed as report form. Refer to description of "report form" section.

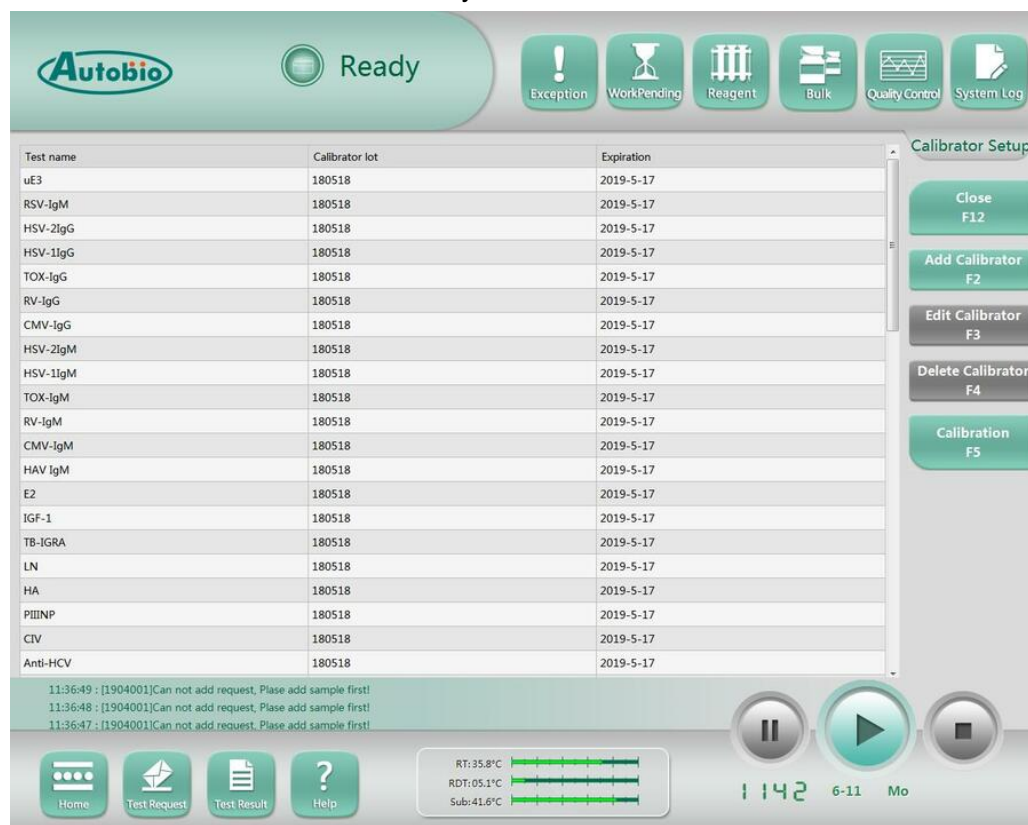
5.7 Update

Update the test results to the newest state, click the refresh button to update.

CHAPTER 6 CALIBRATION

6.1 Calibrator

Open “Calibrator Setup” interface by clicking its icon on the “Test & Analysis” interface. Get access to “Test & Analysis” interface from The Main Menu.



Interface Description

Button Name	Definition
Close(Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu interface if you didn’t open “Test & Analysis” interface before.
Add Calibrator(Labeled F2)	Select it, “Calibrator” dialog window pop up for adding new calibrator.
Edit Calibrator(Labeled F3)	Select one calibrator, and click “Edit Calibrator” button. Then, “Calibrator” Dialog window of selected calibrator pop up for re-editing. “Edit Calibrator” button void without calibrator selected.
Calibration(Labeled F5)	Go to the calibration results interface

6.1.1 Add Calibrator

1. Press "Add Calibrator" Button, "Calibrator" Dialog window pop up for adding new calibrator.

Barcode :	Test name	Standard	Concentration	Units
Calibrator:				
Lot:				
Expiration:				
Standard:				

2. Scan or input the bar code of calibrators.
3. Scan or input the bar code of standards.
4. Click "OK" Button.

6.1.2 Edit Calibrator

Calibrator

Barcode :
Calibrator:
Lot:
Expiration:
Standard:

Test name	Standard	Concentration	Units
uE3	S0	0	ng/mL
uE3	S1	0.2	ng/mL
uE3	S2	0.4	ng/mL
uE3	S3	1	ng/mL
uE3	S4	4	ng/mL
uE3	S5	8	ng/mL

1. Select the calibrator you want to edit, and click "Edit Calibrator" button to display "Calibrator" Dialog window.
2. Scan or input the bar code of calibrator.
3. Click "OK" button.

6.2 Request Calibration

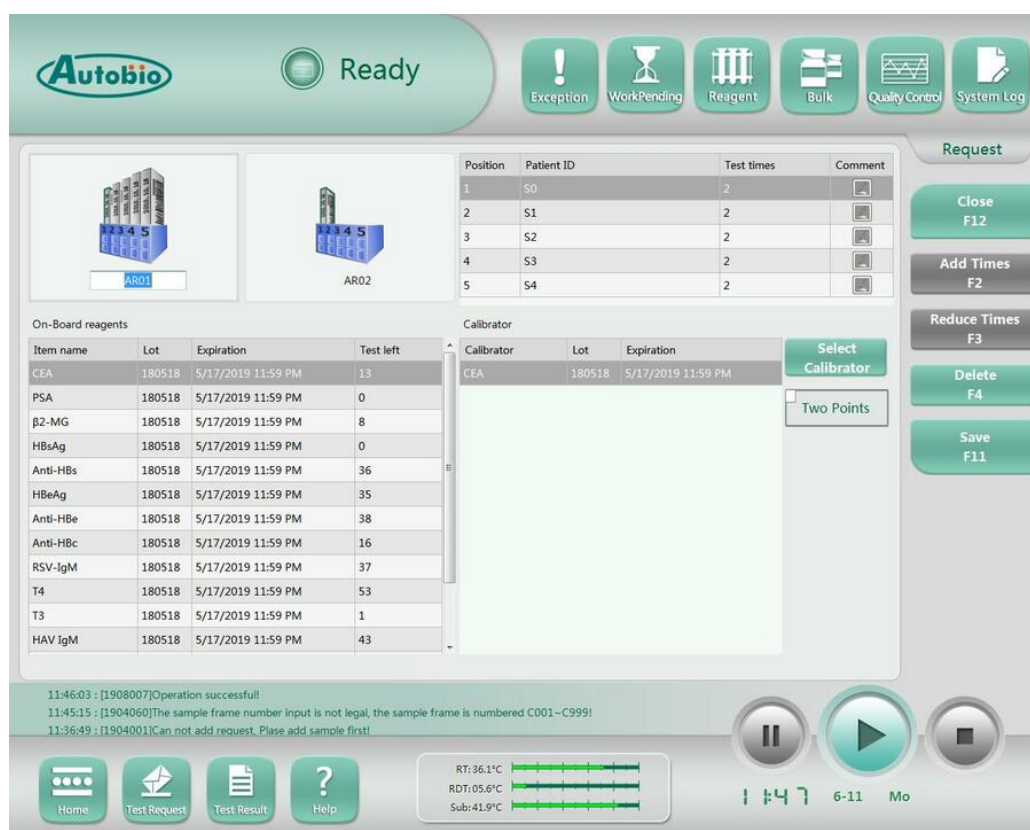
6.2.1 Enter "Request Calibration" interface

Open "Request Calibration" interface by clicking its icon on the "Test & Analysis" interface. Get access to "Test & Analysis" interface from the Main Menu.



CAUTION:

1. Each sample rack has 5 samples at most, but one set of calibration requests has 5-7 standards. So, it's possible to load two sample racks for calibration test.
2. In the selection of calibrator, only the one which has the same batch with its corresponding reagent is allowed to be selected.



Interface Description

Button Name	Definition
Sample rack Code	Click “?” icon below the picture of Sample Rack on the left side to activate the inputting column of sample rack code
Position	Automatically number from 1-5.
Calibrator	Fixedly mark calibrators from S0 to S6.
Test Times	“Test times” is changeable. Qualitative assay calibration can be executed 1-3 times, and each calibrator’s assay times can be defined separately. Quantitative assay calibration times can be executed 1-2 times, but, each calibrator’s assay times have to be same. Default test times for qualitative and quantitative assay calibration are 2.
Comment	Add the description of sample.
Item Name	Differentiate different test items.
Lot	Manufacture date of reagent
Expiration	Expire date of reagent
Left times	Left times of a reagent
Close(Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu

	interface if you didn't open "Test & Analysis" interface before.
Add Times (Labeled F2)	Click it once, Test times of selected calibrators will be added by 1 time.
Reduce Times (Labeled F3)	Click it once, Test times of selected calibrators will be reduced by 1 time.
Select Calibrator	Add the selected "Calibrator" information to "On Board Reagents"
Delete (Labeled F4)	Delete the current selected sample rack and its sample information involved. When sample rack don't exist, the button void.
Save (Labeled F11)	Save the newly added calibration to the system.

6.2.2 Add Sample Racks & Calibrators

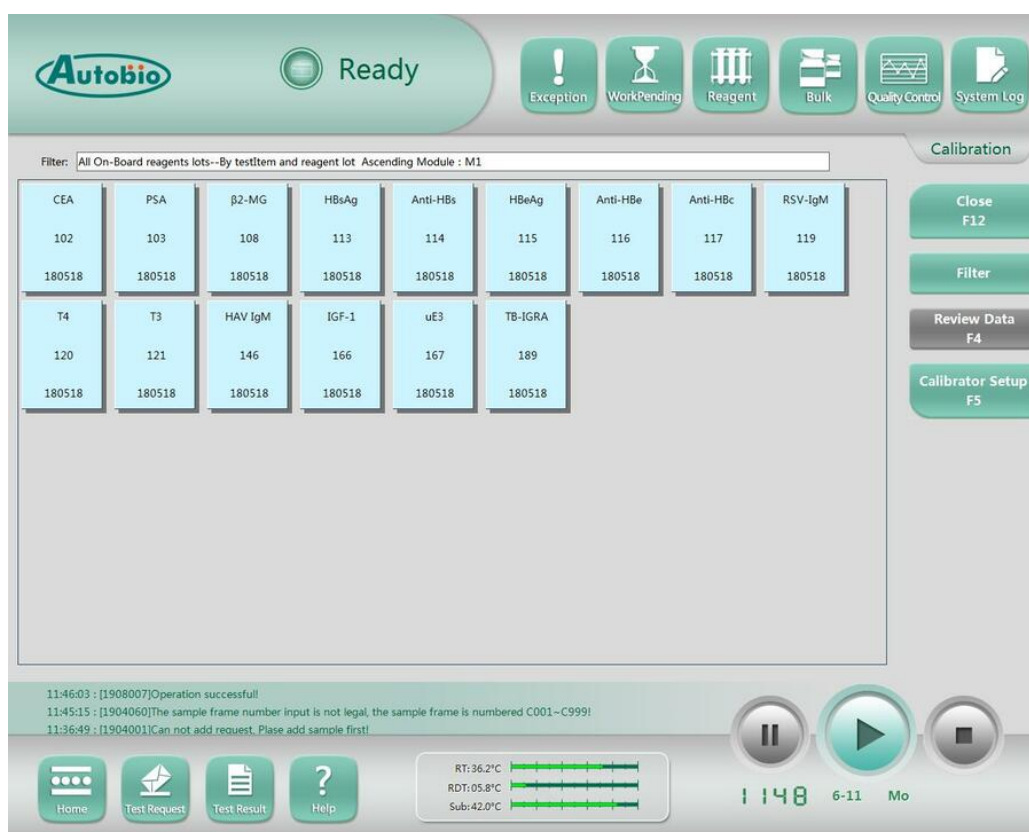
1. Dispense each calibrator to sample tube.
2. Load these sample tubes on sample rack according to its series number.
3. Select the relevant reagents on board from "On Board Reagent" column.
4. Click "Select Calibrator" button.
5. Scan or input the code of sample rack with the calibrators selected for test.

6.2.3 Edit Request Calibration

1. Scan or input the code of first sample rack with calibrators which calibration request needs to be edited on the inputting column.
2. Select the relevant reagents on board from "On Board Reagent" column.
3. Select the calibrators scheduled for test from "Calibration" column.
4. Click "Select Calibrator" button.
5. Scan or input the code of sample rack with the calibrators selected for test.

6.3 Calibration Result

Open "Calibration" interface by clicking its icon on the "Test & Analysis" interface for reviewing the calibration results. Get access to "Test & Analysis" interface from The Main Menu.



Interface Description

Button Name	Definition
Close(Labeled F12)	Exit "Calibration" interface
Filter(Labeled F2)	Filter the calibration results according to the defined condition.
Review Data(Labeled F4)	Review the test data and calibration curve of selected calibration item.
Calibrator Setup(Labeled F5)	Enter into "Calibrator Setup" interface.

6.3.1 Filter

Filter

Condition

All calibrations All On-Board reagents lots

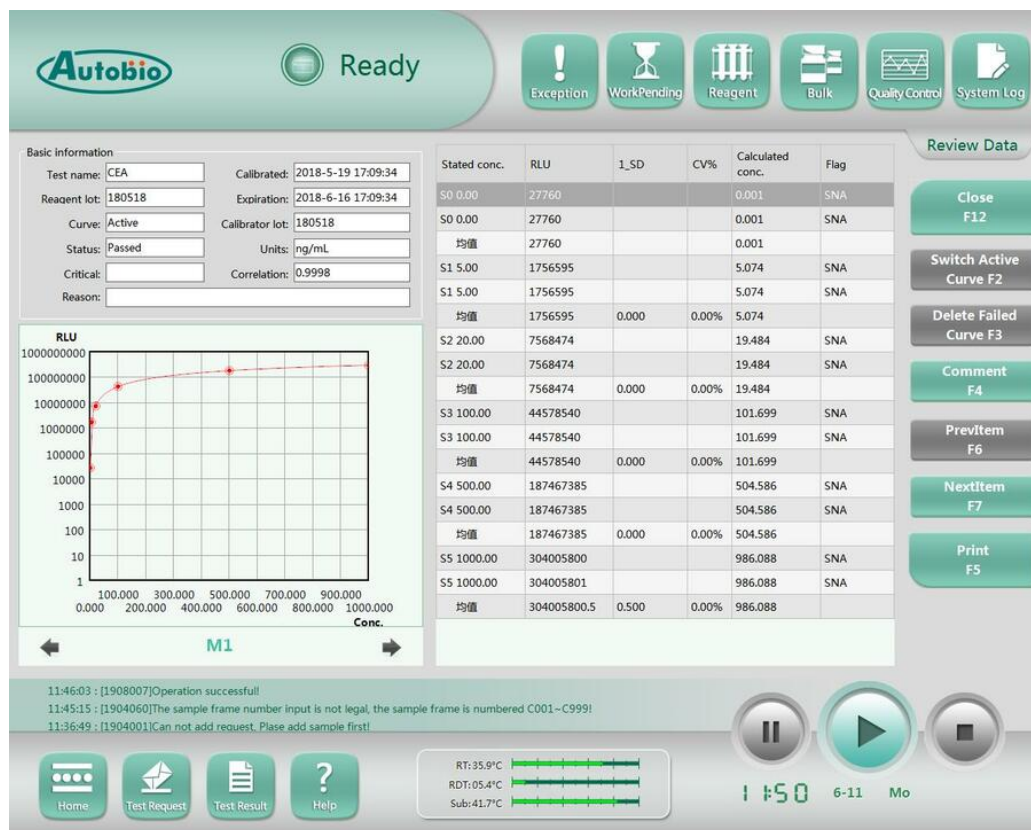
By testItem and reagent lot

Ascending Descending

OK Close

1. Click "Filter" button, Filter setting window pop up.
2. Select the filtering condition.
3. Select the method of sequence.
4. Click "OK" button.

6.3.2 Review Data



Select the calibration, and click “Review Data” button. Then, the “Review Data” interface appears, including basic Calibration information, Calibration data and curve.

Interface Description

Button Name	Definition
Basic Information	1, Basic information: “Test Name”, ”Reagent Lot”, “Calibrated Time”, “Expiration”, “Unit”, “Curve”, “Status”, “Reason”, “Calibrator Lot”, “Critical(cutoff)” 2, “Calibrated Time” means the finished time of Calibration Test. “Expiration” is calculated from the calibrated time plus the effective days, which can be set on ”Calibrator setup” interface. 3, Default options of “Curve” are “inactive” or “active”. 4, Default options of “Status” are “Passed” or “Failed”. 5, “Reason” column includes the following options: “Standard result %CV over valid range”(Judged by LD program), “Data insufficient ” (Judged by LD program), “HCV”, “Bad Fit”, “Not Fit”.
Calibration Data	1,Include “Stated Concentration”, “RLU”, “1_SD”, “CV%”, “Calculated Concentration”, “Flag” 2, Each test value for different time is shown on different row, and the

	mean value is also shown below. 3, "No Value" display in case of no calculated concentration.
Calibration Curve	"Curve" is drawn by the positions of calibrators on the graph based on their concentration and RLU value (Qualitative Assay has no calibration curve). X axis means concentration. Y axis means RLU value. Minimum/Maximum Values on X axis are defined as minimum/maximum concentration value of calibrators. Minimum/Maximum Values on Y axis are defined by test item parameter graph. Calibration curve is plotted according to curve fitting. No calibration curve will be generated for qualitative Assay. So, no curve existed.
Close(Labeled F12)	Exit the current interface, and back to "calibration" interface.
Switch Active Curve (Labeled F2)	Define the current Curve as active curve.
Delete Failed Curve (Labeled F3)	Delete the failed curve.
Comment (labeled F4)	Add comment to the current calibrated curve.
Print (Labeled F5)	Print the current calibration result, including basic information, calibration data and curve.
	Switch to previous calibration result.
	Switch to next calibration result.

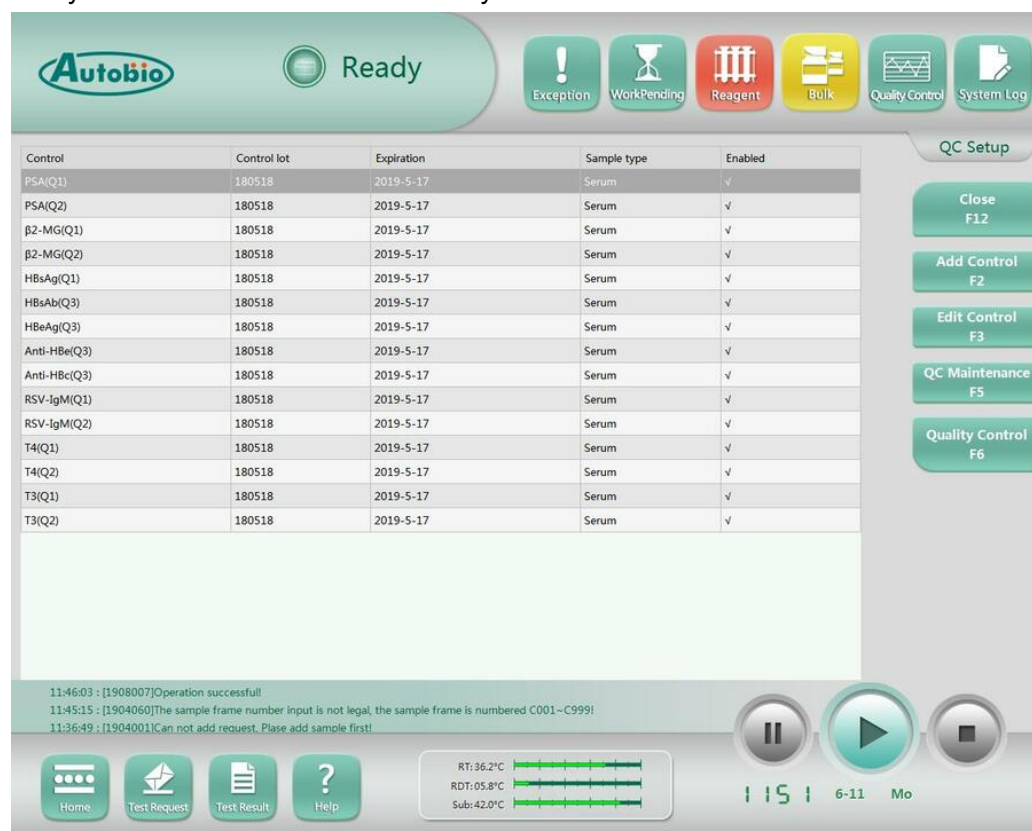
CHAPTER 7 QUALITY CONTROL

Quality controls are very critical to be tested to ensure the validity of calibrated curve. The calibrated curve will become void for a while because of mechanical wear or other factors. Through evaluation of quality control assays, the validity of calibrated curve will get qualified.

User need to carry out quality control every day to ensure the test results is reliable.

7.1 Quality Control

Open “QC Setup” interface to manage QC by clicking its icon on the “Test & Analysis” interface. Enter “Test & Analysis” interface from The Main Menu.



Interface Description

Button Name	Definition
Close(Labeled F12)	Exit the current interface
Add Control (Labeled F2)	Add new quality control to the system.
Edit Control (Labeled F3)	Edit the current quality control
QC maintenance (Labeled F5)	Setup the following values: Means Value (Target Value), SD, Test Items and other basic data.
Quality Control (Labeled F6)	Switch to “Quality Control” interface.

7.1.1 Add Control

Test name	Mean	SD	Units	1-2S	2-2S	1-3S	4-1S	7x
CA125	1	1	U/mL	☐	☐	☐	☐	☐

1. Enter "QC setup" interface, and click "Add control" button. The dialog window of "Control" comes out.
2. Input the following information of quality controls, Bar code, name of quality control, Lot, Expiration, in case quality controls are not manufactured by AUTOBIO.
3. Only input Bar code of quality controls that are manufactured by AUTOBIO.
4. Select the QC rule on dialog window of "Control".
5. Click "OK" button.



NOTE

QC Rule:

1. $1\pm 2s$ (1-2s) – Flag when one point is more than 2 SD from the mean.
2. $1\pm 3s$ (1-3s) – Flag when one point is more than 3 SD from the mean.
3. $2\pm 2s$ (2-2s) – Flag when 2 consecutive points are more than 2 SD from the mean in the same direction. Only the second point is flagged
4. $4\pm 1s$ (4-1s) – Flag when 4 consecutive points are more than 1 SD from the mean in the same direction. Only the fourth point is flagged.
5. 7x (7x) – Flag when 7 consecutive points are on the same side of the mean (for example, 7 points are above the mean). Only the seventh point is flagged.

7.1.2 Edit Control

1. Select "Edit Control" on the interface of "QC Setup".
2. Select "QC Rule" included in the QC item.
3. Edit QC name
4. Click "OK" button for saving.

7.2 QC Maintenance

Click “QC Maintenance” button on the interface of “QC Setup”, and the interface of “QC Maintenance” appears. Here, administrator privileges can maintain third-party QC.

The screenshot displays the "QC Maintenance" interface. At the top, there's a status bar with "Autobio" logo, a "Ready" indicator, and several function buttons: Exception, Work Pending, Reagent, Bulk, Quality Control, and System Log. Below this, the main area is divided into sections. On the left, a "Control number list" shows a single entry "100:CEA". The center section contains input fields for "Control number", "Control name", "Item name", "Item number", "Sample type", and "Units". Below these are "Basic data" fields for Mean (0.10000), SD (0.10000), Upper limit (0.00000), and Lower limit (0.10000). The "Item selection" table lists items 100 to 108 with their names (CA50, CA125, CEA, PSA, f-PSA, CA19-9, Ferr, AFP, β2-MG), sample types (all Serum), and units (U/mL, ng/mL, μg/L). On the right, a vertical column of buttons includes "Close F12", "New F2", "Delete F3", "Add Item F4", and "Delete Item F5". The bottom status bar shows a log of operations, a large digital display "1154 6-11 Mo", and temperature readings: RT: 36.2°C, RDT: 05.8°C, Sub: 42.0°C.

Interface Description

Button Name	Definition
Close(Labeled F12)	Exit the current interface and back to the interface of “QC Setup”.
New(labeled F2)	Add the new QC.
Delete(Labeled F3)	Delete the current QC selected as well as its test items included.
Add Item(Labeled F4)	Add the selected item in the column of “Item Selection” to the current QC
Delete Item(Labeled F5)	Delete the selected item included in the current QC.

7.2.1 New Control

1. Enter into the “QC Maintenance” Interface, and click “new” button for adding new control.

2. Input "Control number" & "Control name"
3. Select the first Item included in the new control under the column of "Item Selection"
4. Select the relevant item on the column of "Control Number List"
5. Input "Mean" value & "SD" value
6. Select "Add Item" in case the controls include other items.
7. Repeat step 4 & 5 above
8. Repeat step 1 to 7 in case of adding more controls.

**NOTE**

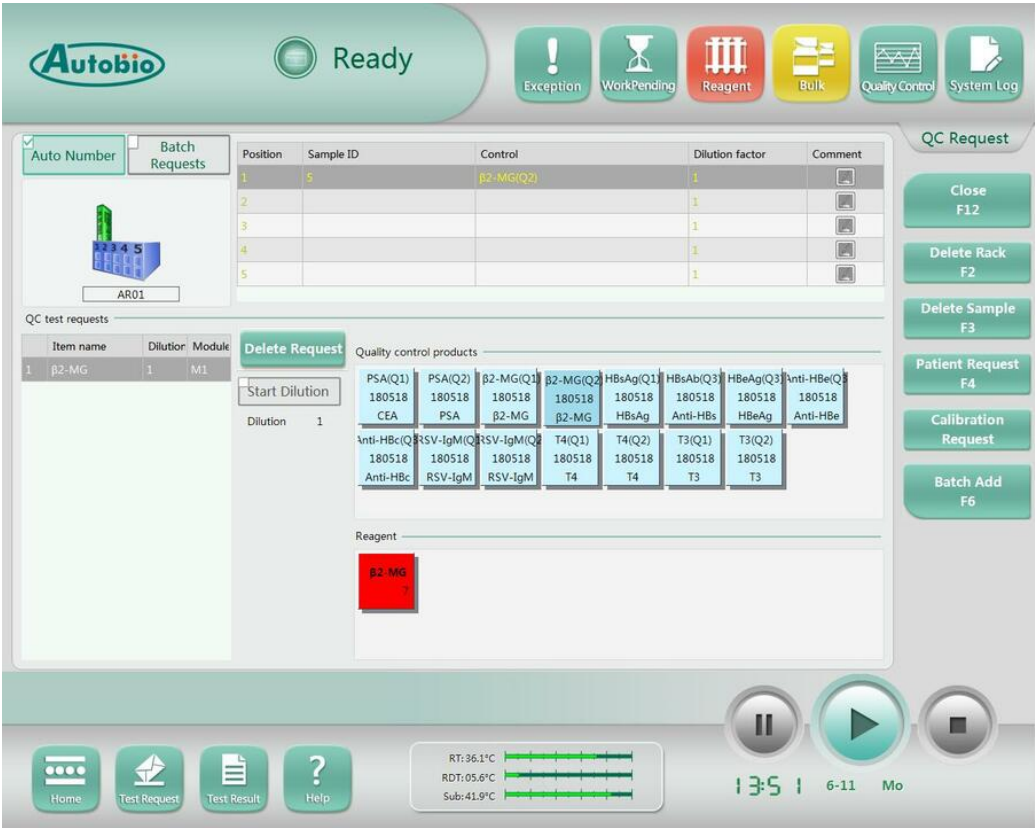
1. "Control Number" has to be 3 digits number from 100-999.
2. "Control Number" doesn't allow to be repeated.
3. "Control Number" & "Control Name" cannot be empty.
4. Control has to include one test item at least.

7.2.2 QC Request

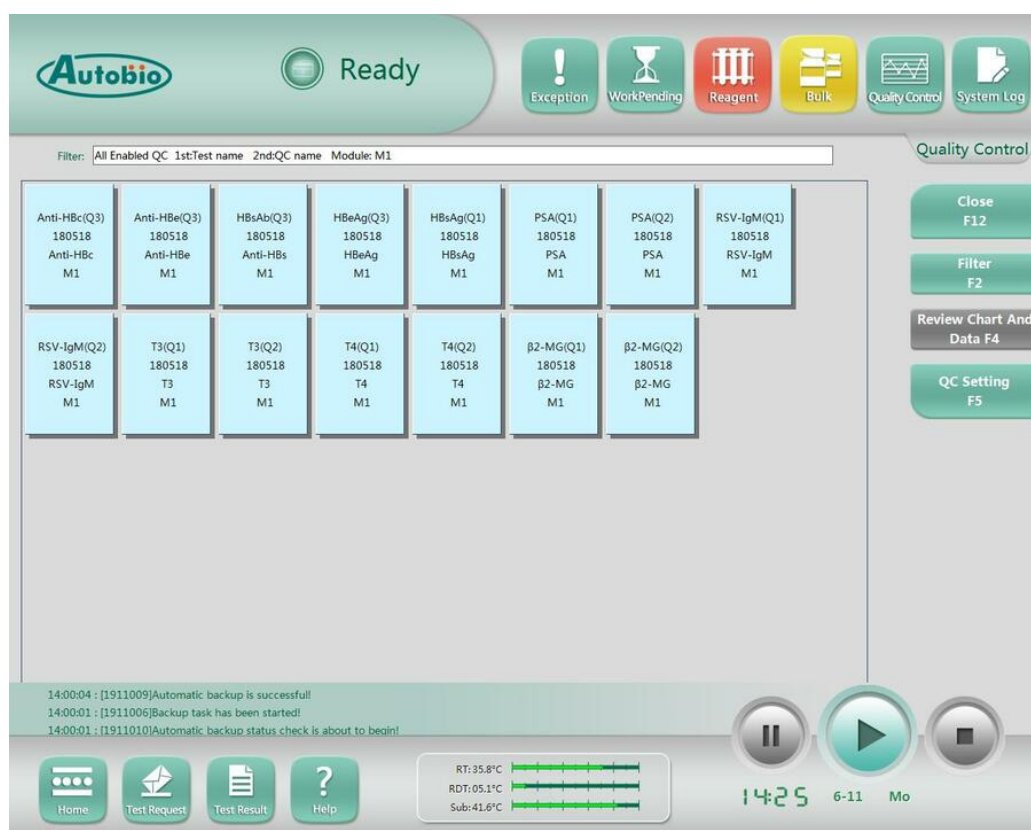
Enter "Quality Request" interface by clicking its icon on the "Test & Analysis" interface. Enter "Test & Analysis" interface from Main Menu.

**CAUTION:**

1. When the sample ID is edited, test item of the selected QC will be added automatically.
2. It is required to re-click 'Reagent Add Request' after delete 'Quality Request'.



7.3 QC Result



Enter “Quality Control” interface to review Quality Control results by clicking its icon on the “Test & Analysis” interface. Enter “Test & Analysis” interface from The Main Menu.



NOTE

The color of normal Quality Control result is blue, but for abnormal or failed Quality control result, the color will become red.

Interface Description

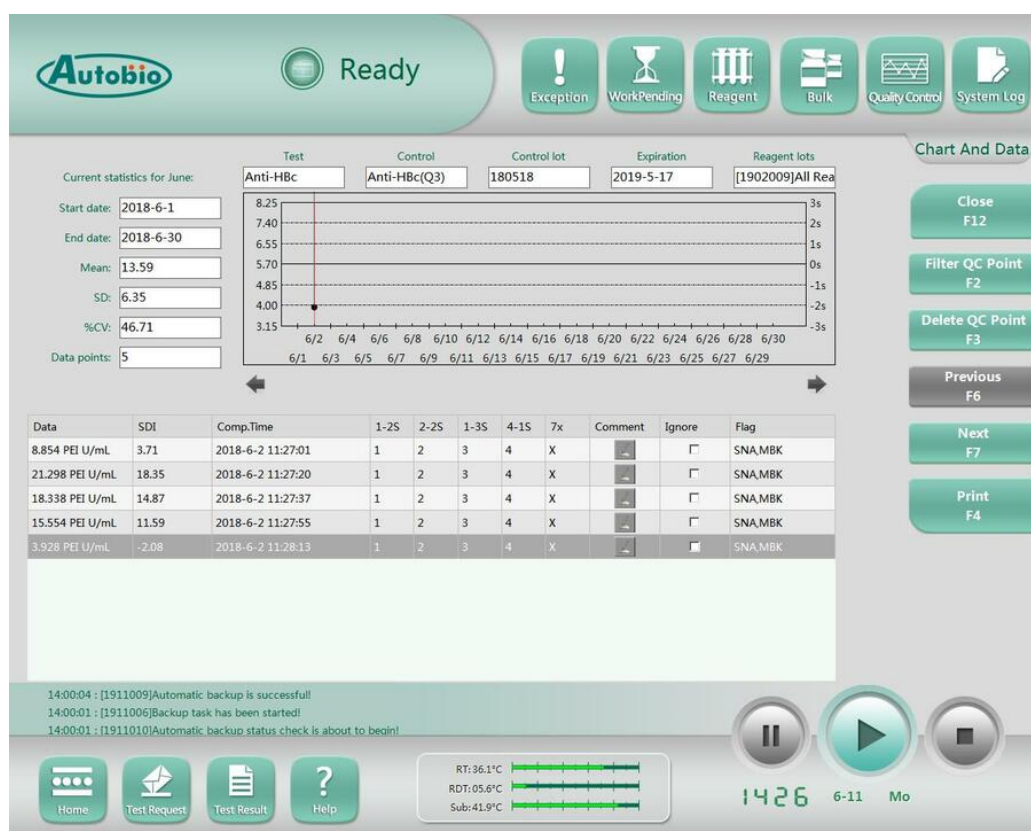
Button Name	Definition
Close(Labeled F12)	Exit the current interface and display its previous “Test & Analysis” interface or directly back to the main menu interface if you didn’t open “Test & Analysis” interface before.
Filter(Labeled F2)	Filter the quality control results according to the defined condition.
Review Chart and data (Labeled F4)	Review result data and QC drawing of selected Quality control. The button void without selecting Quality control.
QC Setting(Labeled F5)	Click to switch directly to the QC setting interface.

7.3.1 Filter

The 'Filter' dialog window is used to filter and sort QC results. It includes a dropdown for 'Test/Quality control' and two columns for sorting options ('First sort' and 'Second sort').

1. Click "Filter" button on the interface of "Quality control", the dialog window come out for filtering QC results.
2. Select filter condition.
3. Select "Sort Order".
4. Click "OK".

7.3.2 Chart and Data



Click “Review Chart and Data” button for reviewing QC result and Diagram on the interface of “Quality Control”.

Interface Description

Button Name	Definition
Basic information of Quality Control	1, Basic information: “Start Date”, “End Date”, “Control”, “Control Lot”, “Expiration”, “Reagent Lots”, “Mean”, “SD”, “%CV”, “Data Points”. 2, “Mean”: Average value of all test datum involved in calculation. “SD”: Standard deviation of all the tests datum involved in calculation. “%CV”: Coefficient variation of all the tests datum involved in calculation. “Data Point”: Records count that participate in calculation.
Test data list of Quality Control	1, Verify the test data according to QC criteria selected. In case of the breach of QC criteria, the criteria breached will be marked in red or “*”. As for the breach of the criteria of “2-2S”, “4-1S”, “7X”, only the last data is marked as invalid. 2, Test data list only shows the test results of quality control with data, and only mark two QC results maximum which are out of range. 3, For Qualitative Tests, the data is “S/CO” Value (Actually RLU/cutoff

	value). Others are concentration value.
QC Diagram	1, Levey-Jennings QC diagram shows a graph of all QC data tested. 2, Y axis shows Test Result Data. Range: -3S to 3S. Step:1S 3, X axis shows date from the beginning to the end. Step: 1 day. Only 31 days show on the whole interface. For reviewing the results of other days, just click  &  button. 4, The horizontal grid values as 'x' and ± 1, ± 2 and $\pm 3S$, and the real line show the mean value.
Close (Labeled F12)	Exit the current interface and returns to "Quality Control" interface.
Filter QC Point (Labeled F2)	Filter the QC results through "Date" and "Reagent Lots".
Delete QC Point (Labeled F3)	Delete the QC Points selected.
Ignore	The QC point ticked as "ignore" won't be used to calculate the SD and %CV value, and the QC Diagram won't show this QC point.
Print (Labeled F4)	Print QC results, including QC basic information, test datum and QC diagram.
	Previous day
	Next day

1. Filter QC Point: Filter the QC results through "Data" and "Reagent Lots". The condition of "Date" includes "all dates", "Last week", "Last month", "Defined Date ("Start day", "End day")". The condition of "Reagent Lots" includes "All Lots", and "Each Reagent Lots".
2. "Delete QC Point": Only administrator User has the right to delete the QC point. Password is required while common user wants to delete QC point.
3. "Ignore": The QC point ticked as "ignore" won't be used to calculate the SD and %CV value, and the QC Diagram won't show this QC point.
4. "Comment": Add comment for QC point. The symbol of  means the relevant QC point has comment, and  means the relevant QC point doesn't have comment.

CHAPTER 8 CONSUMABLE MATERIALS

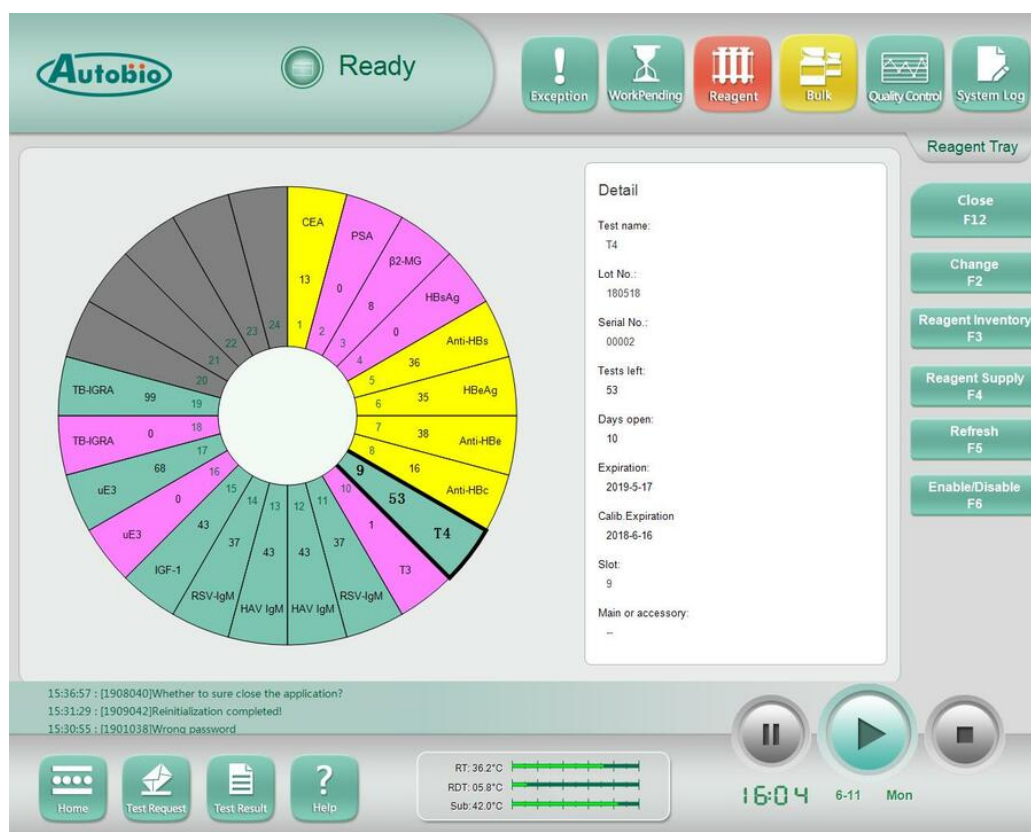


CAUTION

1. The test result is unreliable when using expired consumable materials. Before using reagent, please ensure the consumable materials is within its expiration date.
2. The lid of the reagent bottle need to be pricked before use .so that the reagent can be pipetted normally.

8.1 On-Board Reagent

Select “Test&Analysis” in the main interface menu to enter the “Test&Analysis” wizard window, and click “Reagent Inventory” button to check the reagent data on the machine.



interface Description

Button Name	Description
Close (Labeled F12)	Exit the current interface and return to “Test & Analysis” interface.
Change (Labeled F2)	Execute the operation of replacing the reagent
Reagent Inventory (Labeled F3)	Switch to the reagent manage menu

Refresh(Labeled F5)	Refresh on-board reagents
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8.1.1 View the reagent data

1. Select the reagent slot on the graph.
2. The details information of the reagent display on the right side of the graph.

8.1.2 Change Reagent

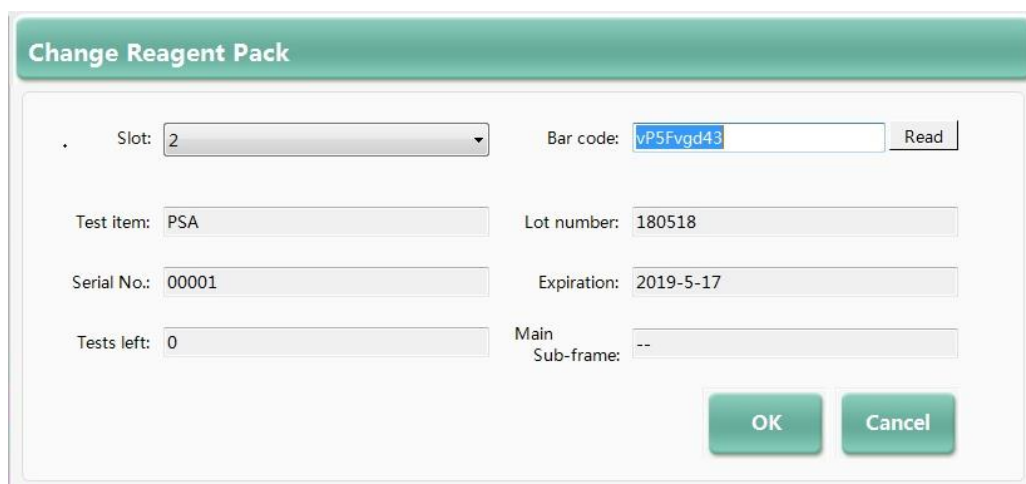
1. Open the cover of the instrument.
2. Open the cover of the reagent supply deck
3. Load or unload the reagent you need, and then close the cover of the reagent supply deck and the instrument successively.
4. Initialize the instrument, to make it in "Ready" state.
5. Click 'Change' button to switch to 'prepare to change reagent' window.
6. Active 'On board reagent change completed'
7. Click 'OK' button in the 'prepare to change reagent' window.
8. The instrument will scan the new reagent. All the reagents on the machine will be displayed in the 'reagent list' window.

Change Reagent Pack						
Slot	Barcode	Test Name	Lots	Serial No.	Expiration	Tests Left
1	uxGxDERPd	CEA	171009	00012	10/8/2018 11:59 PM	100
3	xhIf3a7F	β2-MG	170309	00211	3/8/2018 11:59 PM	100
5	89fabXzvz	P	170102	01050	1/1/2018 11:59 PM	100
6	zzUft1n0b	HBsAg	161128	05712	11/27/2017 11:59 PM	100
7	xhVf3ZPjx	β2-MG	161221	00219	12/20/2017 11:59 PM	100
8	zUzft0nzZ	HBsAg	161128	05699	11/27/2017 11:59 PM	100
9	89faZXzXF	P	170102	00129	1/1/2018 11:59 PM	100
10	xhV3fZoPL	β2-MG	161221	00243	12/20/2017 11:59 PM	100
11	zzUft0fgz	HBsAg	161128	02515	11/27/2017 11:59 PM	100
12	0R38SY2JP	Anti-HBs	170116	06308	1/15/2018 11:59 PM	100
13	0D9mRGavT	Anti-HBs	170526	00016	5/25/2018 11:59 PM	100
14	8af9zXcVB	P	170102	01267	1/1/2018 11:59 PM	100

On-Board[21]	New[21]	Unload[19]	Error[3]	Manual Login	OK
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1. Select "New Reagent" button to view the new reagent list.
2. Select "Unload Reagent" button to view unload reagent list.
3. Select "Prepare to change" button.

4. If the instrument cannot identify the bar code of the reagent correctly, you can click “Manual Login” button to supply the reagent bar code manually.



The image shows a software dialog box titled "Change Reagent Pack". It contains several input fields and buttons. The "Slot" is a dropdown menu set to "2". The "Bar code" field contains "vP5Fvgd43" and has a "Read" button next to it. Below these are fields for "Test item" (PSA), "Lot number" (180518), "Serial No." (00001), "Expiration" (2019-5-17), "Tests left" (0), and "Main Sub-frame" (--) with a "Main" label above it. At the bottom right are "OK" and "Cancel" buttons.

Field	Value
Slot	2
Bar code	vP5Fvgd43
Test item	PSA
Lot number	180518
Serial No.	00001
Expiration	2019-5-17
Tests left	0
Main Sub-frame	--

5. Click 'OK' button.

8.1.3 On-Board Change Reagent

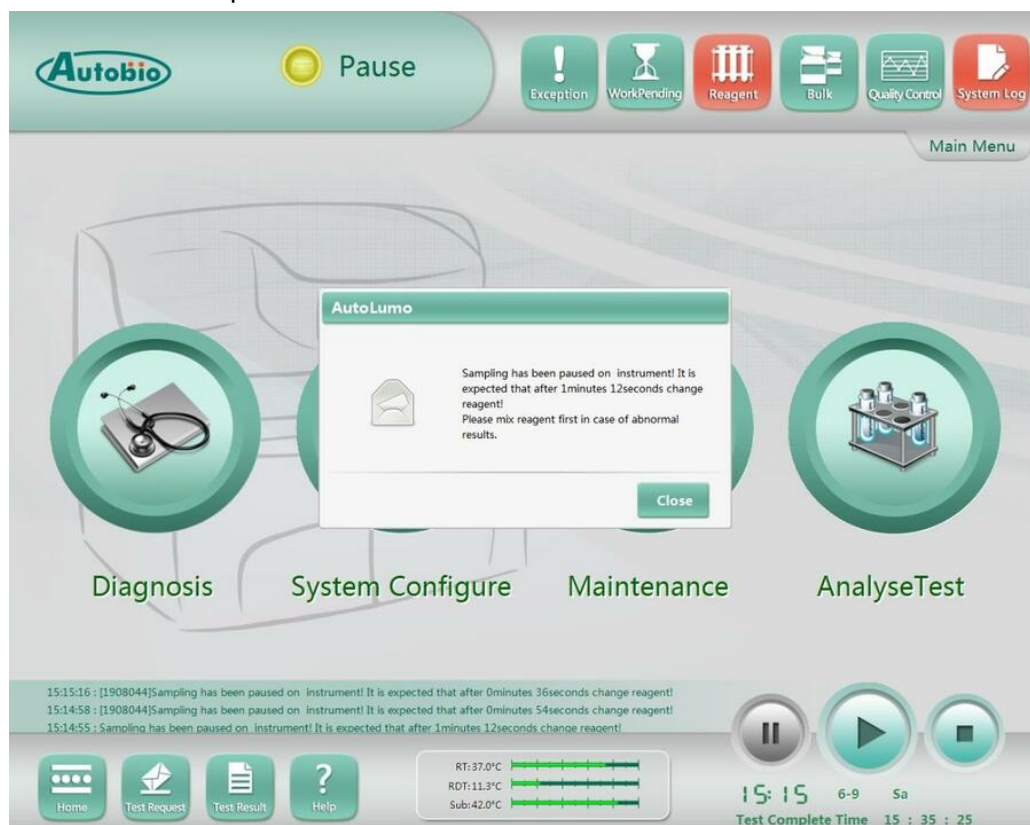
Appointment to change reagent can be initiated when the instrument is in the status of “Ready” or “Running”. The instrument can enter into the status of “Reagent Replacement” directly if the reagent replacement function was initiated in the status of “Ready”. The instrument will enter into the status of “Pause” and stop adding samples if the reagent replacement function was initiated when the instrument is in the status of “Running”, meanwhile, count down for reagent replacement will be showed on client software and Nixie tube of instrument. The instrument can enter into the status of “Reagent Replacement” after the countdown is completed.

1. Press “Appointment” button for more than 3s in the status of “Ready” or Running”

As showed in the picture:



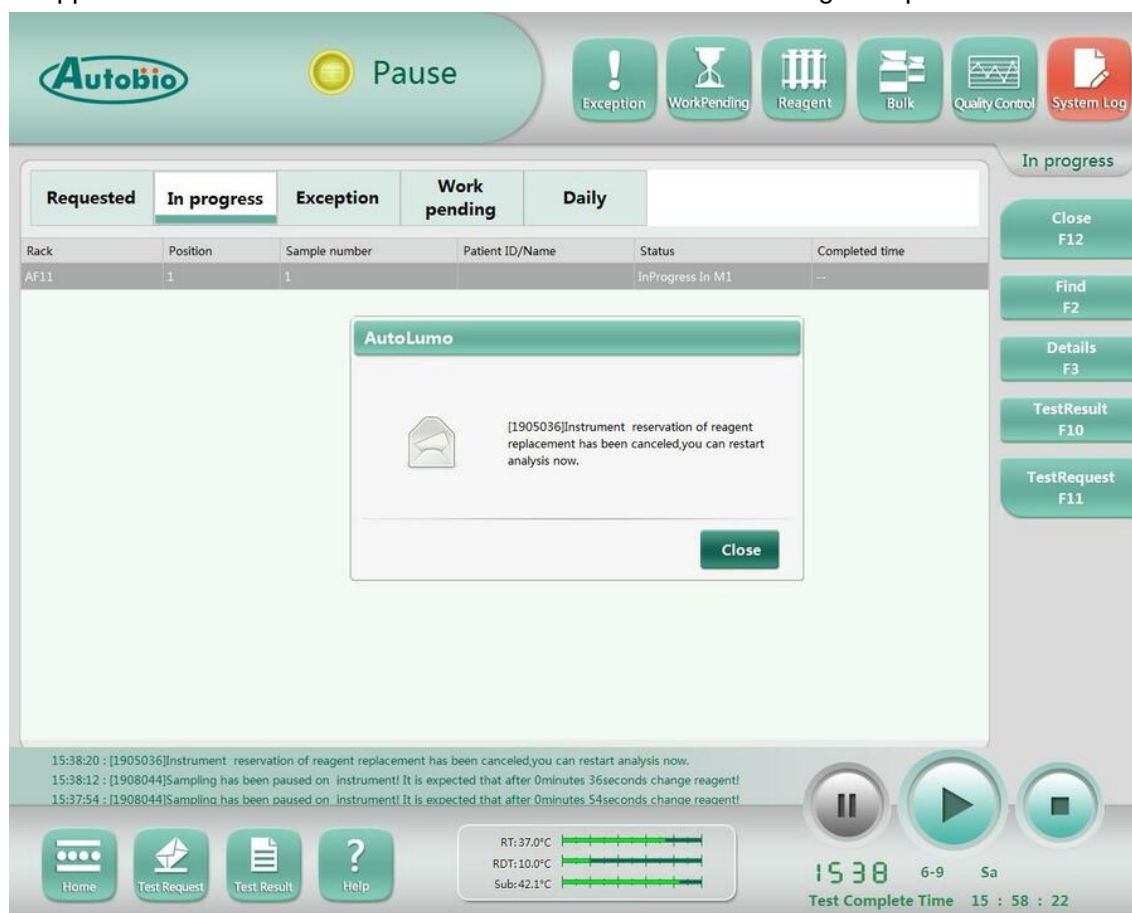
2. Sampling will be paused after initialize 'Appointment to change reagent' function and the expected time of changing reagent is displayed on the instrument.
As showed in the picture:



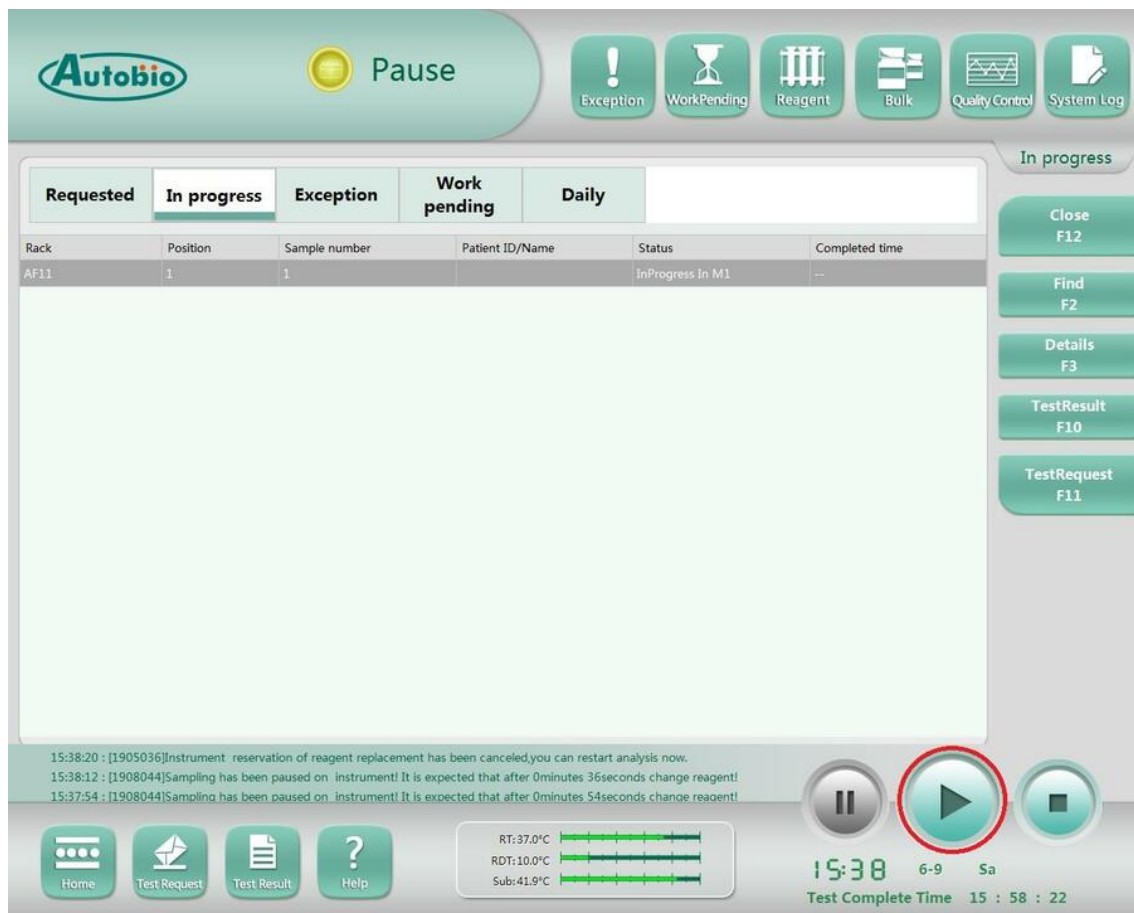
3. When the appointment indicator lights up, it is showed on Nixie tube that after XXmins and XXs change the reagent.



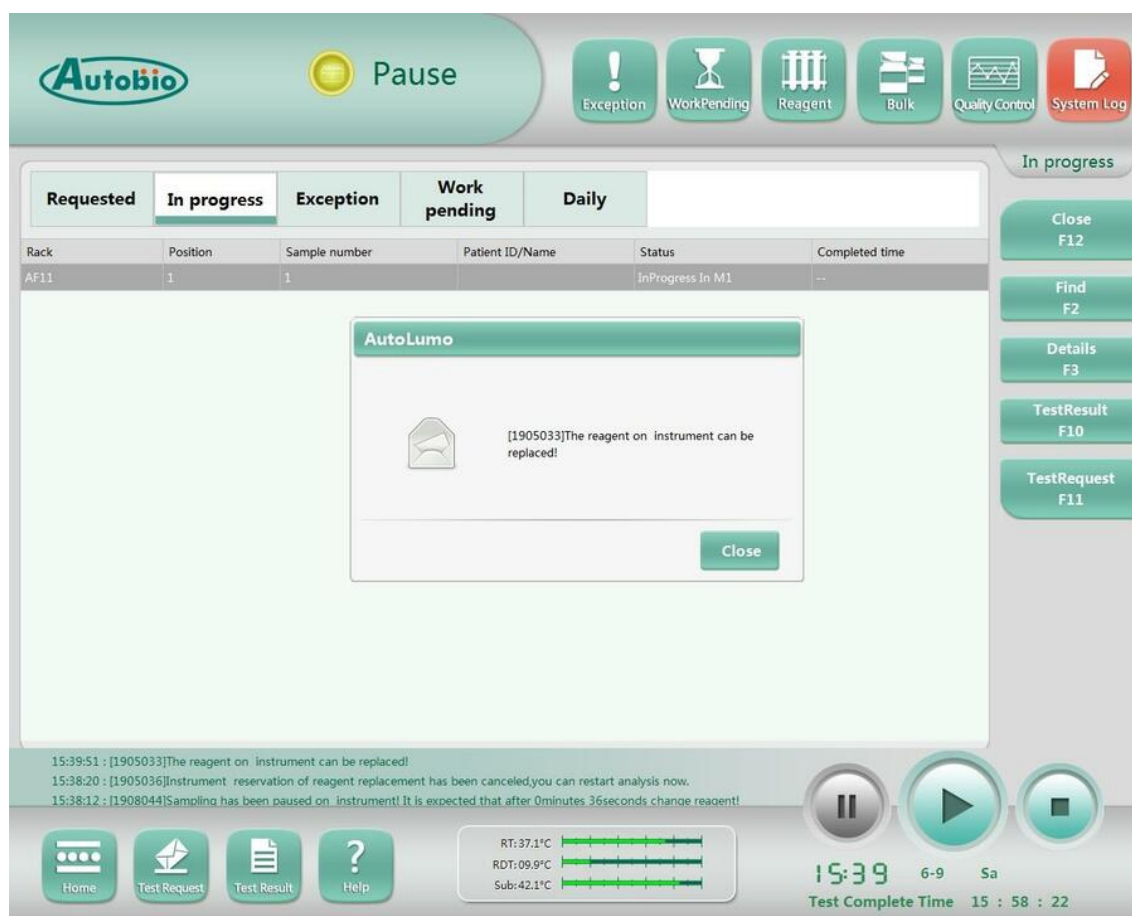
4. When the count down showed on Nixie tube is not up to 00:00 , press the “Appointment” button for more than 3s to cancel the online reagent replacement.



5. After reagent replacement is cancelled, the instrument is still in the status of “Pause”, which requires user to click “Start” button to continue the test.



6. User can change the reagent when it is indicated on the customer software that “The reagent on instrument can be replaced” and the countdown is up to 00:00.



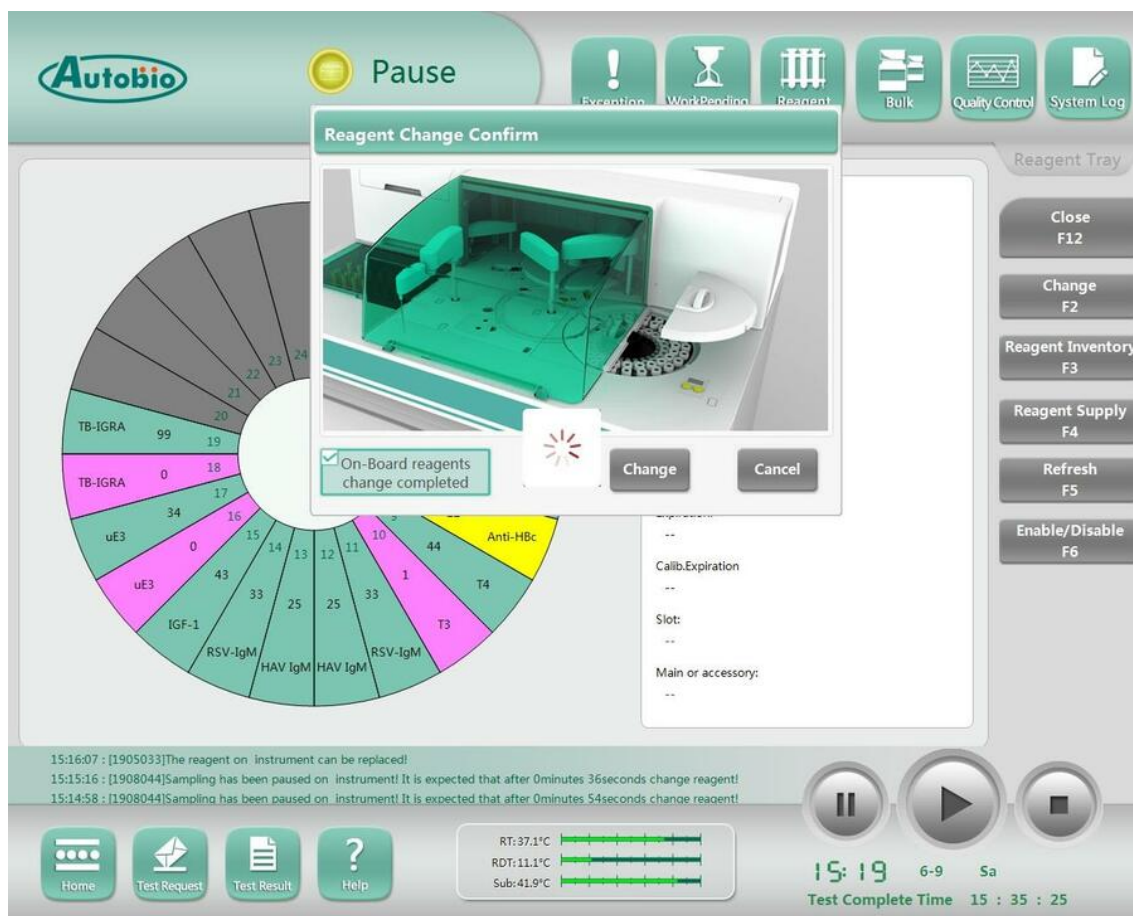
7. Open the cover on left side of reagent tray to find the position for replaced reagent or new reagent and put them into the tray.



8. Press "Rotate" button to turn reagent tray if the position for new reagent can not be found in view. Press for once, the tray turn one-third of circle (eight reagent position).



9. Bar code reading for slot number of the reagent can be started when the operator clicks the 'Appointment' for more than 3s after the reagent being put into the tray and cover being closed.



10. The reagent information will be showed on interface after scanning is

completed.

Change Reagent Pack

Slot	Barcode	Test Name	Lots	Serial No.	Expiration	Tests Left
1	uxGxDERPd	CEA	171009	00012	10/8/2018 11:59 PM	100
3	xhUf3a7F	β2-MG	170309	00211	3/8/2018 11:59 PM	100
5	89fabXzvz	P	170102	01050	1/1/2018 11:59 PM	100
6	zzUft1n0b	HBsAg	161128	05712	11/27/2017 11:59 PM	100
7	xhVf3ZPjx	β2-MG	161221	00219	12/20/2017 11:59 PM	100
8	zUzft0nzZ	HBsAg	161128	05699	11/27/2017 11:59 PM	100
9	89faZkXF	P	170102	00129	1/1/2018 11:59 PM	100
10	xhV3fZ0PL	β2-MG	161221	00243	12/20/2017 11:59 PM	100
11	zzUft0fgz	HBsAg	161128	02515	11/27/2017 11:59 PM	100
12	0R38SY2JP	Anti-HBs	170116	06308	1/15/2018 11:59 PM	100
13	0D9mRGavT	Anti-HBs	170526	00016	5/25/2018 11:59 PM	100
14	8af9zXcVB	P	170102	01267	1/1/2018 11:59 PM	100

On-Board[21] New[21] Unload[19] Error[3] Manual Login OK

15:19:56 : [1905059]When the reagent was replaced, 3 position scanning status was detected inconsistent
 15:16:07 : [1905033]The reagent on instrument can be replaced!
 15:15:16 : [1908044]Sampling has been paused on instrument! It is expected that after 0minutes 36seconds change reagent!

Home Test Request Test Result Help

RT: 37.0°C
 RDT: 11.0°C
 Sub: 42.0°C

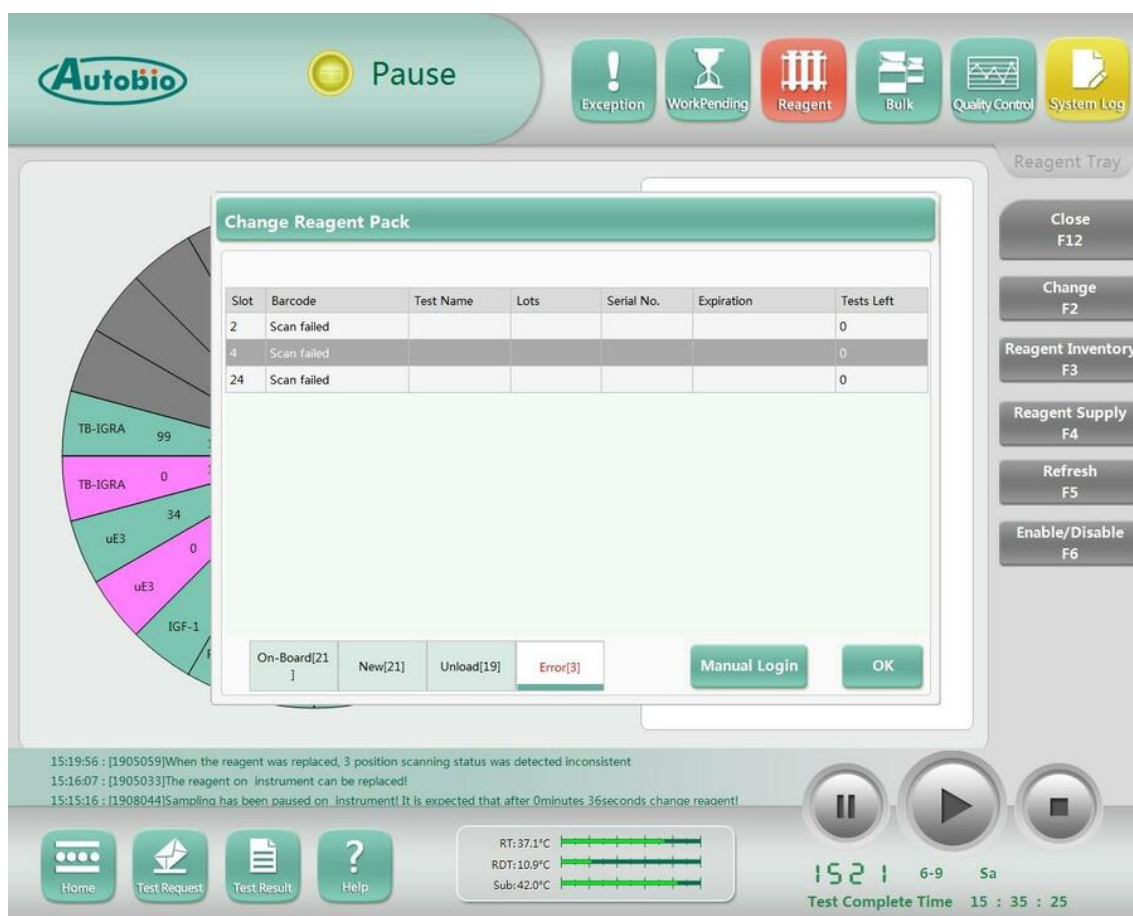
15:20 6-9 Sa
 Test Complete Time 15 : 35 : 25

11. If the slot number was scanned incorrectly, the indication will be showed on the interface.

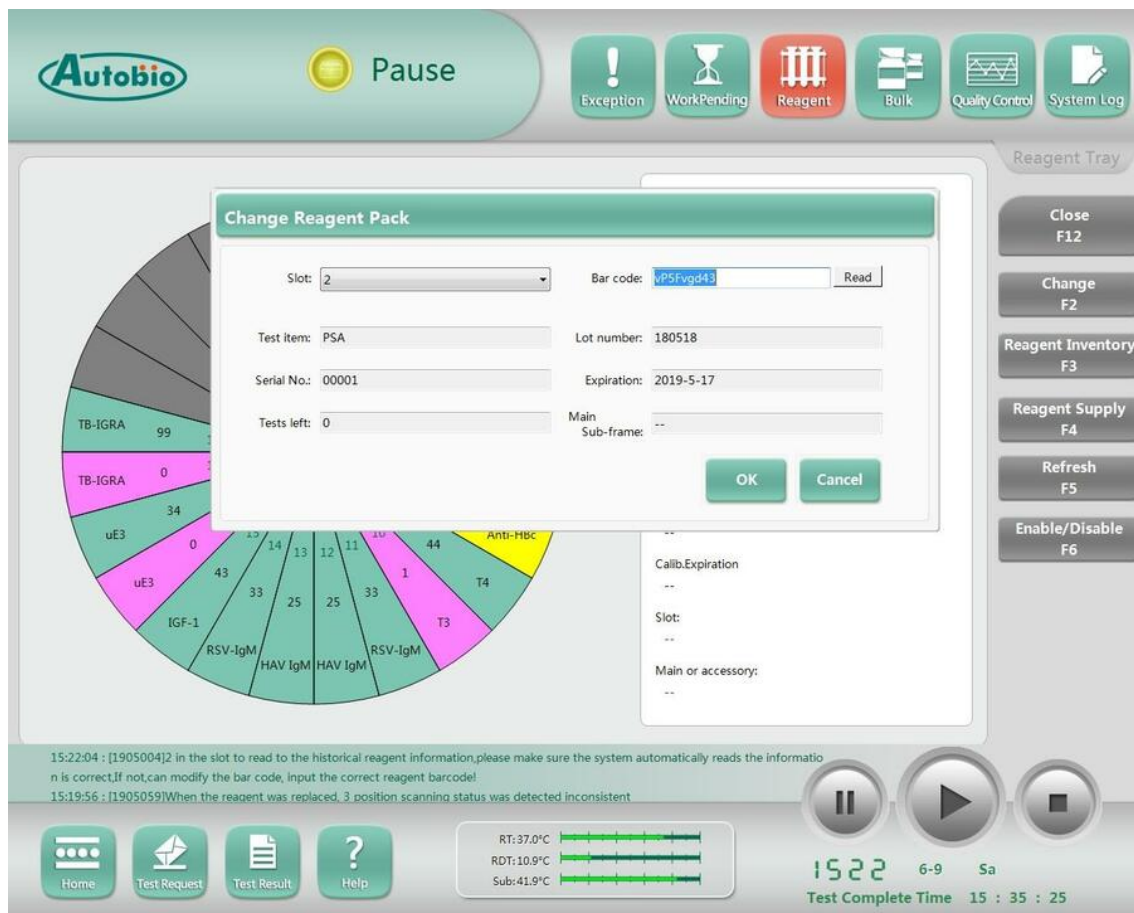
Change Reagent Pack

i The slot number [14] was scanned incorrectly. is it ignored? Ignore the replaceable reagent.

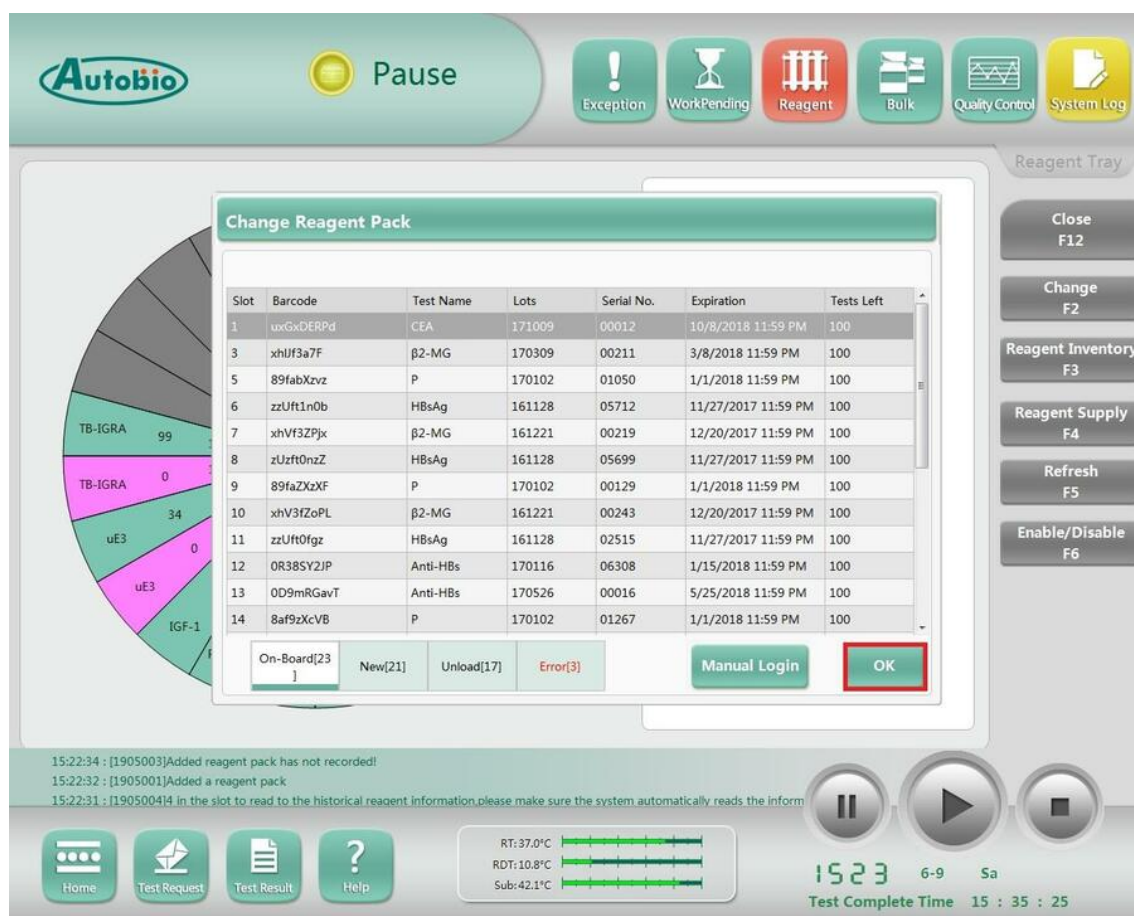
Yes No



12. User can use hand-held bar code reader to scan the reagent and add reagent, please refer to 8.1.2 for steps.



13. After scanning or replenishing of registering reagent completed, click 'confirmation' button and complete the reagent replacement.

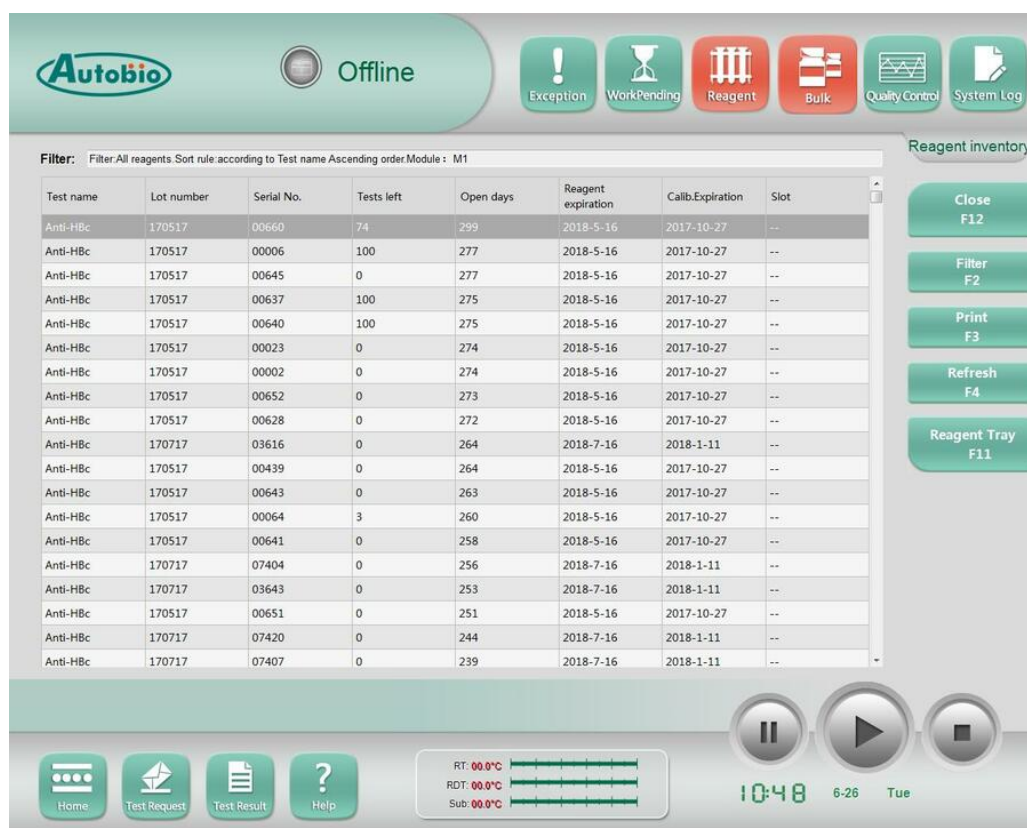


⚠ CAUTION

1. 'Appointment' function can not be initiated when the instrument is under the status of 'Reinitialization', 'Daily Clean', 'Daily Maintenance', 'Substrate Tube Washing', 'Implementing Diagnostic Function' and 'Changing Reagent Kit'
2. Mixing the reagent adequately by manual before putting the reagent into tray during On-Board Change Reagent.

8.2 Reagent Inventory

Select 'Reagent Inventory' button to enter reagent list management window from the reagent management interface.



Note:

1. If the reagent is unload already, the slot will display '--'
2. If the reagent is expired or its calibration is expired, it will be displayed in red color.

Interface Description

Button Name	Description
Close(Labeled F12)	Exit the current window and return to the secondary menu (Test & Analysis, maintenance and Diagnosis) or directly back to the main menu if you didn't open the secondary menu before.
Filter(Labeled F2)	Set the filter factor to filtrate reagent records
Print(Labeled F3)	Print filtrated reagents and their detailed information
Refresh(Labeled F4)	Refresh on-board reagents
On-board Reagent(Labeled F11)	Display all reagents on the machine

8.2.1 Filter

1. Click 'Filter' button to enter the filter factor window.

Filter

☒ All Reagents
☐ On-Board Reagents
☐ Unload Reagents
☐ Unexpired Reagents
☐ Expired Reagents
☐ Calibration Valid Reagent
☐ Calibration Invalid Reagent

Sort by

Sort Direction

OK Cancel

2. Select filter factor
3. Select sort factor
4. Select sort direction
5. Click 'OK'

8.3 Reagent Alarm

In the main menu, click the "reagent" on the interface to enter the reagent alarm form, and check the reagent alarm information.

Reagent Supply

Slot	Test name	Value
1	HBsAg	100
2	HBsAg	100
3	HAV IgM	100
4	HAV IgM	17
5	HAV IgM	17
6	Anti-HBc	100
7	Anti-HBs	100
8	HAV IgM	100
9	Anti-TP	100
10	HBsAg	100
11	Anti-HBe	100
12	Anti-HCV	100
13	Anti-HIV	100
14	HBsAg	70
15	Anti-HBs	63
16	HBsAg	57
17	Anti-HBe	94
18	HEV-IgM	5
19	Anti-TP	79
20	Anti-HIV	93
21	Anti-HBc	53
22	HEV-IgM	100
23	Anti-HCV	100
24	HEV-IgG	93

Reagent details

Test name	Lots	Serial No.	Tests left	Opened days	Expiration	Calibration expiration	Slot	Main or accessory

Reagent alarm information

Slot	Test name	Lots	Tests left	Status
2	HBsAg	171102	100	Reagent valid days open left serious alarm
3	HAV IgM	180211	100	Reagent valid days open left serious alarm
4	HAV IgM	180211	17	Reagent valid days open left serious alarm
5	HAV IgM	180211	17	Reagent valid days open left serious alarm
6	Anti-HBc	171215	100	Reagent valid days open left serious alarm
7	Anti-HBs	171013	100	Reagent valid days open left serious alarm
8	HAV IgM	180211	100	Reagent valid days open left serious alarm

Reagent Supply Actions:

- Close F12
- Reagent Inventory F2
- Reagents Tray F3
- Change F4
- Enable/Disable F6
- Disenable R1 Mixing F7
- Enable R1 Mixing F8

Bottom Bar:

- Home, Test Request, Test Result, Help
- RT: 00.0°C, RDT: 00.0°C, Sub: 00.0°C
- 10:54 6-26 Tue

**CAUTION**

1. It indicates serious alarm if the button background is red and general alarm if that is yellow

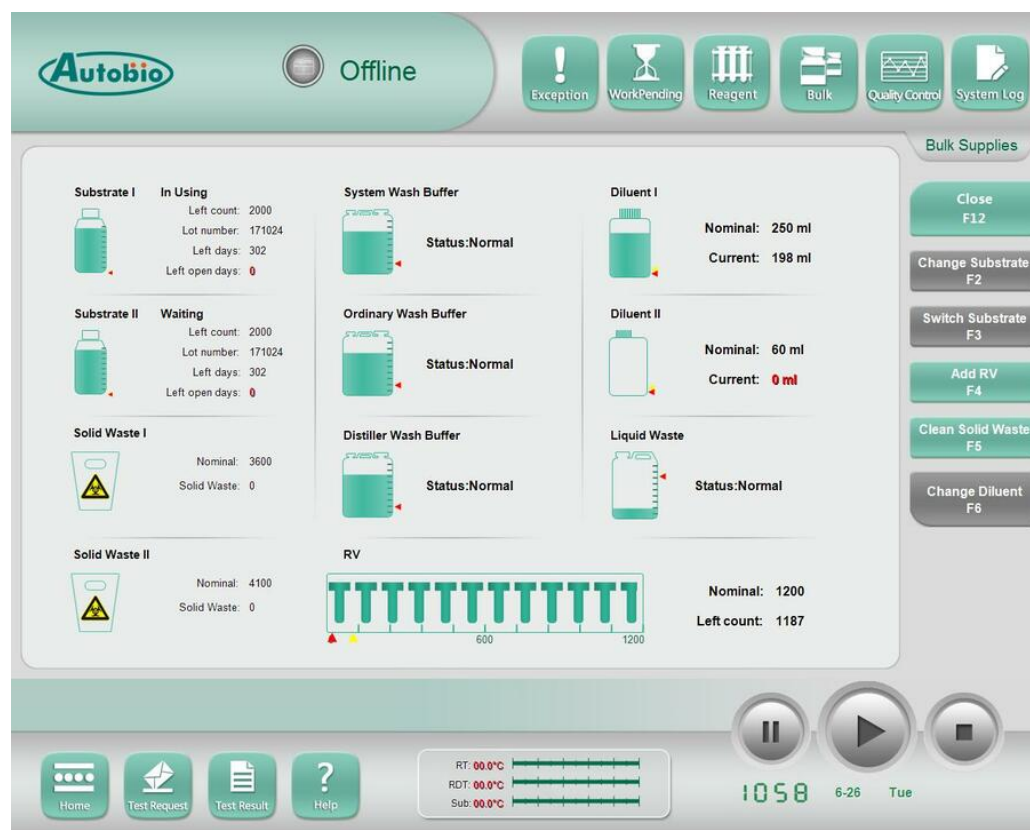
Interface Description:

Button Name	Description
Close	Drop out the interface and return to secondary navigation interface (Detect Analysis).
Reagent Inventory	Return to reagent kit list management interface.
Reagents Tray	Display all existing reagent kit.
Reagent Change	Carry out the operation of reagent change on the instrument.
Enable/Disable	Enable or disable the operation of reagent kit on the instrument.
Disenable Mixing	Disenable reagent mixing function.
Enable Mixing	Enable reagent mixing function.

8.3.1 Check The Reagent Alarm Information.

1. It will display details information while clicking the button which represents reagent kit (If there is an exception, the abnormal information column is marked in red or yellow). If there is alarm information for the reagent kit, the alarm information will be automatically oriented in the list.

8.4 Bulk



interface Description

Button Name	Description
Close(Labeled F12)	Exit the current window and return to the secondary menu (Test & Analysis, maintenance and Diagnosis) or directly back to the main menu if you didn't open the secondary menu before.
Change Substrate(Labeled F2)	Change the CLIA substrate.
Add RV(Labeled F4)	Place the reaction vessel to the cup supplier deck
Clean Solid Waste(Labeled F5)	Change the bag contains wasted reaction vessel
Change Diluent(Labeled F6)	Change the diluent liquid

8.4.1 Change Substrate

1. Select "Change Substrate" button switch to "Change Substrate" windows.
2. There are two types of substrate for **AutoLumo A2000 Plus**, select one of the substrate displayed on the interface you need (if you change substrate II, click the check box of substrate II displayed on the interface).
3. Scan or enter the bar code on the substrate bottles.
4. Open the cover of the instrument.

5. Take down the old substrate bottles and change into the new one. Put the old substrate bottles into biological waste bags.
6. Install the new substrate bottle.
7. Close the cover of the instrument.
8. Click 'OK'.
9. After change completed, instrument will wash pipe automatically.



CAUTION

1. The system will prompt user to change the substrate if the substrate is not enough when running. At this point the instrument is no longer planning a new test, but the current test will still continue.
2. Only in "Ready" state the 'Change Substrate' button is active, please flush substrate pipelines carefully after the change operation.



Warning

In the process you will be exposed to potentially infectious materials, please execute the standard laboratory procedures for handling and disposal the biological hazardous material. Take appropriate methods to protect hands, eyes and face.

8.4.2 Add RV

1. Click 'Add RV' button, switching to "Add RV" dialog.
2. Open the top cover of the cup supplier deck
3. Add new reaction vessel.
4. Close the cover of the cup supplier deck and the instrument's.
5. Click 'OK'.

8.4.3 Clean Solid Waste

1. Click 'Clean Solid Waste' button, switching to "Clean Solid Waste" dialog.
2. Open the bottom cover of the instrument.
3. Take down the container of solid waste and put the waste into biological waste bags
4. Install the container of solid waste.
5. Close the bottom cover of the instrument.
6. Click 'OK'.



CAUTION

The operation of clean solid waste should be executed when the instrument is power off, or the system is ready.

**Warning**

In the process you will be exposed to potentially infectious materials, please execute the standard laboratory procedures for handling and disposal the biological hazardous material. Take appropriate methods to protect hands, eyes and face.

8.4.4 Change Diluent

1. Click 'Change System Wash' button, switching to 'Change System Wash' dialog.
2. Open the top cover of the instrument.
3. Take down the old wash bottle and change into a new one. Put the old one into biological waste bags.
4. Install the new bottle.
5. Close the cover of the instrument.
6. Click 'OK'.

**CAUTION**

When doing dilution experiments, please make sure that dilution position has placed sufficient diluent, if diluent shortage during the test, instrument will alarm automatically, right now you need to add diluent manually, and click "change diluent" in the interface of bulk, the system will detect automatically, and cancel the alarm.

CHAPTER 9 SYSTEM SETTING

9.1 System Setting



Tip: System Setting requires administrator privileges

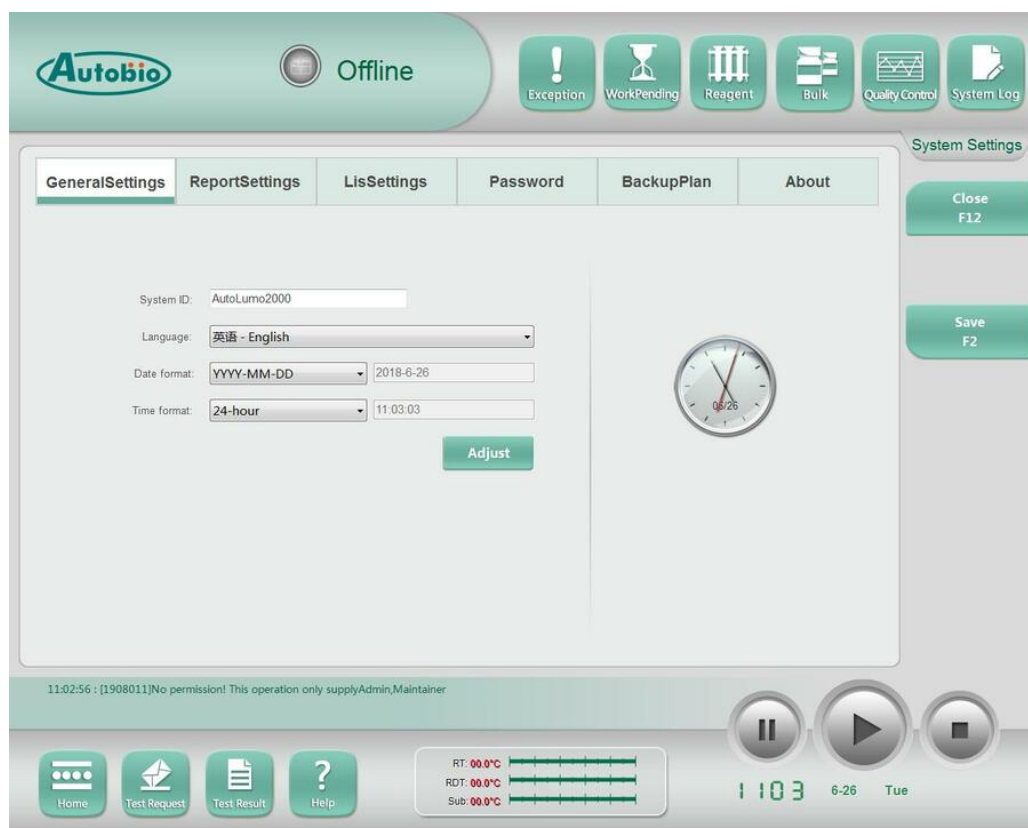
The default user is common operator, and you have to enter the password when click system setting. Password is no need if login the software with administrator privileges.

Interface Description

Button Name	Description
General Settings	General Settings includes system identification, date, time and language
Report Settings	Set the header information of the report, the printer selection and so on.
LIS Communication Settings	Set the LIS Communication parameters
Administrator Password	Reset administrator password
Backup Plan	Set the data backup parameters
About	Display the version information of the software

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes.

9.1.1 General Settings



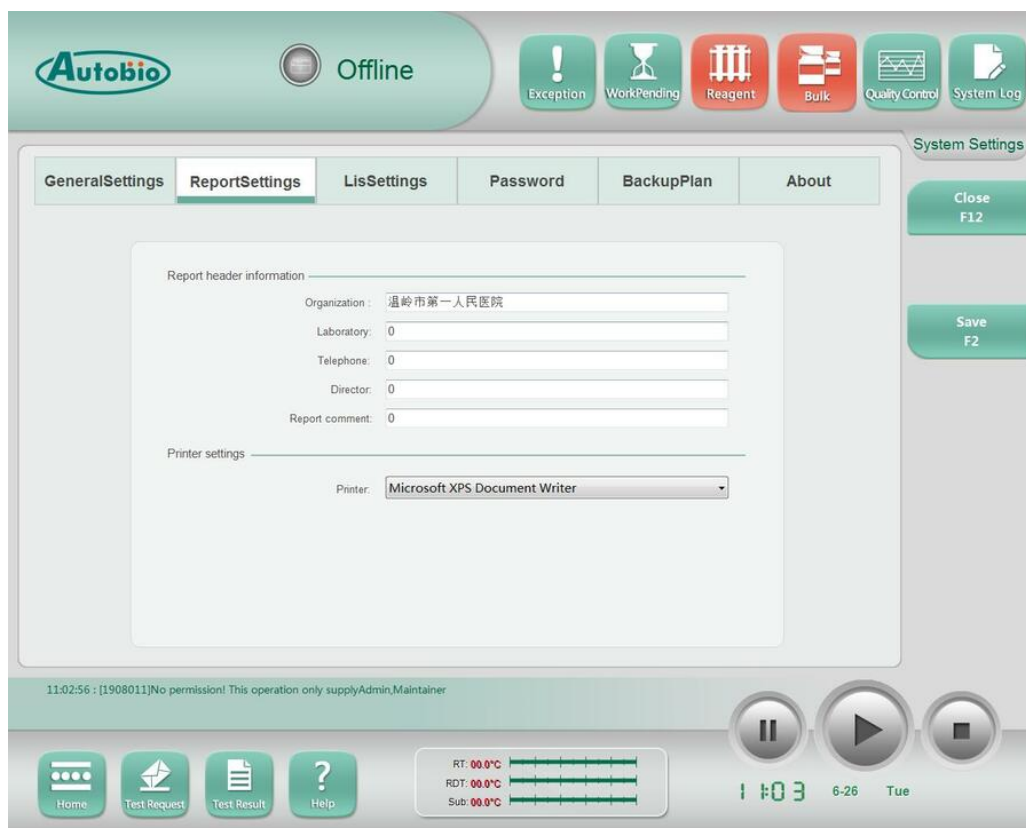
The default interface is the general settings when enter the system setting.

Interface Description

Button Name	Description
System ID	Set the System identification (1-12 characters)
Language	The Language of the software includes: (Chinese(Simplified, PRC)-Chinese (Simplified, PRC)、English-English)
Date Format	Set the date format (YYYY/MM/DD、YYY.MM.DD、YYY-MM-DD、MM/DD/YYYY、MM.DD.YYY、MM-DD-YYY)
Time Format	Set the time format(24-hour and 12-hour)

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes.

9.1.2 Report Settings



Interface Description

Button Name	Description
Organization	Set the name of organization
Laboratory	Set the name of laboratory
Telephone	Set the telephone number
Director	Set the name of director
Report comment	Set the comment of the test report
Printer	Select the printer
Report	Select report type, click "Format Adjustments" if it need to change

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes.

9.1.3 LIS Communication Settings

Autobio Offline

Exception WorkPending Reagent Bulk Quality Control System Log

GeneralSettings ReportSettings **LISSettings** Password BackupPlan About

System Settings

Close F12

Save F2

Receive LIS request

LIS interface: ☒ ON ☐ OFF

Automatically send to LIS

Result type: Send all results

Send mode: By test

LIS communication settings

Protocol type: Autobio

Communication type: RS232

Encoding type: UTF-8

Port: COM1

Baud rate: 9600

Data bits: 8

Parity: None

Flow control: None

Stop bits: 1

11:02:56 : [1908011]No permission! This operation only supplyAdmin,Maintainer

Home Test Request Test Result Help

RT: 00.0°C RDT: 00.0°C Sub: 00.0°C

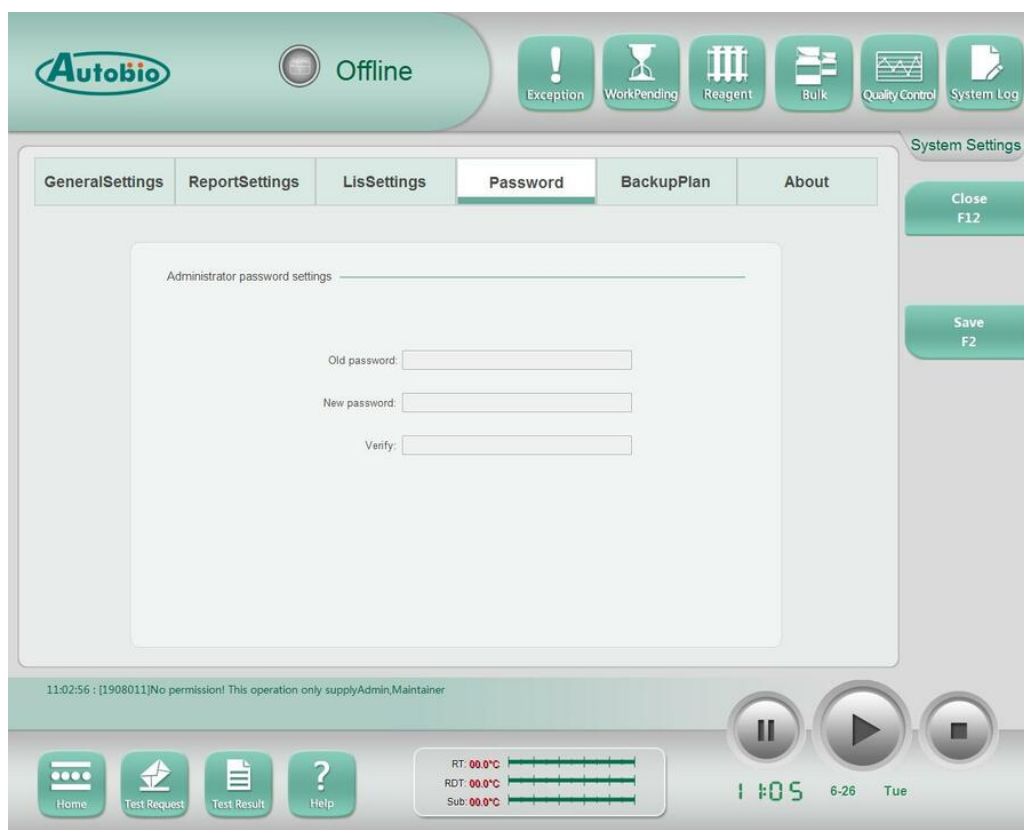
1104 6-26 Tue

Interface Description

Button Name	Description
LIS Interface 【ON】	Open the LIS communication interface
LIS Interface 【OFF】	Close the LIS communication interface
Auto send to LIS---Result Type	Select send all results or close the function
Send mode	Select the mode of send to LIS(By test or by sample)
Port	Set the LIS communication interface(COM0、COM1、COM2、COM3、...)
Baud Rate	Set the Baud Rate of LIS communication interface
Date Bits	Set the data bits of LIS communication
Parity	Set the parity parameters of LIS communication
Flow Control	Set the flow control of LIS communication
Stop Bits	Set the stop bits of LIS communication
LIS communication path	Set LIS export data location

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes

9.1.4 Administrator Password

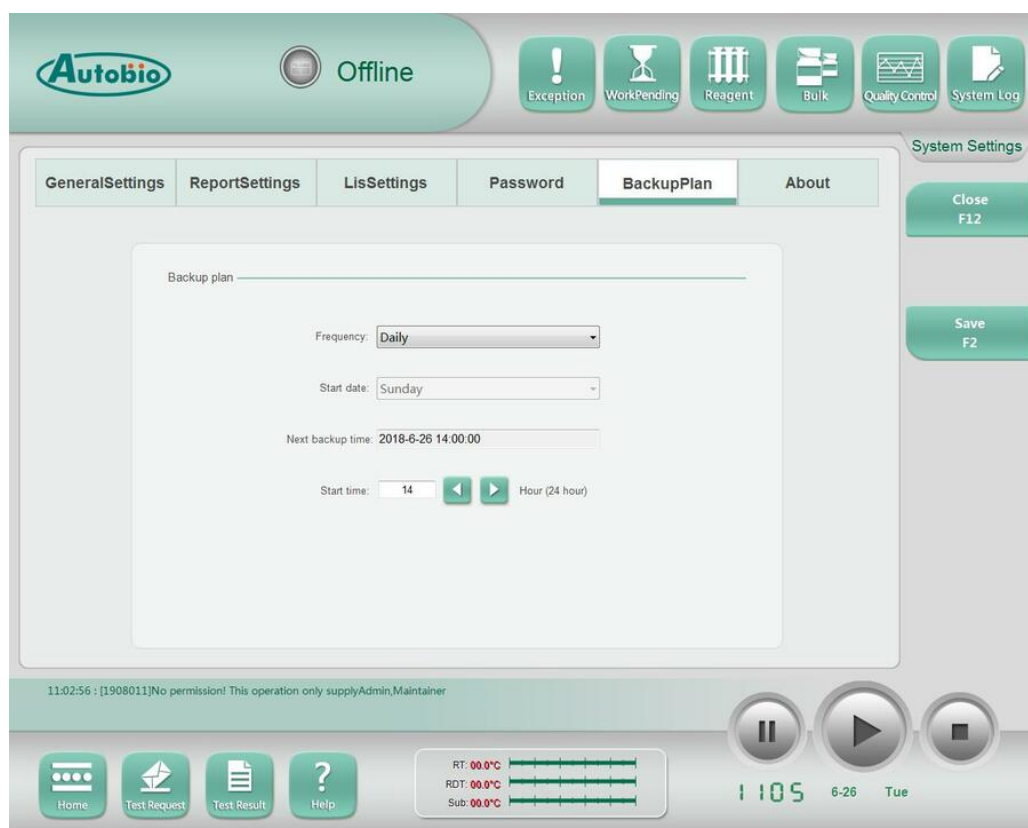


Interface Description

Button Name	Description
Old Password	Enter the old password of administrator
New Password	Enter the new password of administrator
Verify	Reenter the new password of administrator

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes

9.1.5 Backup Plan



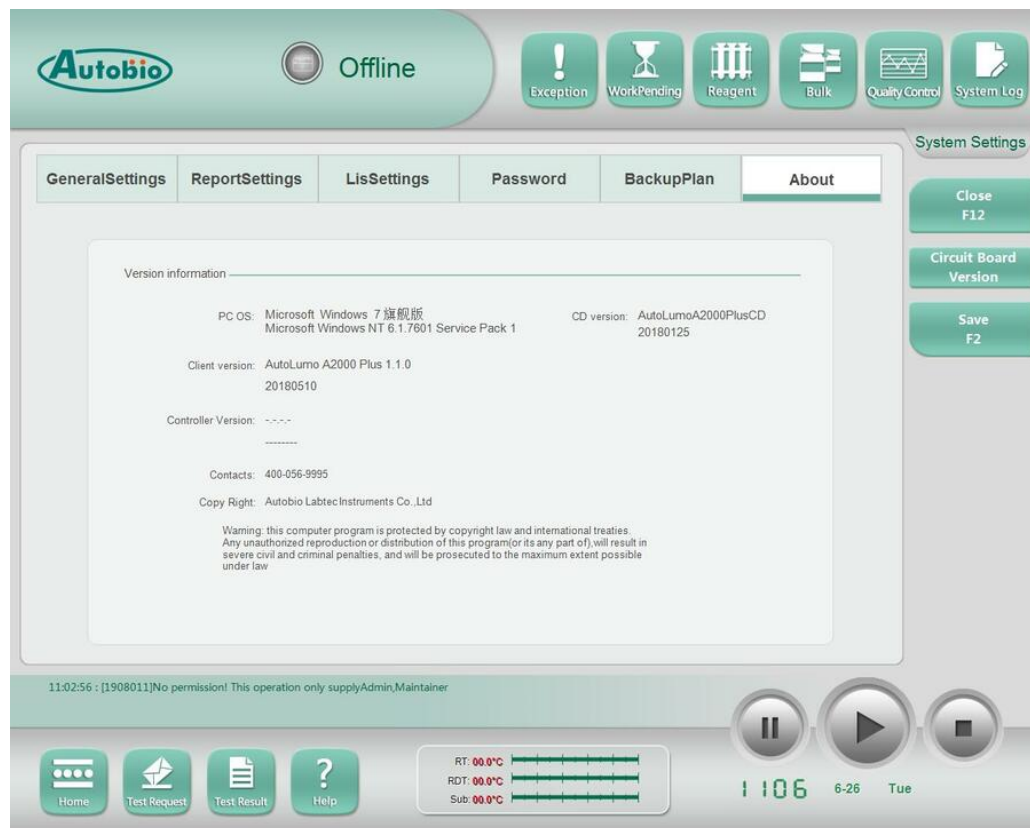
Interface Description

Button Name	Description
Frequency	Set the frequency of data backup(Daily or Weekly)
Start date	Set the begin date of data backup(Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday)
Start time	Set the start time of data backup(0-24 hour)
Next Backup Time	Display the next time of data backup

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the changes.

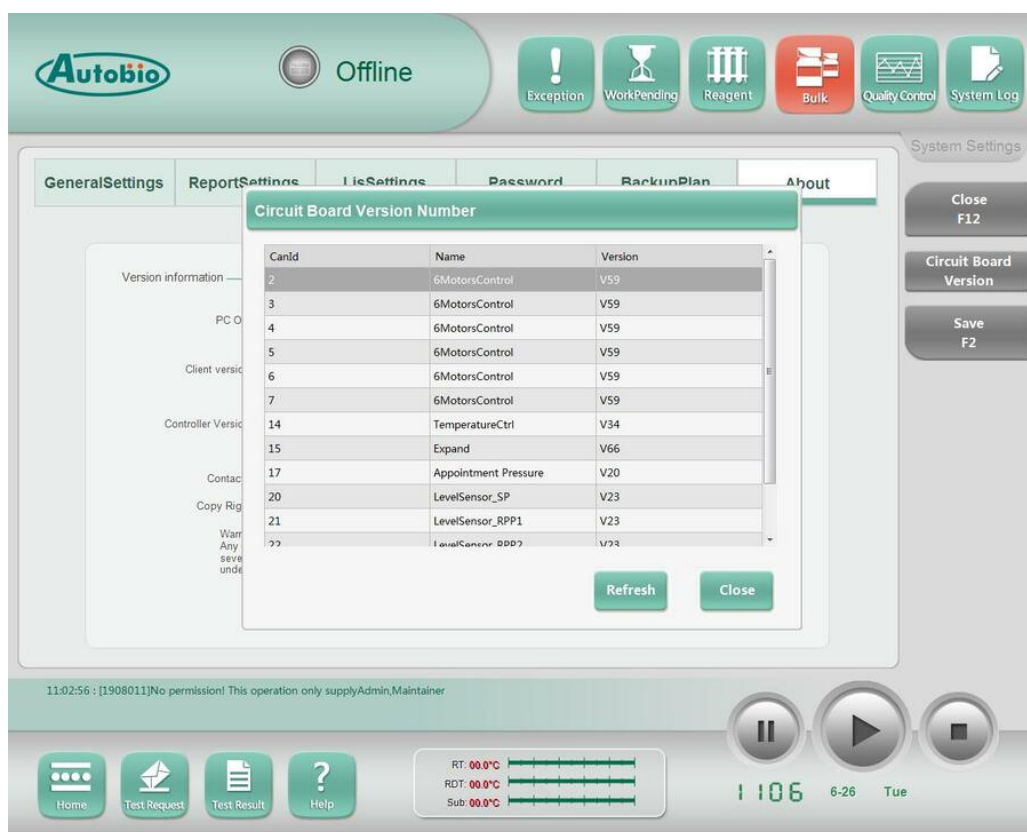
9.1.6 About

Browse the version of the software.



9.1.7 Circuit Version Number

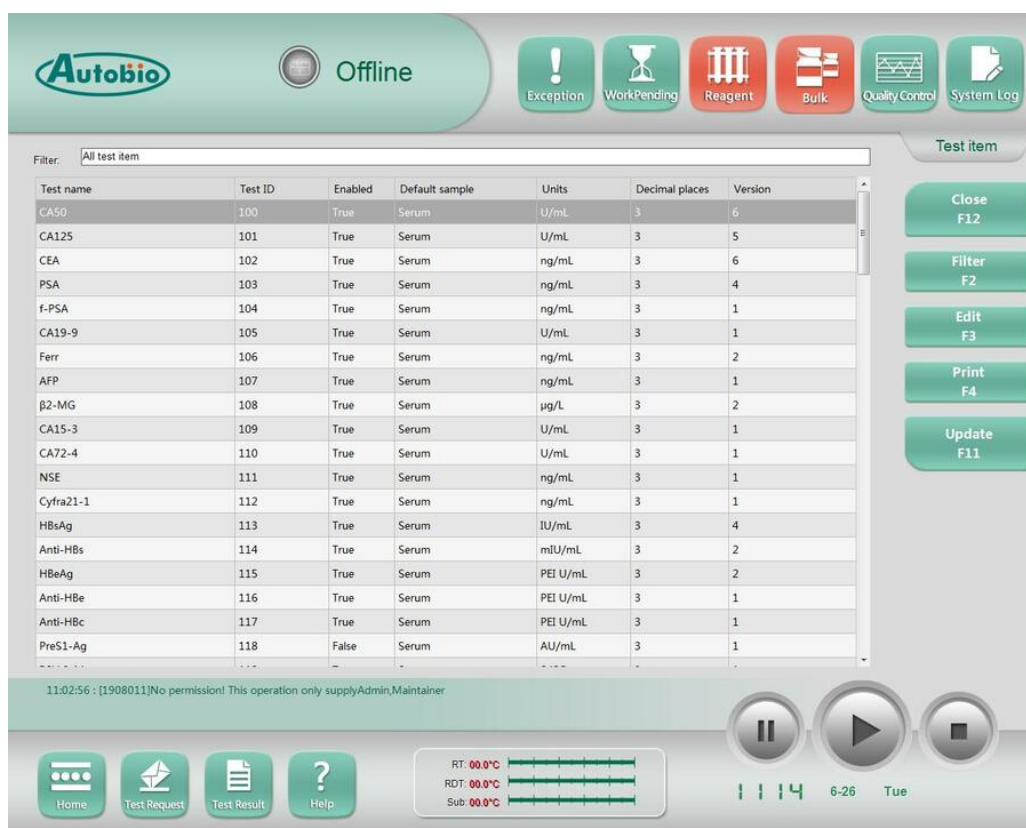
Within the permission of maintenance staff, the user can obtain the information of version number and CanID number in client software.



9.2 Test Items



Note: It requires administrator permission to operate the test item.



Interface Description

Button Name	Description
Filter test list	Display test information by filter
Filter(Labeled F2)	Set the filter factor to filtrate the test
Edit (Labeled F3)	Edit the parameters of test items.
Print(Labeled F4)	Print selected test
Update (Labeled F11)	Update test items.

Click 'Close' if you want to cancel the operation.

9.2.1 Filter

Click 'Filter' button to appear this window.

Button Name		Description
Range	All Tests	Enter the name and search in the all tests
	True	Enter the name and search in the enabled tests
	False	Enter the name and search in the disenabled tests
Sort		Select the parameters for sorting test (Test name, Test ID, Enabled, Default Sample, Units, Decimal Places and Version.
ASC		Tests will be list ascending according the filter
DESC		Tests will be list descending according the filter

1. Click 'OK' to confirm filter or 'Cancel' to abolish the operation.

9.2.2 Edit

Edit Test Item

Basic Settings

Name: CA19-9
Full Name: 糖类抗原CA19-9
ID: 105
Places: 3
Units: U/mL
☒ Enabled

Normal Range

Min: 2
Max: 1000

Reference Range

Range1: 0-35 U/mL
Range2:
Range3:
Range4:
Range5:
Range6:

Critical Range

Critical1: U/mL
Critical2: U/mL
Below Critical1:
Between Critical1 and Critical2:
Up Critical2:

OK

Cancel

Interface Description

Button Name		Description
Basic Settings	Name	Display test item name and can not be edited
	Full Name	Display full name of test item and can not be edited
	ID	Edit test item ID number and the number range is from 100 to 999.The repeat test item number is not allowed to enter.
	Places	Set the places number of the test results (0-5), exceeded the range will be handled by rounding up or rounding down.
	Units	Set data units of the test results (IU/mL、mIU/mL、NCU/mL、ng/dL、ng/mL、pg/mL、U/mL、μIU/mL、pmol/L)
	Enabled	Set enable the test or not.
Normal Range	Min	The minimum value of the normal range.
	Max	The maximal value of the normal range.
Reference Range		Set the reference range for printing the test result (0-60 characters) in range1 to range 6  Note: Reference range will not be printed in the current report.
Qualitative Judgment	Gray upper	Set the value equal to or higher than the critical value
	Gray lower	Set the value equal to or lower than the critical value

	Less than gray lower judgment result	Set the result when the result is less than the gray lower
	Between judgment result in gray	Set the result when the result is in the gray range
	More than gray upper judgment result	Set the result when the result is bigger than the gray upper
Critical Range (quantitative)	Critical value 1	Define critical value 1 of quantitative item.
	Critical value 2	Define critical value 2 of quantitative item.
	Less than critical value 1's results	Set the result when it is less than critical value 1.
	results between critical value 1 and critical value 2	Set the result when it is between critical value 1 and critical value 2.
	More than critical value 2's results	Set the result when it is bigger than critical value 2

1. Click 'Save' after achieving the settings.
2. Click 'Close' if you want to cancel the settings.

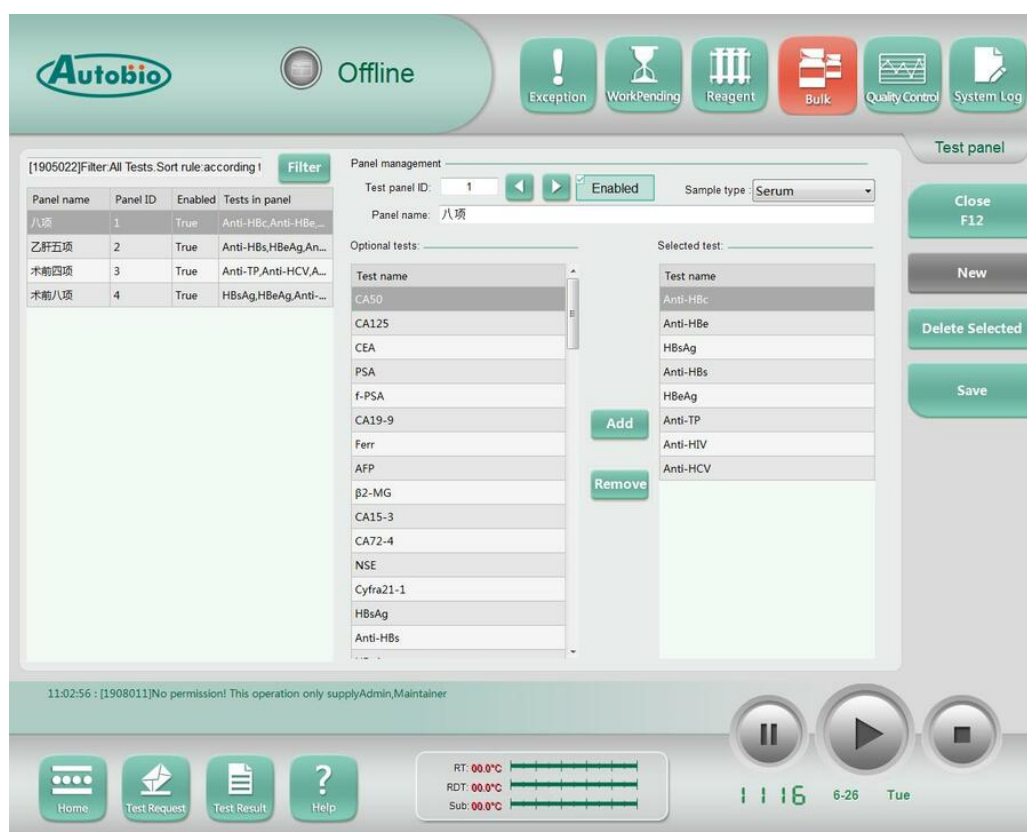
9.3 Test Panel



Prompt: Test Panel requires administrator

privileges

Select 'Test&Analysis' in the main interface menu to enter the Test&Analysis wizard window. Click Test Panel to enter this window.



Interface Description

Button Name	Description
New	Create a new test panel
Delete Selected	Delete the selected test panel
Save	Save after modifying test panel content
Test Panel ID	The ID cannot be changed if it is existed already.
Enabled	Enabled the selected test panel or not.
Sample Type	Select the type of sample.
Panel Name	Enter a name for a new test panel or edit the name of an existed test panel.
Optional Test	Display all tests of current sample type
Selected Test	Display the tests which are added into the test panel.
Add	Add the test into a test panel(multi-test and repetitive are allowable)
Remove	Remove the selected test.

1. Click 'Save' after achieving the settings.
2. Click 'Cancel' if you want to cancel the changes.
3. Click 'Close' if you want to close the window.

9.4 Derived Results



Prompt: Derived Results requires administrator

privileges

This function is used to derive a new result through four arithmetic calculate (addition, subtraction, multiplication, and division) according to one or more existing test results. As a new test item, the test result will create a new data in accordance with derivation rules after patient sample is completed with all tests.

Interface Description

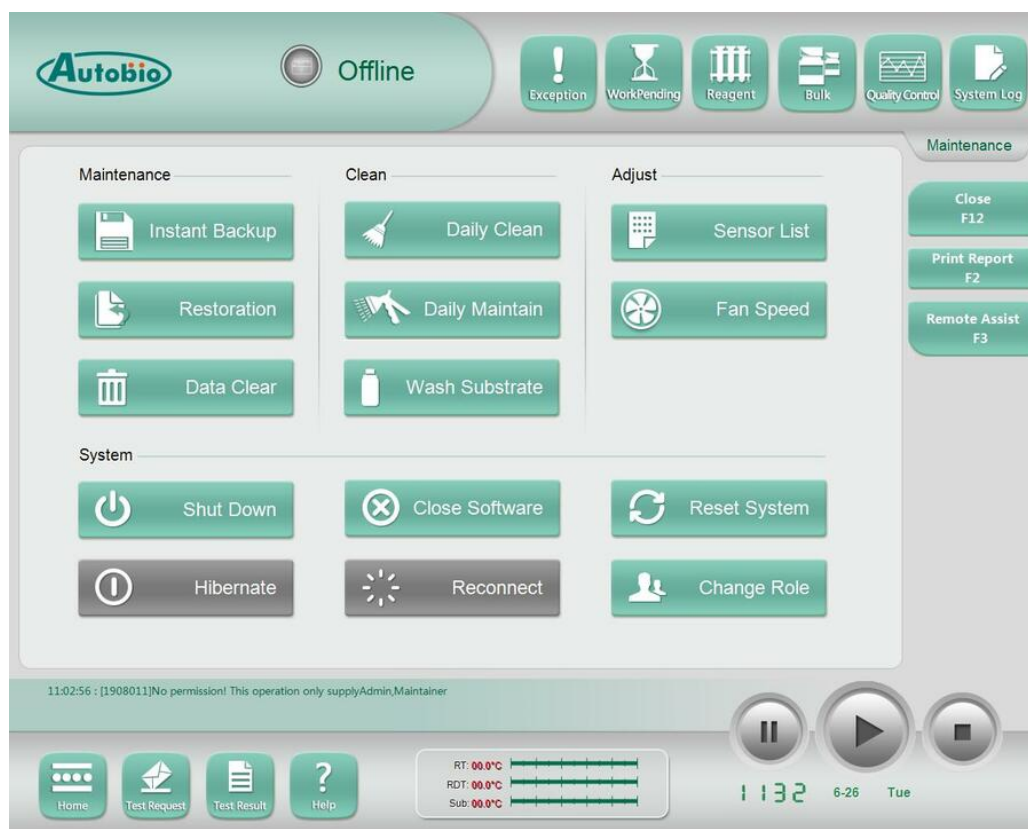
Button Name		Description
Derived Results		Display the existing derivation result information.
Derived Formula	Name	Set the name for derived formula.
	Units	Set the units for the test results.
	Enabled	Enabled the selected Derived Formula or not.
	Decimal Place	Set the decimal place of test results (0-5), exceeded the range will be handled by rounding up or rounding down.
	Formula	Enter or edit the derived formula.
Symbol		The button for derived formula.
Test Item		Display current tests item which can be selected.

Save As	Save current formula, and edit a new formula.
Clear Formula	Clear the formula.
Save	Save current derived formula.
Delete	Delete the selected derived test result

CHAPTER 10 MAINTENANCE

This menu provides some essential maintenance items for normal working of the instrument, and please doing the maintenance step according to the menu. The damage caused by improper maintenance is not guaranteed.

Select Maintenance button in the main menu to enter this window.



Note:

Some command buttons may be not displayed due to different management privileges. If you want to adjust the items, please contact with our engineers.

10.1 Maintenance



Prompt: Maintenance of the three operation

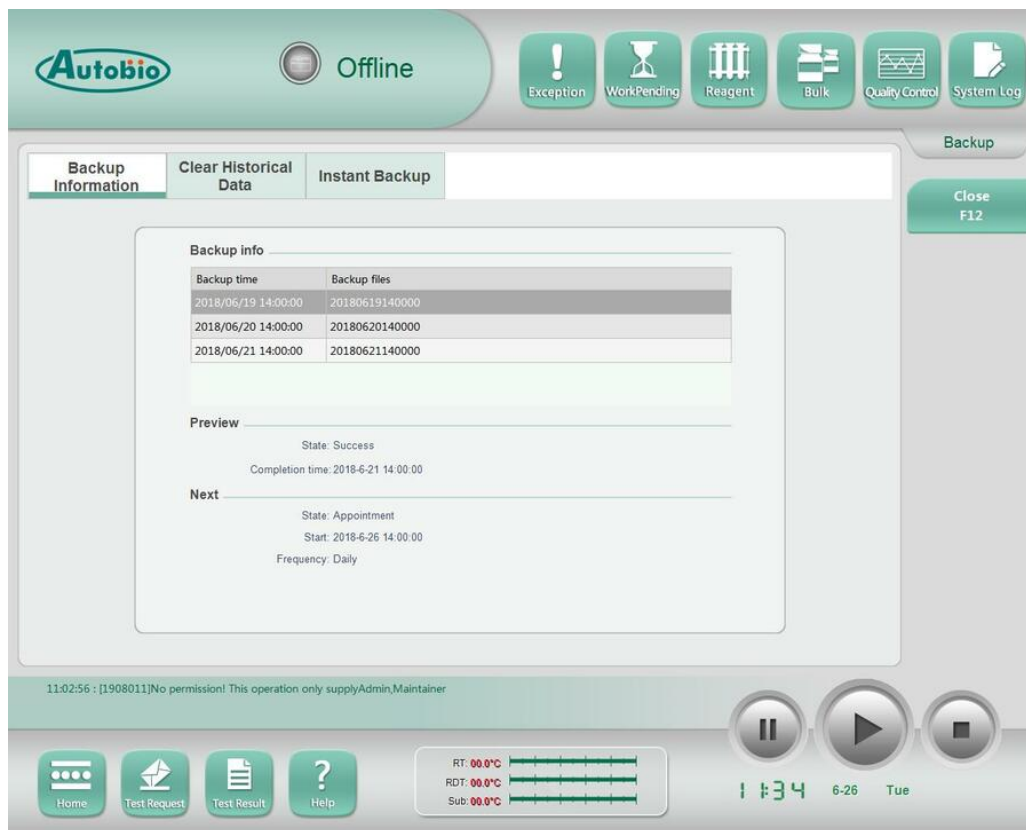
requires administrator privileges

The maintenance menu includes Immediate Backup, Recover System and Clear Historical data.

Instant Backup:

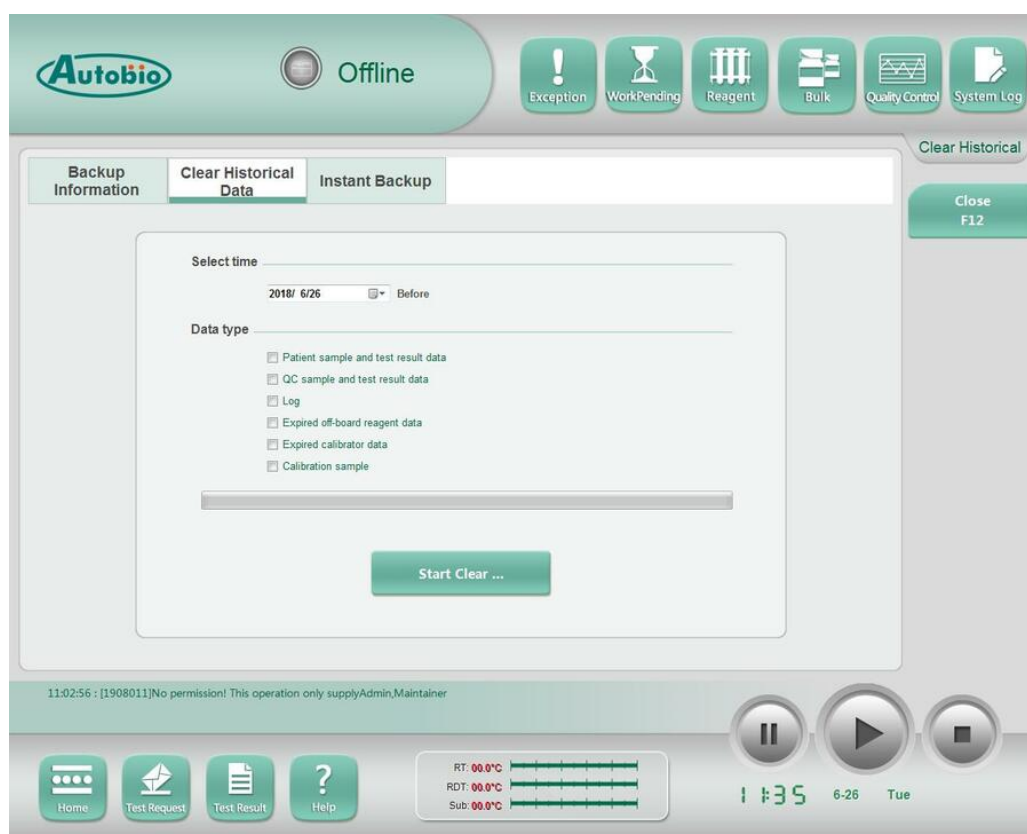
Backup the database to prevent data loss due to database corruption or misuse. The backup path can be designated.

You can check the detail information about the database in 'Backup' command after the backup is completed. It includes backup status, backup time and backup path. See the figure blow.



Data Clear:

The historical data can be deleted in the Data Clear menu by selecting time and data type. This operation should be careful as the data cannot be recovered.

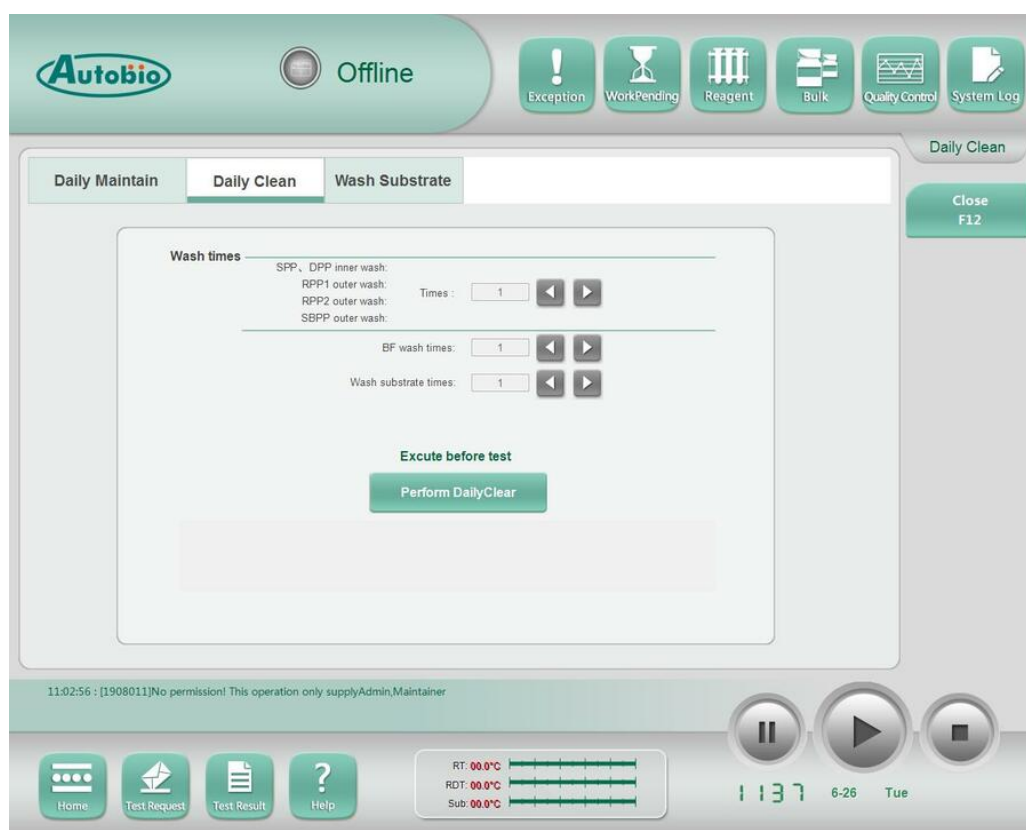
**Restoration:**

Choose restoration and it will give a dialog box asking whether to recover the entire system, select Yes or No. Recovery system will initialize the system, and therefore the database backup should be carried out first to prevent database corruption or data loss cause by incorrect content.

10.2 Clean

Daily Clean

Cleaning procedures should be performed by operator before starting the test every day. Due to the substrate tip has preheating function, it is recommended to clean at least 20 times before the start of the experiment.



Enter the times for washing and then click 'Perform Routine Clean', the progress of cleaning will be displayed at the below of the window. Do not try to finish the process of the clean, to avoid database damage or failure of the operation.

Notice

The substrate may affect test results if it is in high temperature for a long time, as the substrate tips have preheating function. The operation of substrate spit out should be executed strictly.

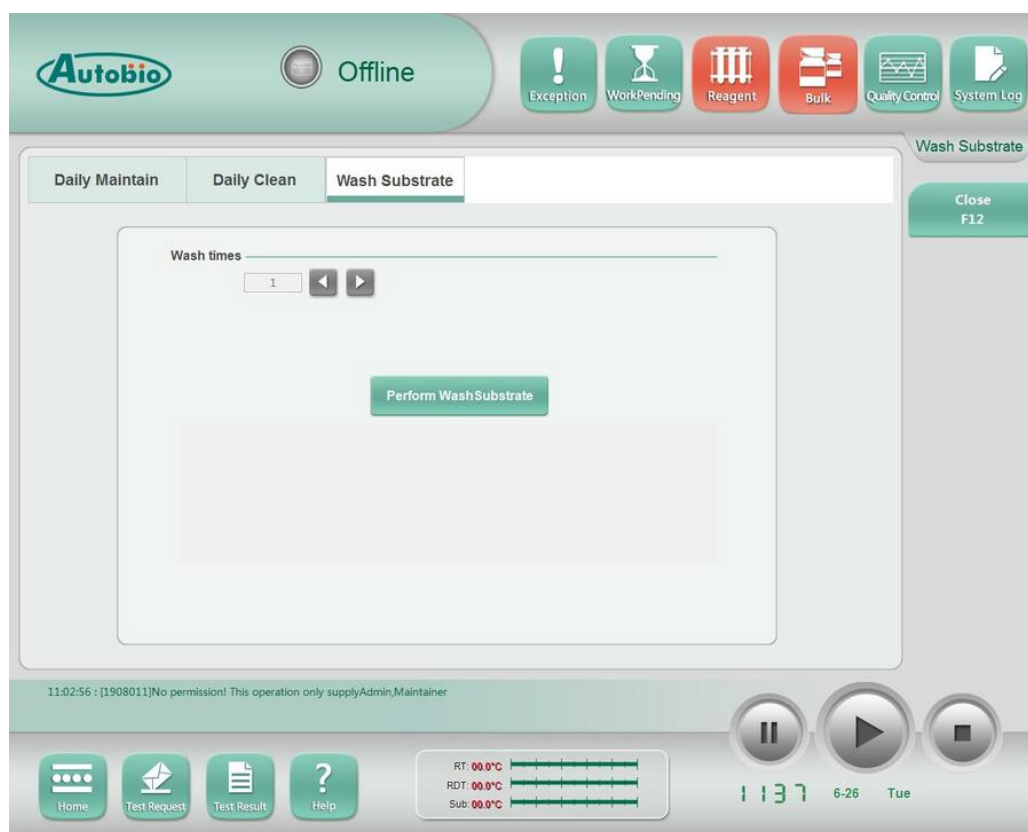
Routine Clean

It is suggested to run the cleaning procedures after finishing the test every day. The instrument should be cleaned in the longest idle date if your laboratory is continuing to run.

Wash Substrate

Replacement of substrate in two ways:

1. When changing a different batch of substrate, first pull the substrate tube out from the substrate bottle, then emptied at least 100 times, and wipe outside of the tube with distilled water, followed insert the substrate tube into a new bottle, last flush at least 100 times.
2. To change the same batch of substrate or if the instrument is idle for a long time, please flush at least 50 times. See the figure blow.



10.3 Clean Inside

Clean inside once every three days by operator:

Turn off the instrument first.

Open the cover of instrument.

Wipe the sample deck, reagent deck and RV supplier station with a cloth dipped water and neutral detergent.

Wipe the sample pipettor, #1 reagent pipettor, #2 reagent pipettor and substrate pipettor with a cloth dipped grain alcohol.

Close the cover of the instrument after cleaning and checking the safety device is normal.

10.4 Clean Outside

Clean outside once every three days by operator:

Turn off the instrument first.

Wipe the instrument housing with a dry cloth dipped neutral detergent.

 **WARNING**

In the process you will be exposed to potentially infectious materials, please execute the standard laboratory procedures for handling and disposal the biological hazardous material. Take appropriate methods to protect hands, eyes and face.

Acid and alkaline lotion is harmful to eyes and skin, please following the correct laboratory procedures for processing.

Turn off the instrument before operation to prevent electric shock.



CAUTION

To ensure the accuracy of experimental results, suggest that change the sample pipettor once every half a year, change the liquid pipeline once every half a year.

10.5 Debug

Sensor List:

Display port status of each sensor, convenient for you to troubleshoot and feedback problems.

10.6 System

Shutdown:

In case of relocation the instrument or shutdown it for more than 5 days, please close the software and computer.

Low Power:

In the condition of do not use the instrument for a long time, select instrument hibernated to save energy. At dormant state, instrument cannot run testing.

Reconnect:

Restart software and computer.

Close software:

Close the software.

Reset system:

Perform initialization of the control system and the operating system in the state of "Not Ready". During the process, the instrument will not be able to operate.

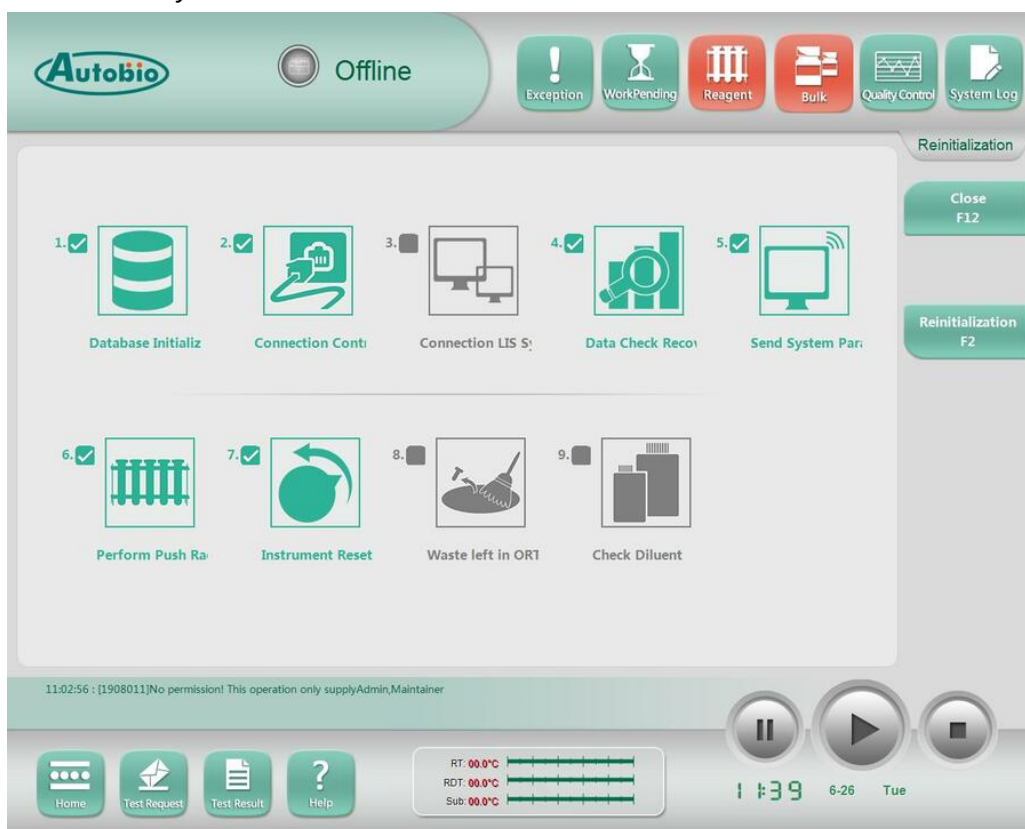
Change role:

Different user permissions switch.

10.7 Reinitialization

When the instrument cannot work normal caused by fault or other problems, you can reinitialize the software and the operating system. Select the items you need

and click reinitialization button. The default items are marked and you can change them manually.



Notice

Before the reinitialization you should perform each needle returns to the initial position, in order to reduce the risk of striker.

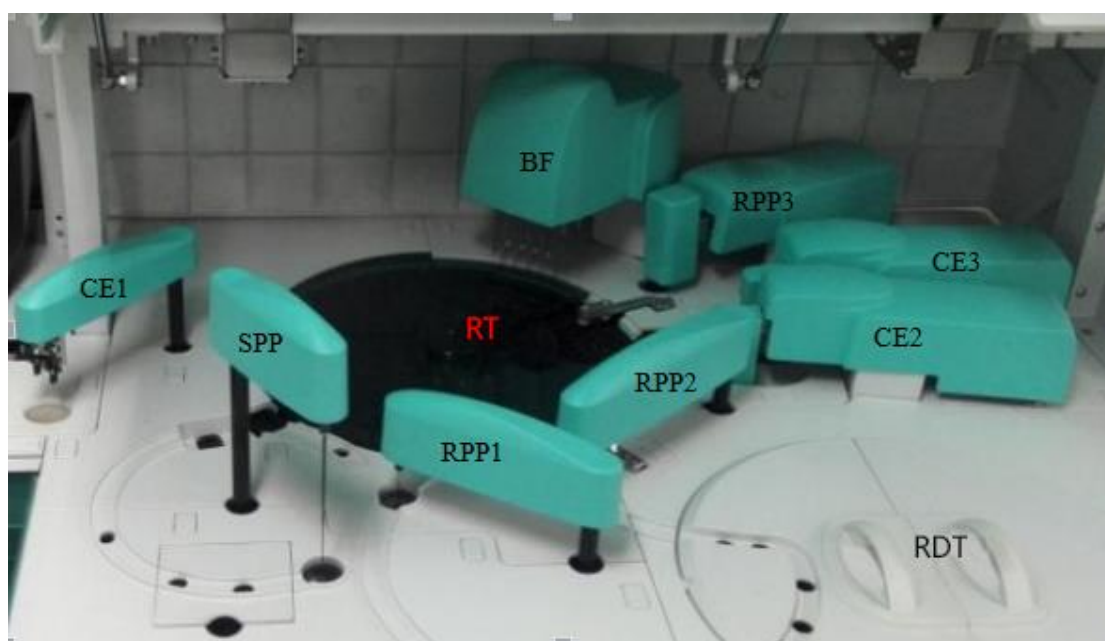
CHAPTER 11 TEST OPERATION

11.1 Preparation

11.1.1 Instrument Position Check

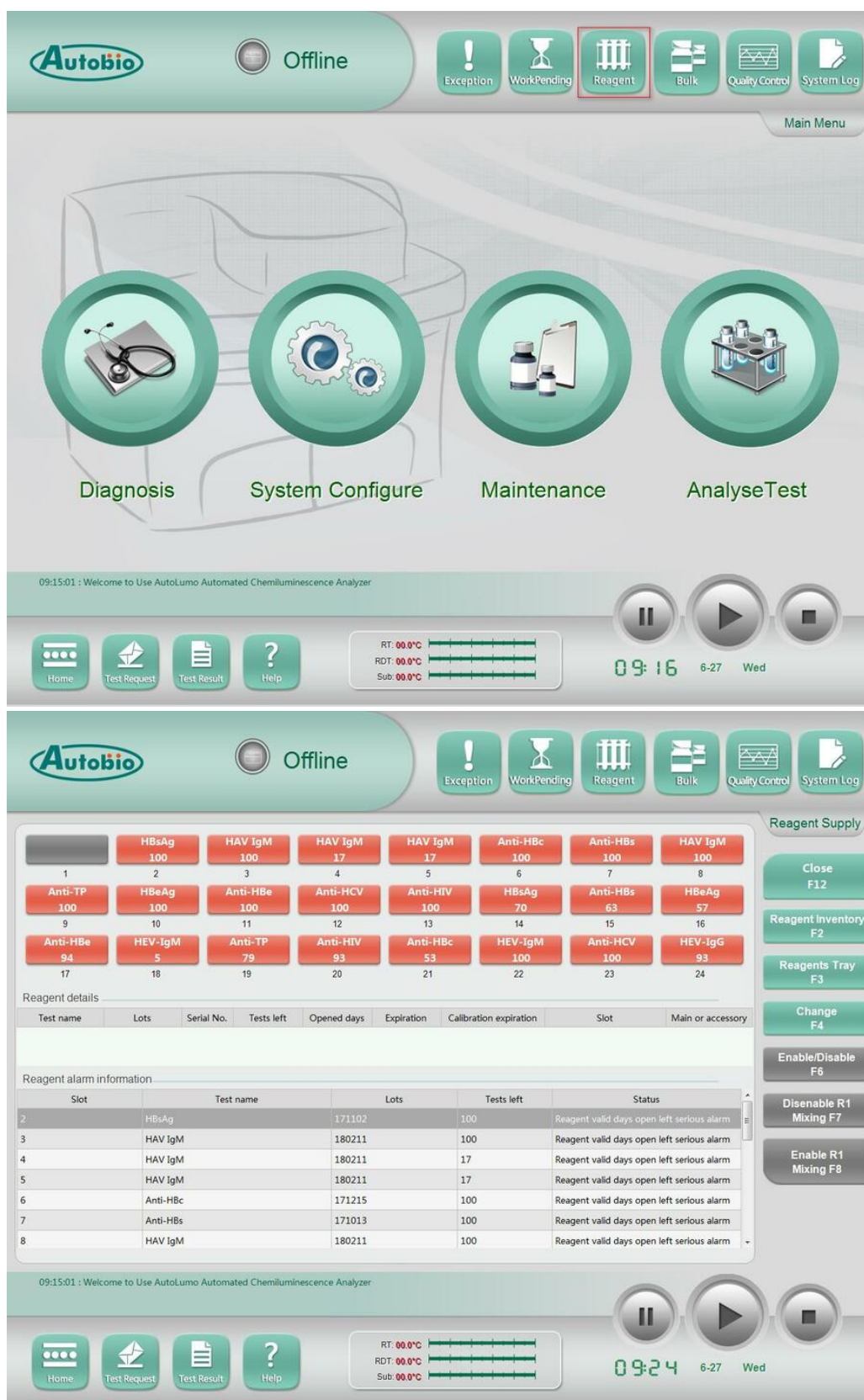
Open the front cover to confirm:

1. Open the reaction tray cover and confirm whether there is a reaction vessel on the plate. If there is a reaction vessel, please clean it manually.
2. Confirm the SPP, RPP1, RPP2 and RPP3 is located above the washing station. If they are not on the position, remove them manually.
3. Confirm that CE1, CE2 is located above the discarding vessel, CE3 is above measurement position; If they are not on the position, remove them manually.
4. Confirm that all covers of reaction tray, reagent tray, instrument front and rear are well closed.
5. Confirm that the emergency switch button is not pressed.



11.1.2 Reagent Check

1. Click "reagent" button to enter reagent alarm interface.



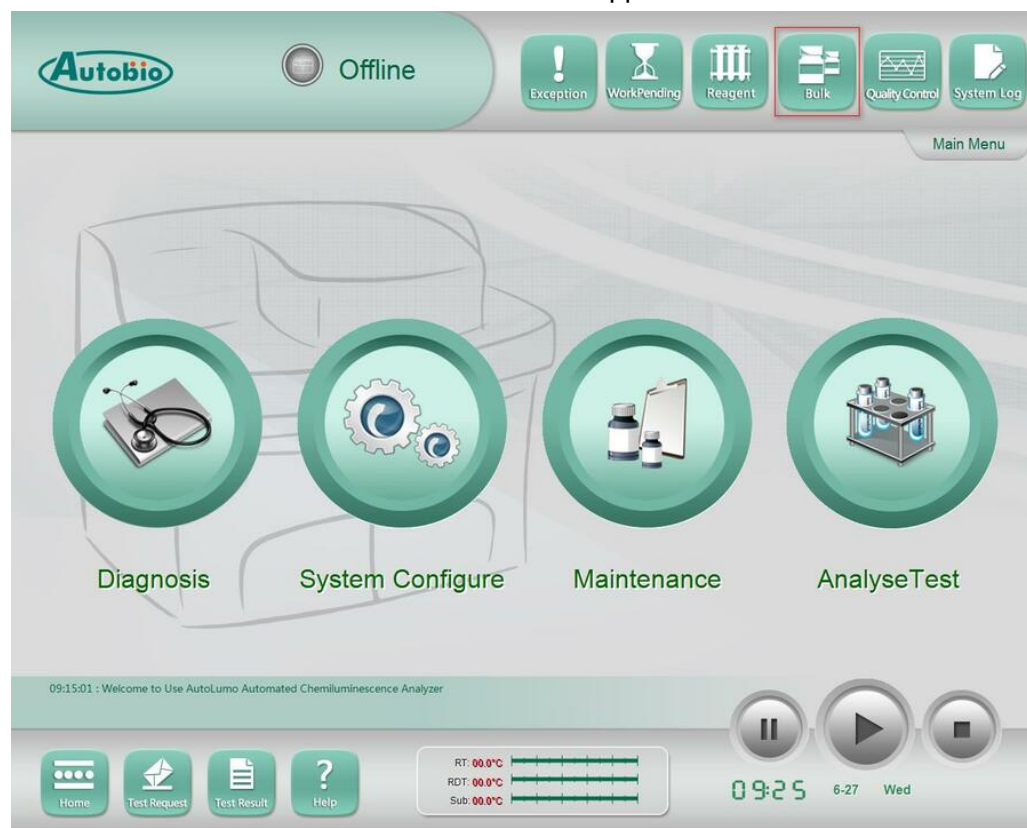
2. Check whether the reagent kit on RDT is sufficient for test, if not please add the reagent. (Refer to chapter 8)

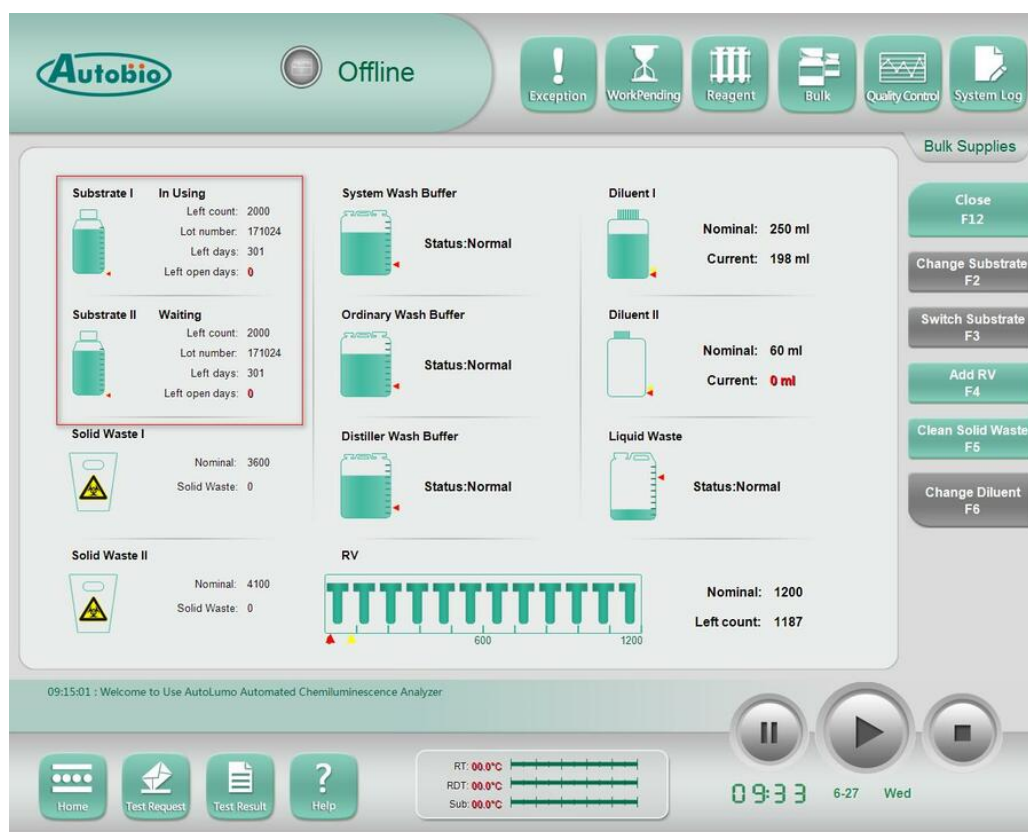
Notice:

- ◆ Be sure to remove the cap sticker from the newly added reagent.
- ◆ Confirm whether there is bubble in the new added reagent bottle. Please remove the bubble by the sample gun if available.
- ◆ The reagent requires mixing for 30 minutes before test.
- ◆ Calibrate before use the new lot of reagents.
- ◆ Confirm the reagent change before test starts. The instrument will automatically scan the reagent by click "Reagent Kit Change" (Refer to Chapter 8.1).

11.1.3 Chemiluminescent Substrate Check

1. Click the "Bulk" button and enter into "Bulk Supplies" interface





2. Check whether left count of Substrate I and Substrate II can meet the requirements of test. Please add new substrate if they are not sufficient (Refer to the section of 8.3.1).

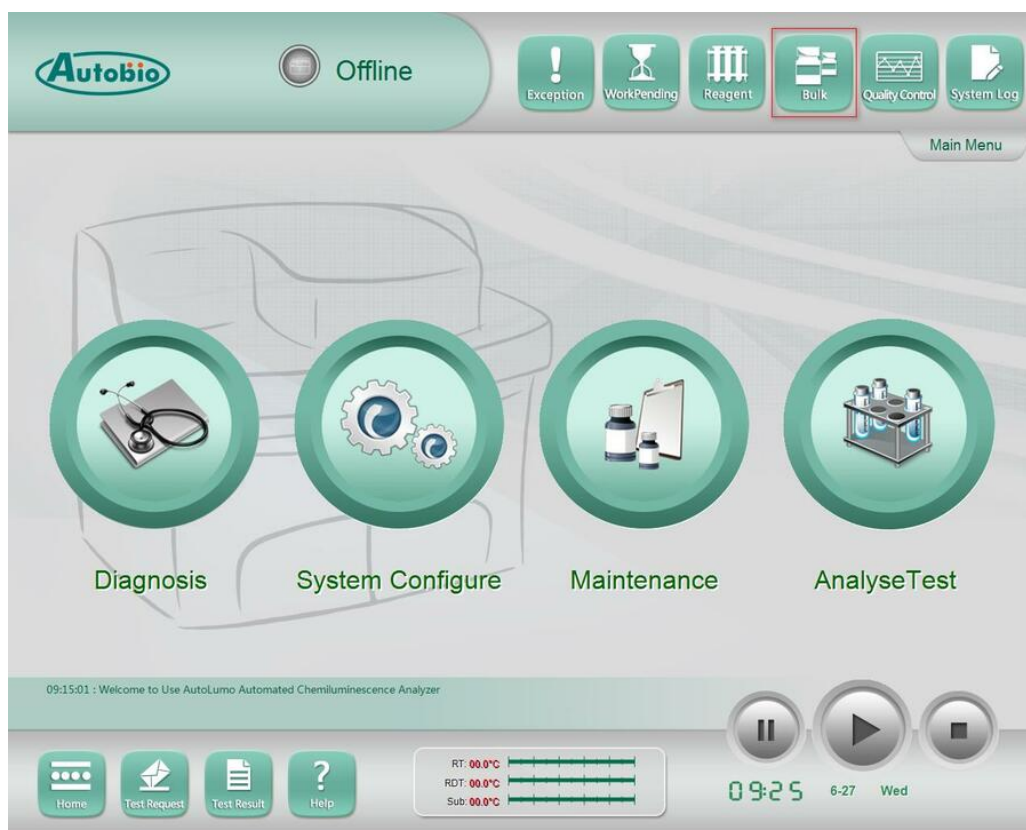
Notice:

- ◆ Do not place the substrate A and substrate B in contrary. Please refer to the color of bottle cap.
- ◆ Please check whether the pipette inside of the substrate bottle is bent and use medical gloves to straighten it before place the substrate.

11.1.4 Consumables Check

11.1.4.1 RV Check

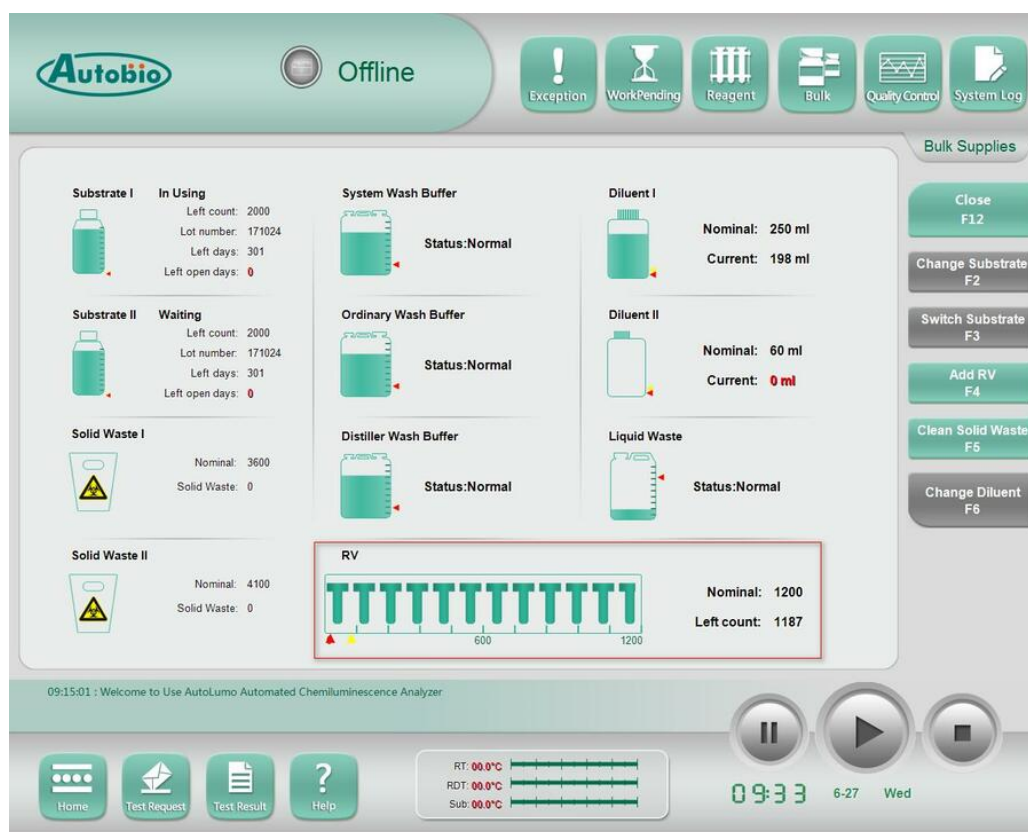
1. Click 'Bulk' button and enter into 'Bulk Supplies' interface.



2. Check whether the left count of RV can meet the requirements of test, if it is not sufficient, please execute the operation of 'Adding RV'
3. Check if there are enough RV in the RV container. Please add RV without delay and execute the operation of 'Adding RV'

Notice:

- ◆ The number of RV in the container can not exceed 1000 and the height can not reach up to two third of the container.
- ◆ After RV is added to the container, operation of 'Adding RV' will be executed.



11.1.4.2 Wash Buffer Check

Check whether there is enough ordinary washing lotion and that in the system washing lotion container. Please add without delay if lotion is insufficient.

Notice:

- ◆ Change the washing lotion before test if they are not used for a long period and clean off the container.
- ◆ Test is suspended due to lack of washing lotion. New washing lotion should be added before click 'Start' button.

11.1.4.3 Distilled Water

Check whether there is enough distilled water in the container. Please add if it is insufficient.

Note: If there is distilled water equipment, manual adding is not required.

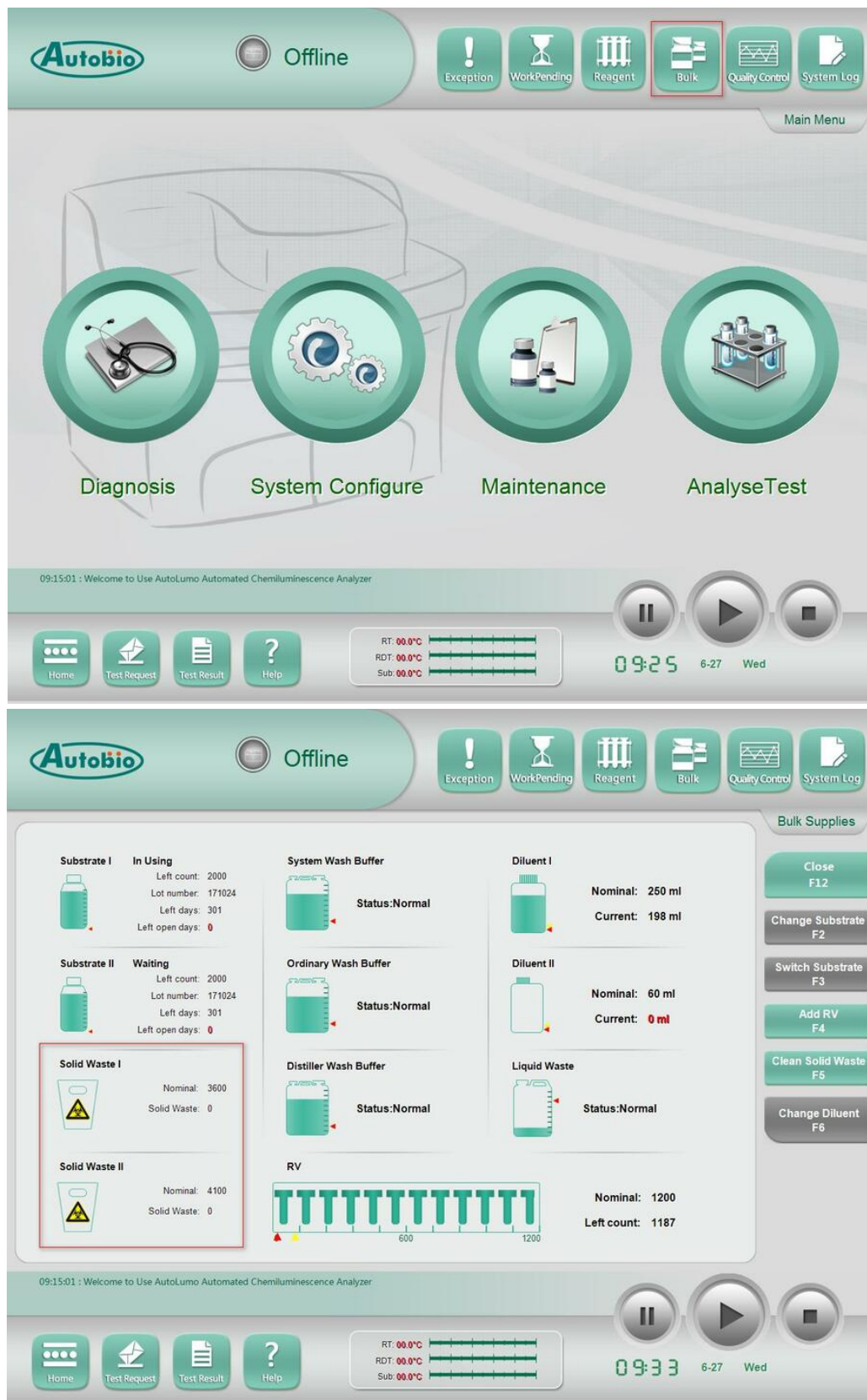
11.1.4.4 Liquid Waste Check

Please clean off waste liquid without delay if there are too much in the waste liquid container.

Note: If there is waste liquid discharging pipe connecting to the instrument, manual cleaning is not required.

11.1.4.5 Solid Waste Check

1. Click 'Bulk' button and enter into 'Bulk Supplies' interface



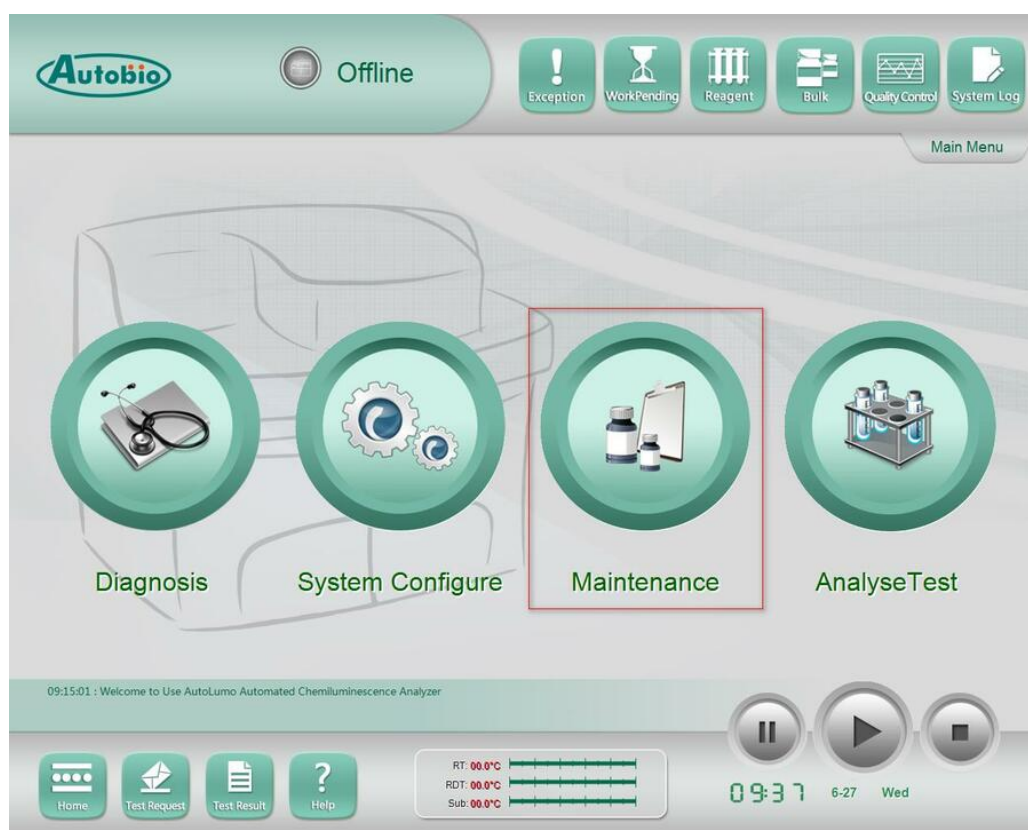
2、 Check whether the solid waste cabinet I and solid waste cabinet II are stacked with solid waste. Please clean off the cabinets if full without delay and execute the right side function of “Clean Solid Waste”

Notice:

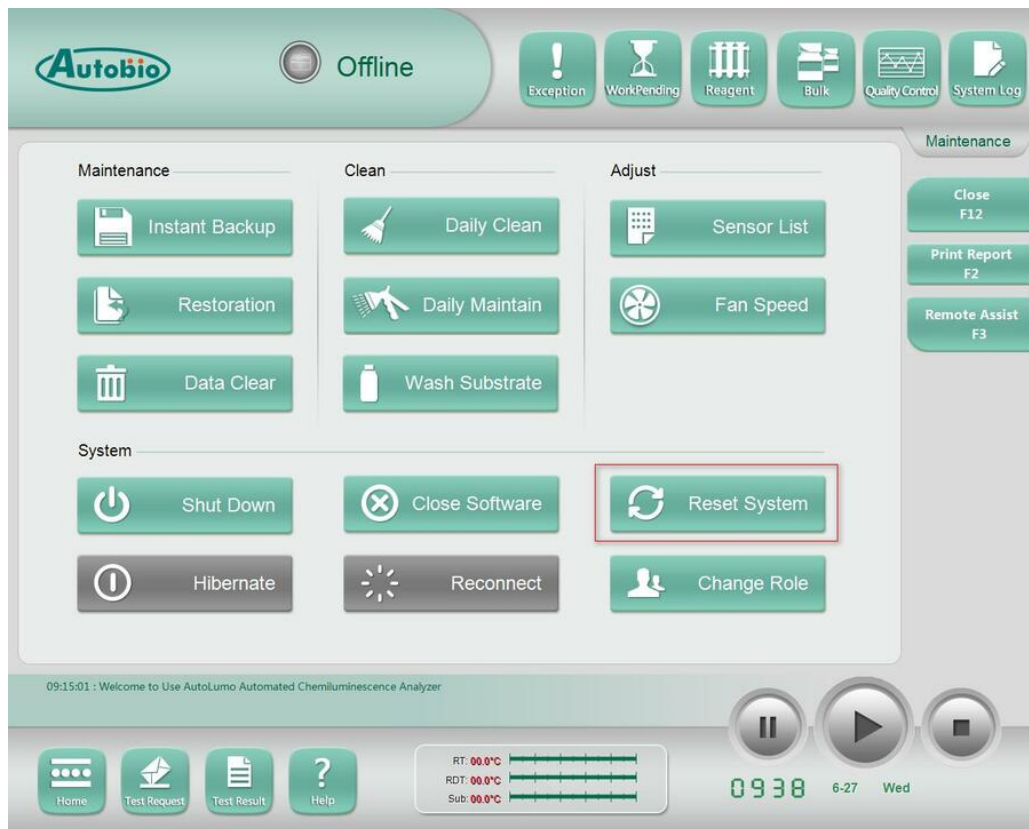
Cleaning off the two solid waste cabinets while executing the operation of “Clean Solid Waste” in the software.

11.1.5 Instrument Initialization and Daily Cleaning

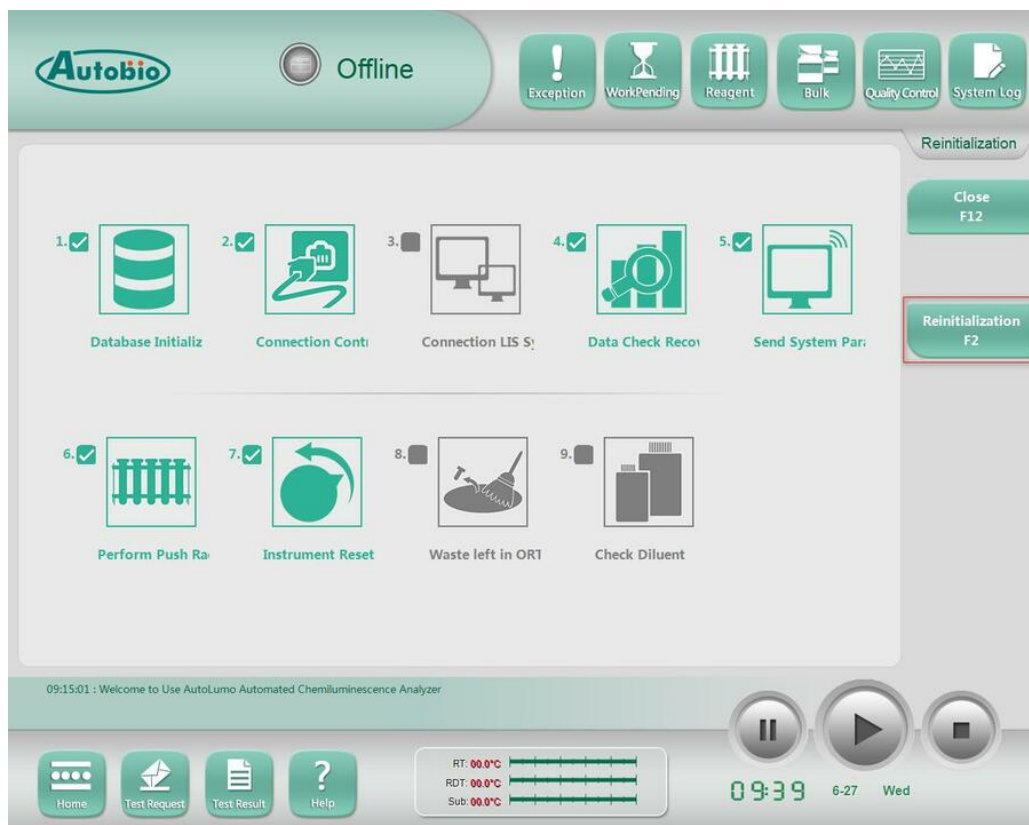
1. Click ‘Maintenance’ in the software interface and enter into Maintenance interface.



2. Click ‘Reset System’ and enter into reinitialization interface.



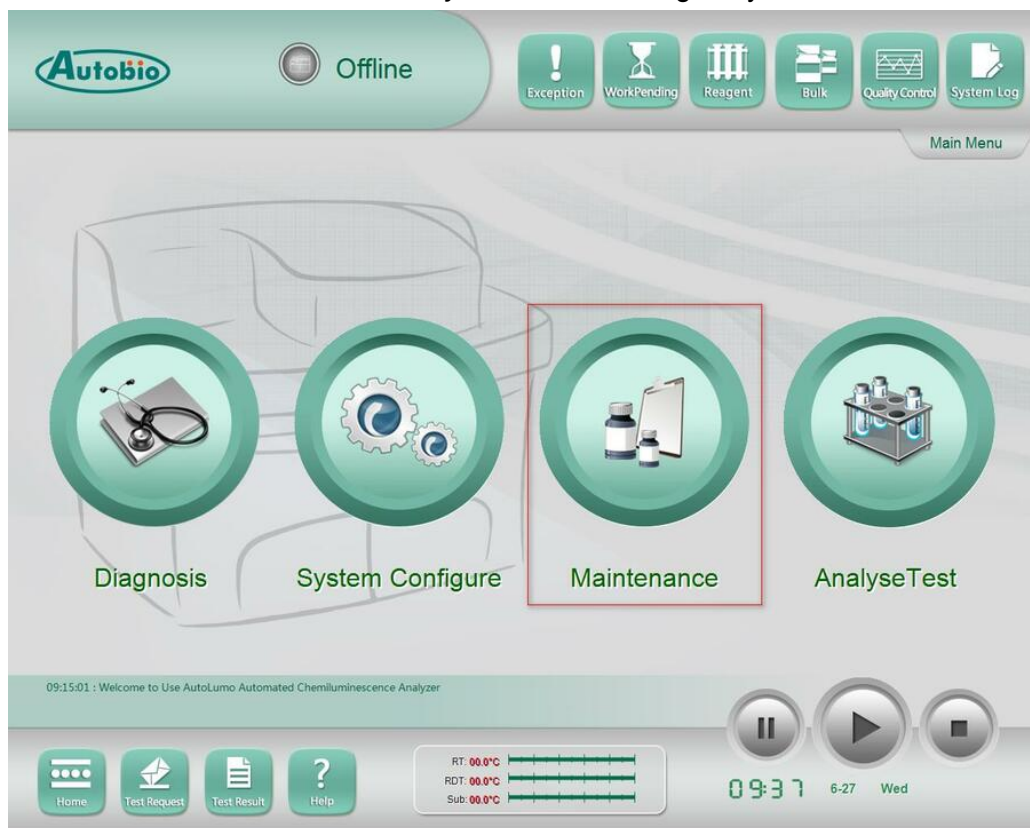
3. Click 'Reinitialization'

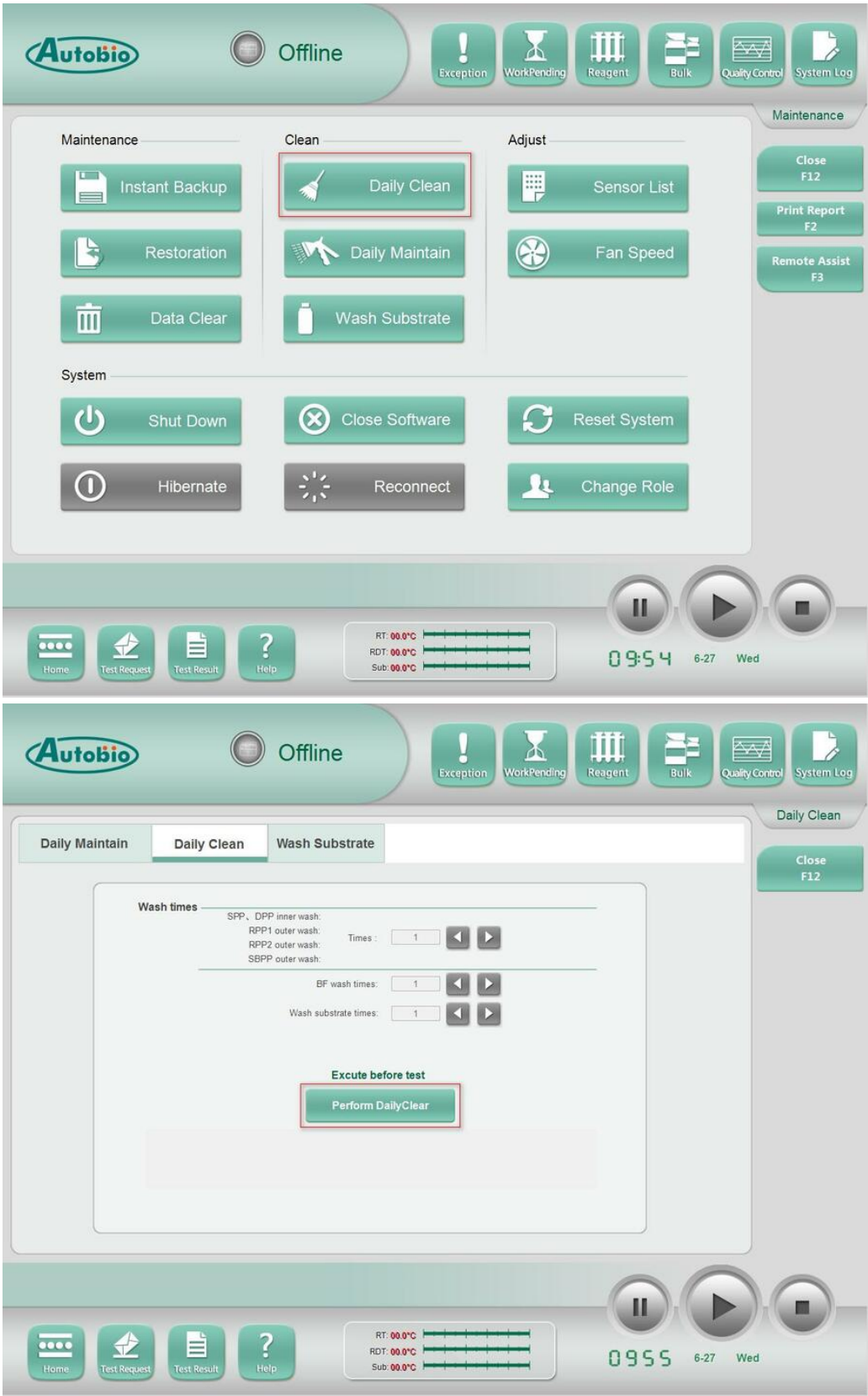


4. It indicates 'Ready' after reinitialization is completed.



5. Execute 'Maintenance' - 'Daily Clean' - 'Executing Daily Clean' in the software.





11.1.6 Patient Test Request

11.1.6.1 Sample Pretreatment

Before the experiment, the sample should be pretreated and the pretreatment should meet the following requirements.

1. The serum is suitable for most test items, and some tests can only use serum, and the sample type is referred to the instructions of the kit.
2. Sample pretreatment is of great importance. The residual fibrous protein in the sample, caused by improper mixing, agglutination time short, sample tube not being placed up right and insufficient centrifugation, may lead to abnormal test.
3. Severe hemolysis, severe lipid blood, turbidity, flocculent or coagulant can not be used, which will cause the magnetic particle to be unable to disperse effectively and seriously affect the determination result.
4. Avoid cross contamination during sample operation and use disposable SPP or sample injection tips.
5. The samples prepared by static separation method cannot guarantee the accuracy of the results, and the samples need to be centrifuged.
6. Samples containing lipid stratification, blood clots, blood silk, protein groups or silk should be centrifuged after removal of impurities.
7. The proposal sample centrifuge procedure:
 - ◆ Keep the sample tube under incubation at 37°C for 30 minutes(if incubation condition can not be realized, keep separate gel tube at 20-30°C ambient temperature for no less than 30 minutes and general tube for no less than 60 minutes) and centrifugation at rotation of 3500-4000 rpm for 10 minutes at least.
8. Sample intensified centrifugation process(Apply for high sensitive sandwich method item)
 - ◆ General tube sample : keep the sample under incubation at 37 °C for 60 minutes, then transfer 10000 g of the serum into centrifugation tube for 10 minutes and extract the upper serum for test.
 - ◆ Pro-coagulation tube sample: keep the sample under incubation at 37 °C for 30 minutes, then transfer 10000g of the serum into centrifugation tube for 10 minutes and extract the upper serum for test.
 - ◆ Anti-coagulation tube sample: transfer 10000g of the plasma into centrifugation tube for 10 minutes and extract upper serum for test.
 - ◆ RCF value is depended on rotation speed and radius gyration(r, calculated by mm). The following formula can be used:

$$RCF(g) = 1.11 \times 10^{-6} (rpm)^2 r$$

9. A sample containing lipids in the upper part after centrifugation need to be transferred to the second sample cup or test tube, which should be carefully manipulated to ensure that the sample does not contain lipid.

10. Check whether there is bubble in the sample. Remove bubbles by using disposable SPP or tips to avoid cross contamination.

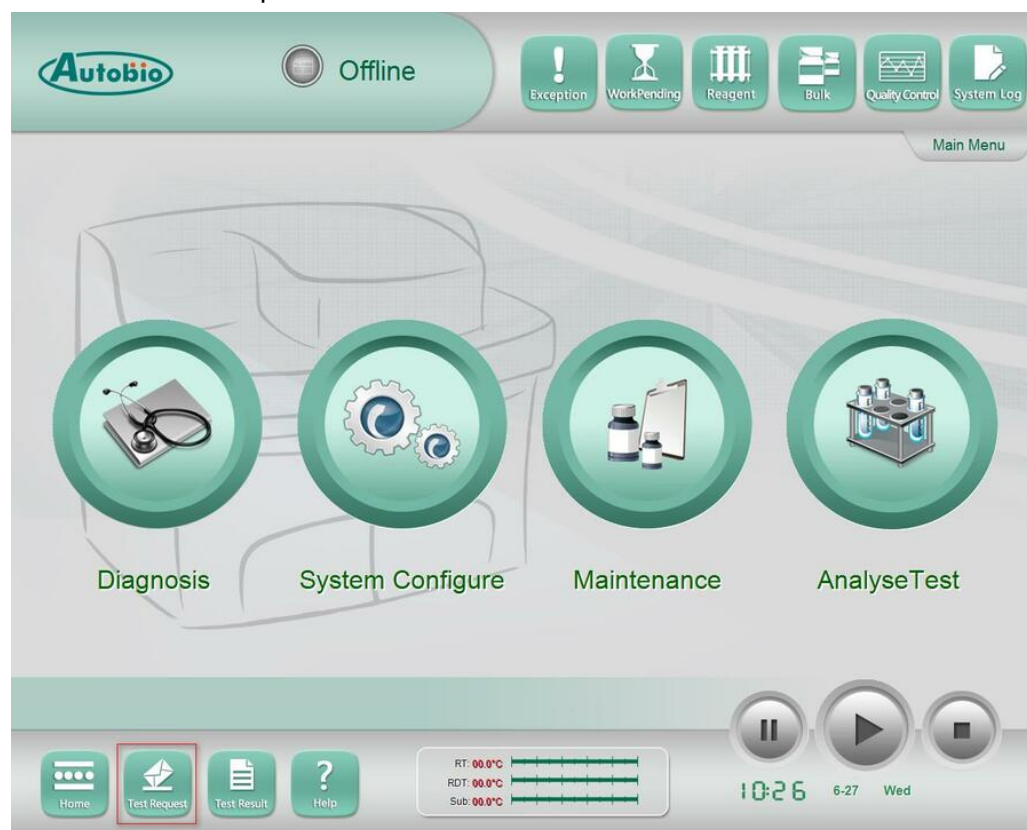
11. If there is any sample which contains blood clot, blood streak, protein group or streak after centrifugation, repeat the centrifugation after remove impurities.

Note: Our company will not bear any liabilities caused by the abnormal test result due to noncompliance with requirements of pretreatment.

11.1.6.2 Edit Sample

11.1.6.2.1 There are two options to enter into the interface of "Test Request":

First: Click 'Test Request' on bottom left and enter into the interface.

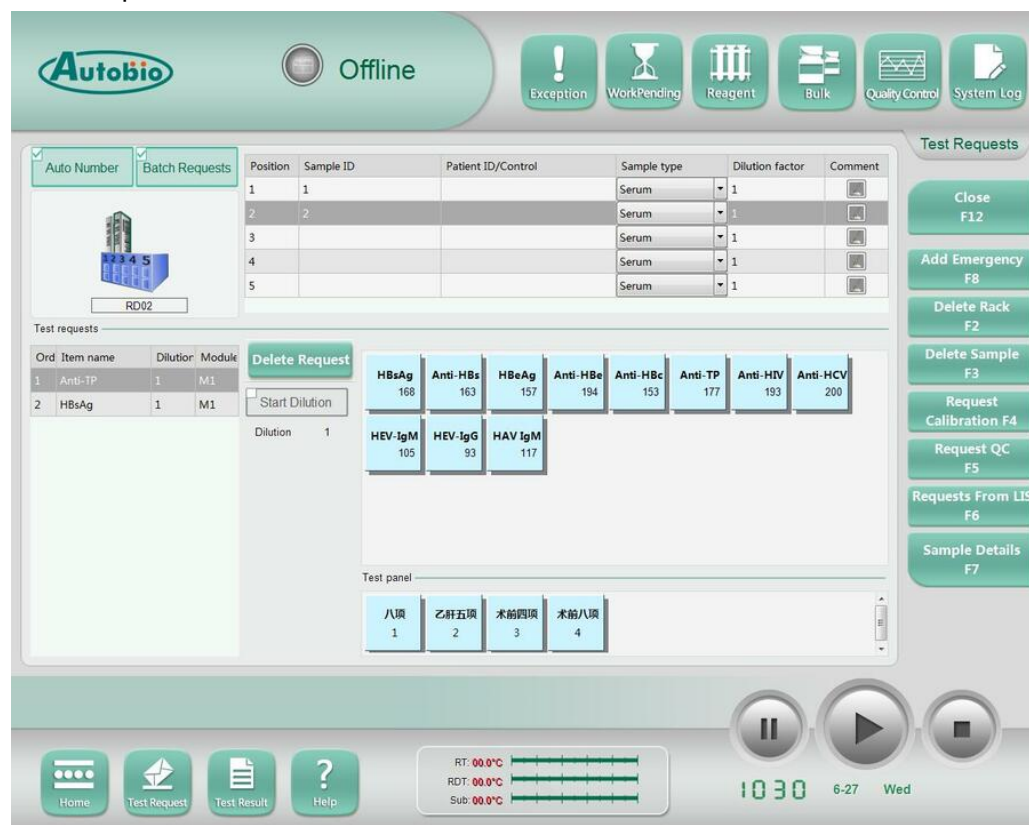


Second: Click on main menu 'Analyse Test'--'Patient Test Request' and enter into the interface of 'Test Request'



11.1.6.2.2 Edit Sample Information (Please refer to 'Chapter 3 Patient Sample' for details)

1. Input sample rack bar code by Bar code Scanner or manual input
2. Input sample number in 'Position'
3. Click required items and item name will displayed on left 'Test Request' area, it means adding is successful; If the item is not required, click 'Delete Request' in 'Test Request' area.



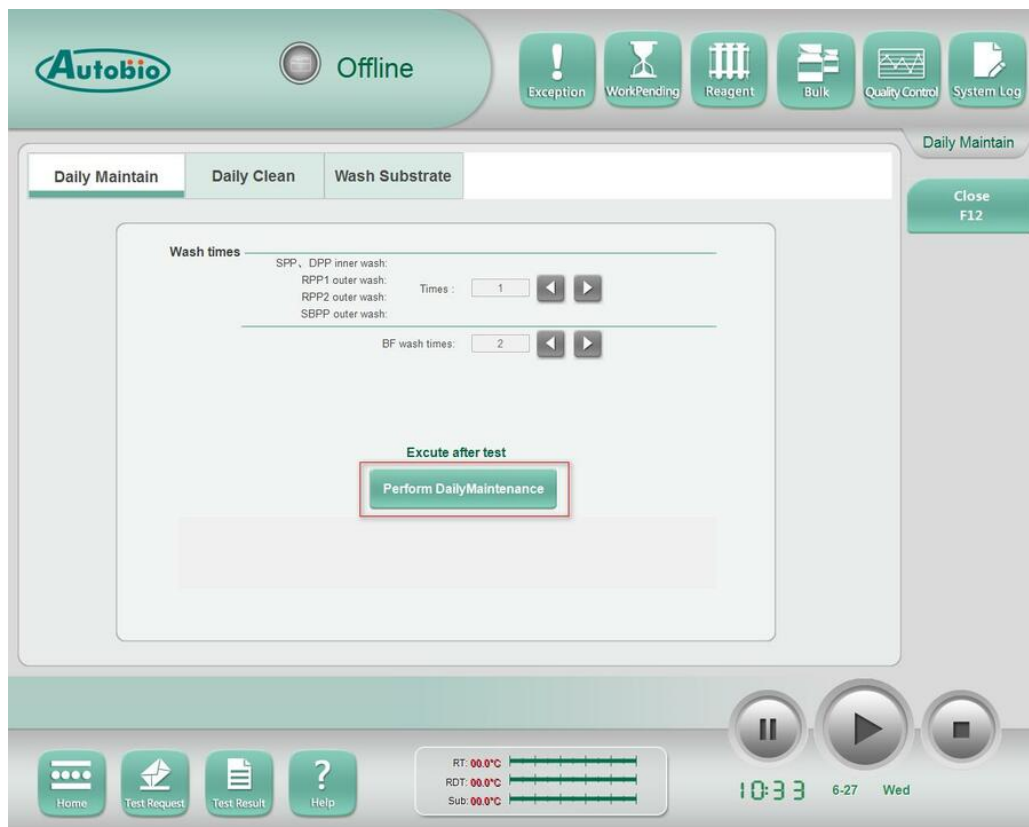
11.1.6.3 Start to test

After completion of sample edit, put the sample rack into SRK and close the SRK cover, click 'Start' button and start to test.

11.1.7 Daily Maintenance

Click on main menu 'Maintenance' - 'Daily Maintenance' - 'Perform Daily Maintenance'.



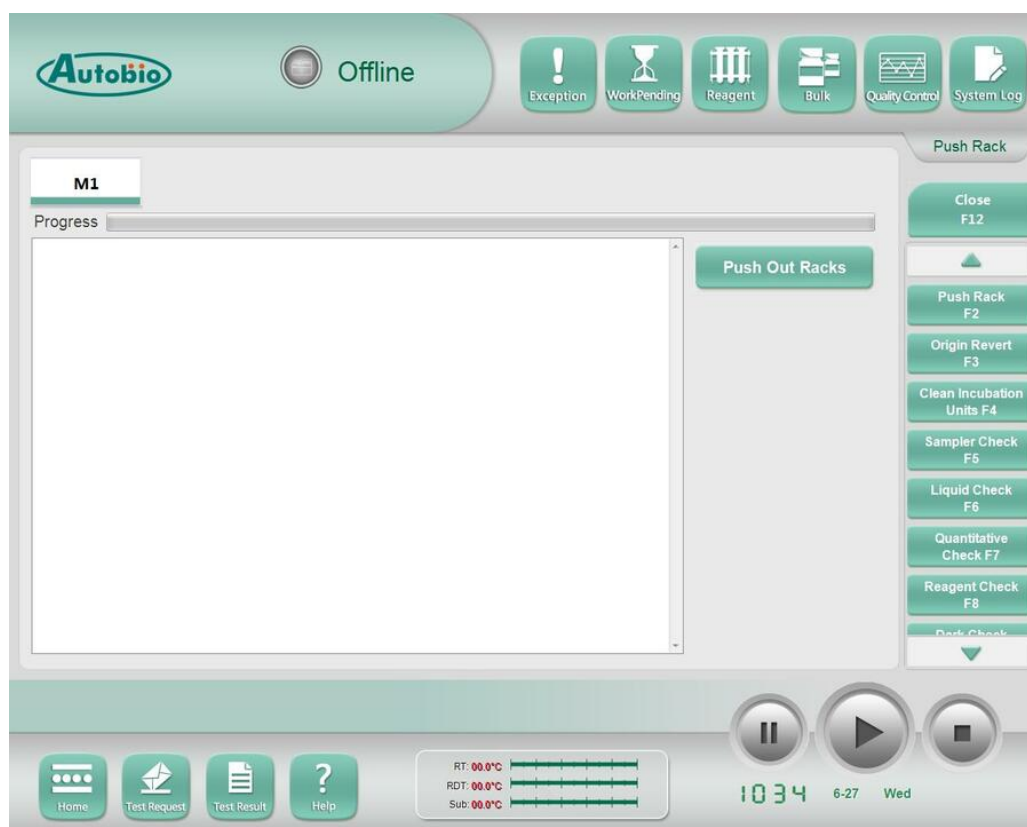


11.1.8 Close the Instrument

Power off the instrument and computer after test is completed (the power switch is on the rear of instrument).

CHAPTER 12 DIAGNOSIS

Click 'Diagnosis' in the main menu entering 'Diagnosis' interface to facilitate your check of system malfunction. The list below is instruction of each button, please read in detail.



Buttons Description:

Button Name	Description
Close	Close the window and return to the main window
Push Rack	Push out the uninitialized sample rack in the fault status
Clean Incubation Units	Clear away all reaction vessels in the reaction stage and preachment tray.
Sampler Check	Check the quantity, position and bar code of sample by scanner inside.
Liquid Check	The sample pipettor and reagent pipettor will detect the liquid level of sample, reagent and diluent to check the liquid detection function.
Quantitative Check	The difference between the test value and set value is measured by the balance to check dispense accuracy.
Reagent Check	Scan the bar code of the reagent to check the information and liquid level
Dark Value Check	Check the dark value of the detection device with nothing inside to confirm the dark value not exceeding the normal

	range.
Light Source Check	Check the repeatability of the light source
Pressure Test	Check whether the sample pipettor is blocked
BF Test	Perform the injection and suction test, to judge the washing problems encountered
AUTOBIO Light Source Check	Perform AUTOBIO light source test, to judge whether the value of light is in compliance with the requirements. Because this operation may cause damage to the PMT, it is necessary for the professional to perform this operation.
Mixer Test	Carries on the mixing test, to judge whether the mixing motor is normal, the mixing intensity is suitable.
Liquid Level Calibration	By sending the command to the control machine, the detection difference of the liquid surface is obtained. According to these differences, the current operation is determined and the threshold value of the detection is obtained
Report Test	Click "PRINT" to print the test report



Note:

Please contact with engineers of AUTOBIO before diagnosis test, make sure whether you can diagnose or adjust those projects or not.

CHAPTER 13 TROUBLESHOOTING

Simple troubles information and solution will be stated as follow, if there are any other problems and errors, please contact with our engineer as soon as possible.

Error 1:

Description:

Reaction vessels is not enough or slide is blocked, please check immediately.

Overtime for supplying of reaction vessels, lacking of reaction vessels.

Error analysis:

1. Software—consumables—warning of lacking of RV.
2. There is not enough RV in the container.
3. Reaction vessels slide get stuck.
4. Reaction vessels slide inlet get stuck.

Solutions:

1. Execute steps from the software: “consumables”—“adding reaction vessels” and click “OK”.
2. Supply reaction vessels into the container in time and click “OK”.
3. Open the cover of the container, clear off the stuck vessels in the slide, and close the cover, then click “OK”.
4. Open the cover of the container, clear off the stuck vessels at the inlet of slide, and close the cover, then click “OK”.

Error 2:

Description:

The lip of the reaction vessels alignment device is open.

Error analysis:

The cover of reaction vessels feeder is not closed.

Solution:

Closet the cover and click “OK”

Error 3:

Description:

Click “OK”, it is displayed on the interface “Disable” for test.

Error analysis:

The cover of the sample loading device is not covered.

Solution:

Re-close the cover of sample loading device and click “OK”.

Error 4:

Description:

Sample line jam

Error analysis:

The sampling line sensor is misfired.

Solution:

After the test is finished, the instrument is reinitialized, and put into the new samples and Click "OK" to start the test.

Error 5:**Description:**

The user presses the emergency stop button on the instrument.

Error analysis:

The emergency stop button is pressed.

Solution:

1. Open the front cover of the instrument, clean off the waste reaction vessels.
2. Re-close the cover of reaction tray, remove each pipette tip and the transfer arm to the home position.
3. Close the front cover, click emergency stop button.
4. Reinitialize the instrument, and reedit the sample, and click "OK" to start the test again.

Error 6:**Description:**

Front cover of the instrument is open, emergency stop.

Error analysis:

Front cover of the instrument is open.

Solution:

1. Open the front cover of the instrument, clean off the waste reaction vessels.
2. Re-close the cover of reaction tray, remove each pipette tip and the transfer arm to the home position.
3. Close the front cover.
4. Reinitialize the instrument, and reedit the sample, and click "OK" to start the test again.

Error 7:**Description:**

Rear cover of the instrument is open, emergency stop.

Error analysis:

Rear cover of the instrument is open.

Solution:

1. Open the front cover of the instrument, clean off the waste reaction vessels.
2. Re-close the cover of reaction tray, remove each pipette tip and the transfer arm to the home position.
3. Close the front and rear cover.

4. Reinitialize the instrument, and reedit the sample, and click “OK” to start the test again.

Error 8:**Description:**

Transfer arm 1 can not put up the RV in right place constantly for 3 times, please check if there is any problems for RV alignment device or transfer arm 1 .

Error analysis:

RV alignment device or transfer arm 1 is abnormal.

Solution:

1. Please refer to Error 1 if RV supply is abnormal.
2. If RV supply is normal, click “PAUSE” button, after the testing items is completed, reinitialize the instrument and start test again.

Error 9:**Description:**

Buffer area is full of unloaded sample racks, please clear off the sample racks unloaded.

Error analysis:

Buffer area is full of unloaded sample racks.

Solution:

Open the cover of sample loading device, take out of the sample racks from buffer area and close the cover, click “OK” and continue the test.

Error 10:**Description:**

The instrument is not initialized and can not be started for test.

Error analysis:

Start the test in the status of “NOT READY”

Solution:

Reinitialize the instrument and click “OK” to start the test.

Error 11:**Description:**

The sensor for the distilled water can not work properly, please check.

Error analysis:

Distilled water is not full in the distilled water bucket in the limit time, the outer join for distilled water is not enough.

Solution:

Supply the distilled water without delay.

Error 12:

Description:

This test is coming to an end, the sample management unit stops sample injection.

Error analysis:

The ongoing test is going to be completed. It is not permitted for sample injection.

Solution:

Click "OK" and continue the test after the ongoing test is completed.

Error 13:**Description:**

When RPP1V driving toward downside, the RPP1V stays above RTT/ORTT while the PTT/ORTT are turning, or RPP1V stays above RDTT while the RDTT is turning.

Error analysis:

The reagent carousel cover is not closed.

Solution:

Circumstance 1: "PAUSE" is displayed on the interface: After the ongoing test is completed, close the reagent carousel cover and front and rear cover of instrument, then reinitialize the instrument and continue the test.

Circumstance 2: "NOT READY" is displayed on the interface, test is stopped:

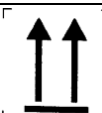
1. Open the front cover of the instrument, clear off the waste reaction vessels in the reaction tray.
2. Close the cover of reaction tray again, move each pipettor pin and transfer arm to HOME position.
3. Close the front and rear cover of the instrument and reaction tray cover.
4. Reinitialize the instrument and reedit the sample, click "OK" to start test.

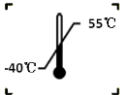
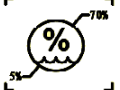
CHAPTER 14 TRANSPORT AND STORAGE

14.1 Transport

The transport of the instrument should be avoided of moisture, water, excessive vibration and extrusion. Handling, loading and unloading should be executed under the guidance of professional engineers. Temperature range between $-40^{\circ}\text{C} \sim +55^{\circ}\text{C}$, and relative humidity is 5% -70%RH .

Essential information of symbols on instrument's package as below:

Symbol	Description
Serial Number: 	This symbol shall be accompanied by the manufacturer's serial number.
Manufacturer: 	This symbol shall be accompanied by the name and the address of the manufacturer
Authorized Representative In The European Community: 	This symbol shall be accompanied by the name and the address of the authorized representative in the European Community
Keep away from sunlight: 	Please keep away from sunlight when transport
Keep Dry: 	Please keep dry when transport
Fragile: 	Contents of the transport package are fragile therefore it shall be handled with care
This way up: 	Indicates correct upright position of the transport package
Do not roll: 	Transport package shall not be rolled

Do not stack:		Stacking of the transport package is not allowed and no load should be placed on the transport package
Temperature limitation:		The upper and lower limits of temperature when transport
Humidity limitation:		Indicate the range of humidity to which the medical device can be safely transport

14.2 Storage

The instrument should be stored in a room with non-chemical, non-corrosive gas, good ventilation, sanitation, temperature range between $-40^{\circ}\text{C} \sim +55^{\circ}\text{C}$, and relative humidity is 5% -70%RH.

CHAPTER 15 EMC INFORMATION

EMC safety instructions:

- a) **AutoLumo A2000 Plus** meets emission and immunity requirements of EN 61326-2-6:2013.
- b) The equipment is in accordance with the class A equipment of CISPR11 design and testing. In the family environment, the device may cause radio interference, the need to take protective measures.
- c) It is recommended to evaluate the electromagnetic environment before the use of **AutoLumo A2000 Plus**.
- d) Prohibit the use of this equipment in a strong radiation source (such as non-shielded RF source) side, as it may interfere with normal operation of the device.

Note 1: AUTOBIO has the responsibility to provide electromagnetic compatibility information of **AutoLumo A2000 Plus** to customers or users.

Note 2: The user has the responsibility to ensure device's electromagnetic compatibility environment, so that **AutoLumo A2000 Plus** can work properly.

APPENDIX 1

Fatal Flag

Flag	Explanation	Correction
CCR	Caused by: •The derived formula contains a test which cannot generate a result. •The system is unable to calculate the result caused by another error.	1. For derived results, review each test result used in the derived result formula. If a result failed. Troubleshoot according to the flag for that result. 2. Review the Event Log and troubleshoot according to the error events with a similar date and item to this event. 3. Repeat test Repeatedly export all testing results from the formula for a certain test. 4. If the problem persists, please contact technical support.
CLT	Detect clot in the sample tubes or in the reaction vessel.	1、 Review the Event Log and troubleshoot according to the error events with a similar date and time to this event. 2、 Repeat the test. 3、 If the problem persists, please contact technical support
IND	• The test result locates at the low dose position of the curve for sandwich assay. The results cannot be separated from the system error as the RLU or concentration is too low. •The test result locates at the low dose position or the high dose position of the curve for competitive assay. The results cannot be separated from the system error as the RLU is too low or	1. Check the event close to the date of the problem happened in the "Event Log" Please contact technical support. 2. Run QC test (Quality Control), and test again. Eliminate this error if the QC result is not in the normal range.

	too high.	
QNS	The sample volume is insufficient.	1. Follow the troubleshooting instruction for the QNS event in the Event Log. 2. Repeat the test(s). 3. If the problem persists, please contact technical support.
RLU	The RLU value is exceeded the acceptable illuminometer measuring range.	1. Review the Event Log and troubleshoot according to the error events with a similar date and time to this event. 2. Repeat the test. 3. If the problem persists, please contact technical support.
SYS	A device error occurred during processing.	1. Review the Event Log and troubleshoot according to the error events with a similar date and time to this event. 2. Repeat the test. 3. If the problem persists, please contact technical support.
TRI	When test culturing, incubator temperature is outside the acceptable range, it signs reaction stage temperature exceeds the normal range.	1. Regularly monitor incubator temperature until into the acceptable range. If the instrument is to be restart, or the cover of instrument is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return to normal, or if the problem persists, please contact technical support.
TRIH	When test culturing, and incubator temperature is outside the acceptable	1. Regularly monitor incubator temperature until into the acceptable range.

	range, it signs reaction stage temperature is too high.	If the instrument is to be restart, or the cover of instrument is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return to normal, or if the problem persists, please contact technical support.
TRS	When the substrate is allotted, and the substrate temperature is outside the acceptable range, it signs substrate temperature exceeds the normal range.	1. If the instrument is to be restart, or the cover of analyzer is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return to normal, or if the problem persists, please contact technical support.
TRSH	When the substrate is allotted, and the substrate temperature is outside the acceptable range, it signs substrate temperature is too high.	1. If the instrument is to be restart, or the cover of instrument is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return

		to normal, or if the problem persists, please contact technical support.
TRR	Reagent storage temperature exceeds the normal range.	1. If the instrument is to be restart, or cover of the instrument is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return to normal, or if the problem persists, please contact technical support.
TRRH	Reagent storage temperature is too high.	1. If the instrument is to be restart, or cover of the instrument is opened, you need to wait at most 10 minutes, so that the temperature returns to normal. If the system power is turned off for a long time, then you need to wait at most an hour, so that the temperature returns to normal. 2. Repeat test when incubator temperature reaches an acceptable range. 3. If the temperature does not return to normal, or if the problem persists, please contact technical support.
QSB	When dispensing substrate, tests display abnormal because of lacking of substrate.	1. Please check the log to find related events. 2. Change substrate. 3. If the problem persists, please contact technical support.
ERR	Undetectable causes abnormal test.	1. Reasons undetectable caused abnormal (unplanned shutdown). Please check the log to find related events. 2. If the problem persists, please contact technical support.

QWB	There is a lack of lotion or distilled water while the test is in progress.	1. Please check the remaining amount of lotion.
QNR	There is a lack of reagent when allotting reagent.	1. Check the amount of reagent remaining after the experiment. 2. If the remaining amount is normal, please contact technical support.
DRK	High dark value.	Please contact technical support

Non-Fatal Flag

Flag	Explanation	Correct operation
CEX	The calibration curve or cut-off value expires.	Repeat the test, and recalibrate the assay.
REJ	The abnormal RLU is rejected, and it won't participate in the calculation.	1. Whether the instrument is abnormal when make calibrate assay in the log. 2. Recalibrate
OVD	The result is the test result after overrun dilution	Software is already automatically multiplied by the dilution factor.
OLR	The diluted result is less than the dilute limit. It is not suggested to do the dilute test.	Without dilution, recommended to re-test.
CRH	Test results of patients exceed the upper limit of the critical range.	This is a valid test result. No corrective action is necessary.
CRL	Test results of patients exceed the lower limit of the critical range.	This is a valid test result. No corrective action is necessary.
EXS	Substrate expires.	1. Please replace the substrate bottle. 2. Repeat test.
GRY	For qualitative or semi-quantitative analysis, test results or quality control results are in gray area.	This is a valid test result. No corrective action is necessary.
LEX	The reagent pack lot is expired.	1. Unload the expired reagent pack and load a new one. 2. Repeat the test(s).

ORH	The result is above the upper limit of the reference range.	This is a valid test result. No corrective action is necessary.
ORL	The result is below the lower limit of the reference range.	This is a valid test result. No corrective action is necessary.
OVR	The calculated concentration is above the highest or most concentrated calibrator. This flag is only used for quantitative and semi-quantitative assays.	- Review the Event Log for error events with a similar date and time to this event. - If events occurred, follow the troubleshoot instructions. - Take one of the following actions: - If events occurred, and you performed troubleshooting procedures already, run quality controls and then repeat the test. - If the quality controls are in range, and the result is reported as greater than the value of the highest calibrator ($>X$), you may be able to dilute the sample. To identify whether the assay allows dilutions, refer to the product instructions. - If the quality controls are not in range, follow the QC troubleshooting instructions. - If no events occurred, take one of the following actions: - If the result is reported as $>X$, you may be able to dilute the sample. To determine whether to allow dilutions, see product instructions. If dilution is allowed, dilute and repeat the test. - If dilution is not allowed, no further action is necessary. - If you have questions about the result or if the problem persists, please contact technical support.
PEX	For the reagent box, the stability time of open box has elapsed. When the	Unload the expired reagent pack and load a new one. If the lot of the reagent pack is different from the

	system first puncture it, which has started testing the stability of the open box.	expired one, recalibrate. 2. Repeat test.
QCF	The results of quality control violate one or more quality control rules.	1. Check "QC Chart and Data" interface, and find which rule not meet. 2. Please operate according to the explanation of quality control troubleshooting.
LQCF	The final quality control of sample tests lose control. (Regardless of any test lose control, follow-up sample test result flag is marked as LQCF.)	1. The project is out of control the last time. Suggest redoing control or calibration.
QEX	Lot of quality control expired, which is valid only for quality control tests.	1. Add a new lot number of quality control which is not expired. 2. Repeat test.
HCV	The %CV of the standard test result is beyond the scope allowed.	1. Repeat test. 2. If the problem persists, please contact technical support.
DRX	The results come from the derivation formula.	This is a valid test result. No corrective action is necessary.

APPENDIX 2

1. Product specific risks may affect service personnel and corresponding protective measures for these risks:

	Risk	Module	Protective Measures
(1)	Waste liquid and discard sample cup may cause biological hazard to service personnel	Other	1. When repairing instrument, service personnel must wear protective mask and glove. 2. Service personnel should get a professional pre-service training
(2)	When repairing instrument some hazardous liquid may splash into eyes or mouth of service personnel and cause biological hazards	Other	1. When repairing instrument, service personnel must wear protective mask and glove. 2. Service personnel should get a professional pre-service training.
(3)	When repairing instrument the moving part may cause a machine hazard to service personnel	Other	1. Service personnel should get a professional pre-service training. 2. Close the cover of instrument during instrument working. 3. Add warning symbol on instrument
(4)	The risks of crushing the hand by the upper, front and back covers.	Frame	1. Service personnel should get a professional pre-service training. 2. Add warning symbol on instrument
(5)	The power indicator may not lightened when the voltage is 12V, which may cause the risk of electric shock.	ELEC	1. Before operating the electrical cabinet, assure that the power switch is off, the Power cord is cut off and the instrument should stay out of operation for 10 minutes. 2. Service personnel should get a professional pre-service training.

2. Verify the safe state after repair:

- (1) To check that the corresponding circuit is connected correctly, the connectors are fixed tightly.
- (2) To check the liquid pipes are connected to the pumps properly and the joints are fastened by nylon wires tightly.

- (3) To clear and fix the circuit and pipes without contacting and friction with moving parts.
- (4) To clean off the reaction cups and bolts left inside the instrument.
- (5) To check and assure that the Power cord is not loose.
- (6) To check and assure that there is no spark and abnormal smell caused by the hot or burned wires in the power box of the instruments.
- (7) Check and ensure all protective earth is fixed tightly in instrument.
- (8) Check and ensure supply mains are fixed tightly.

APPENDIX 3

Sample Tube Barcode Requirements:

Support Barcode Type	CODE 128 (Ordinary)、CODE 39、NW7、ITF、CODE 93
Tube Type	Height: 100 mm or 75mm; Diameter: 13mm
Barcode Dimension Requirements	Length of Barcode: $\leq 70\text{mm}$ (100mm Tube); $\leq 50\text{mm}$ (75mm Tube) Width of Barcode: $\geq 8\text{mm}$; (Bar Height) Bar Width: $\geq 0.15\text{mm}$; (Width of the thinnest bar) Empty Area: $>2.5\text{mm}$, Suggestion 3mm; (The empty area at the two side of Barcode, See figure 1) Width to Narrowness: $\geq 2:1$; (Ratio between wide bar and narrow bar)
Barcode Scanning Area	In case of scanning error, confirm that the empty area at the two edge of Barcode is more than 2.5mm. Stick the label aligning with center line of sample tube. (See figure 2)
Barcode Quality	Barcode should be clean, non-contamination, bar clear, straight and unbroken(See figure 3: Incorrect Barcode).



Figure 1: Know the Barcode

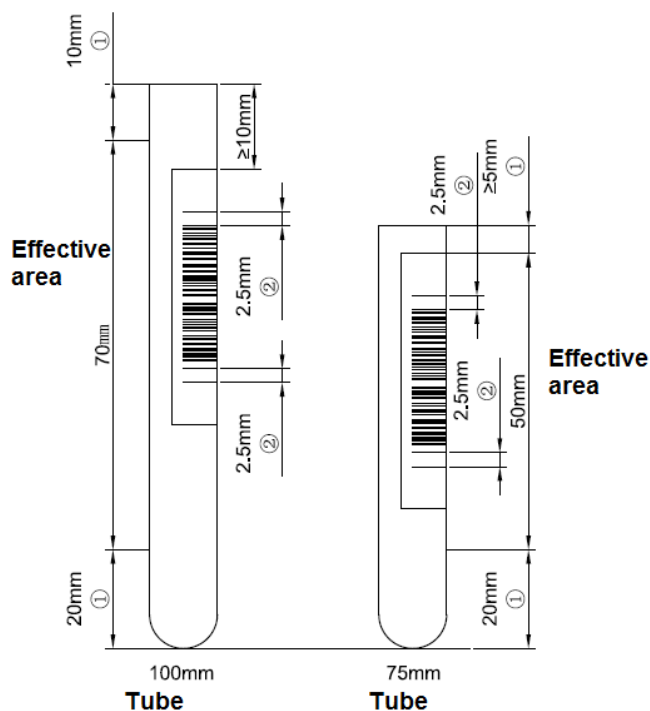


Figure 2: Paste Requirements
At the position of diagram①, blank area should be left at the edge of label. Meanwhile, label should be pasted on sample tube aligning with its center line.
At the diagram② area, any other information can not be displayed in case of bar skinning error.

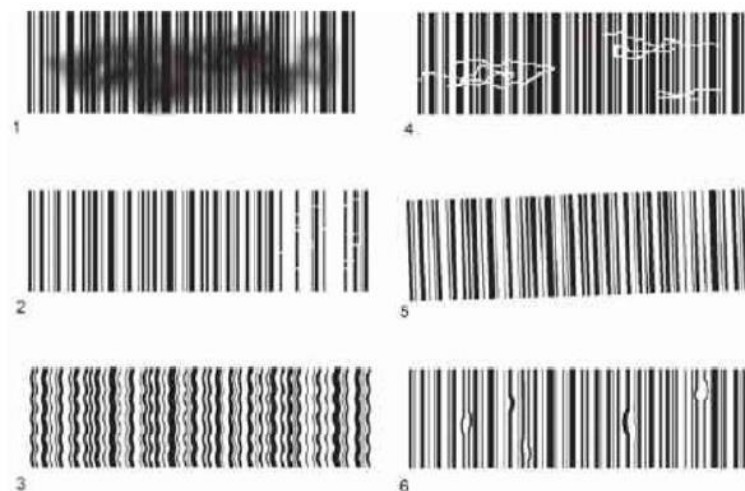


Figure 3: Incorrect Barcode:

1. Barcode smearing or contamination
2. Barcode fading
3. Barcode line wavy
4. Barcode scratching
5. Barcode tilt
6. Blank in the Barcode

Other factors may influence the Barcode:

Print Contrast Signal	PCS Value $\geq$ 75%. Optical index of Barcode symbol. The bigger of PCS value, the better of the optical characteristics. Low PCS value may result distinguishing black and blank bars improperly.
Barcode Coat Requirements	Barcode surface is coated with protective plastic after printing. Mate coat is suggested. The Barcode reader is same with eyes. If the reflected light is too strong, the Barcode can not be read successfully.
Barcode installation distance (Service staff)	Adjust by using of tooling. At the distance of 35mm (from scanning window to Barcode) .

APPENDIX 4

Explanation

There are some English words or letter abbreviation on the instruments to indicate name, position or usage. See the following details:

Noun	Description
PC	Indication computer or interface connecting with computer
POWER	Power supply indication
ON	Instrument Power on
OFF	Instrument Power off
EM.STOP	Emergency Stop Symbol
STAT	Emergency Position, indicating position of emergency sample racks
CC	Position of Wash Buffer
SC	Position of System Wash Container
DW	Position of Distilled Water Container
WL	Position of Waste Liquid Container
SW1	Position of Solid Waste 1
SW2	Position of Solid Waste 2
DW-IN	Distilled Water Inlet
WL-OUT	Waste Liquid Outlet
SG1	Position of Substrate Group 1
SG2	Position of Substrate Group 2



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